

MH1101 Calculus II – Revision Notes

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Abstract

Comprehensive revision notes for MH1101 Calculus II, integrating core definitions, key results, worked examples, and exam-focused techniques.

Course Overview & Topic Map

Topic Area	Chapters	Key Concepts
Integrals	Ch. 1	Antiderivatives; area problem; definite integrals; Fundamental Theorem of Calculus; substitution rule; improper integrals; Mean Value Theorem for integrals
Applications of Integrals	Ch. 2	Area between curves; volumes (disks/washers); cylindrical shells; arc length
Techniques of Integration	Ch. 3	Integration by parts; trigonometric integrals; trigonometric substitution; partial fractions; numerical integration (midpoint, trapezoidal, Simpson's rule)
Sequences and Series	Ch. 4	Limits of sequences; monotone sequences; series
Convergence Tests	Ch. 5	Integral test; comparison tests; absolute vs. conditional convergence; ratio test; root test
Power Series	Ch. 6	Power series; Taylor series; Maclaurin series; binomial series

This revision guide follows the official MH1101 chapter structure while emphasizing problem-solving techniques and exam-relevant intuition.

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How to Use This Document

Structure of the Notes

These notes are designed to be used together with your lectures and tutorials. Each chapter is organised into:

1. **Core Theory** — definitions, statements of key results, and conceptual interpretation.
2. **Mathematical Framework** — formulas, standard manipulations, and common patterns.
3. **Worked Examples** — representative questions with full solutions.
4. **Applications** — modelling or interpretation links (where relevant).
5. **Exam Focus** — high-yield checklists, typical traps and time-saving techniques.

A practical way to revise:

- First pass: read Core Theory + Key Formula boxes to map the chapter.
- Second pass: attempt Worked Examples yourself *before* checking solutions.
- Final pass: review Common Mistake boxes and your own error log before timed practice.

Colour and Box System (Legend)

The document uses a colour-coded system. Each box has a **name** at the top so you can quickly recognise its purpose:

Colour	Box Name	Use This For
Green	Definition / Theorem	Formal statements of definitions and theorems used throughout the course.
Blue	Key Formula	Core identities and methods you should be able to apply quickly.
Brown	Worked Example	Fully worked questions. Try on your own before reading the solution.
Red	Common Mistake	Typical errors, misleading shortcuts and conceptual traps to avoid.
Yellow	Policy Application	Interpretation notes (units, modelling intuition, and “what the answer means”).
Purple	Exam Focus	High-yield checklists, time-saving techniques, and typical exam patterns.
Cyan	Extension	Optional enrichment: elegant viewpoints, stronger generalisations, or structural remarks.

When you print in black and white, you can still identify boxes by their **title text**:

- Look for the name at the top: *Definition, Theorem, Key Formula, Worked Example, Common Mistake, Policy Application, Exam Focus, Extension*.
- In digital version, use the colour + name; in printed version, rely mainly on the name.

1 Integrals

1.1 Antiderivatives and Indefinite Integrals

Definition 1.1: Antiderivative and indefinite integral

Let f be defined on an interval I . A function F is an **antiderivative** of f on I if

$$F'(x) = f(x) \quad \text{for all } x \in I.$$

The **indefinite integral** denotes the family of all antiderivatives:

$$\int f(x) dx = F(x) + C.$$

Theorem 1.1: All antiderivatives differ by a constant

If F and G are antiderivatives of f on an interval I , then $F(x) - G(x)$ is constant on I .

Proof. Since $F'(x) = f(x) = G'(x)$ on I , we have $(F - G)'(x) = 0$ on I . Hence $F - G$ is constant on I . $\square$

Key Formula: Basic antiderivative library

For constants $a \neq -1$ and $k \neq 0$:

$$\int x^a dx = \frac{x^{a+1}}{a+1} + C, \quad \int e^{kx} dx = \frac{1}{k} e^{kx} + C, \quad \int \cos(kx) dx = \frac{1}{k} \sin(kx) + C, \quad \int \frac{1}{x} dx = \ln|x| + C.$$

Worked Example: General antiderivative (power simplification)

Find the most general antiderivative of $\frac{5 - 4x^3 + 2x^6}{x^6}$ for $x \neq 0$.

Solution. Rewrite using powers:

$$\frac{5 - 4x^3 + 2x^6}{x^6} = 5x^{-6} - 4x^{-3} + 2.$$

Integrate term-by-term:

$$\int 5x^{-6} dx = 5 \cdot \frac{x^{-5}}{-5} = -x^{-5}, \quad \int (-4x^{-3}) dx = -4 \cdot \frac{x^{-2}}{-2} = 2x^{-2}, \quad \int 2 dx = 2x.$$

Hence

$$\int \frac{5 - 4x^3 + 2x^6}{x^6} dx = -\frac{1}{x^5} + \frac{2}{x^2} + 2x + C.$$

Common Mistake: Dropping $+C$ (and losing solutions)

Indefinite integrals represent *families* of functions. Omitting $+C$ can remove valid solutions, especially when applying an initial condition later.

Exam Focus: Solving for f given f' (and initial data)

Workflow:

- (1) Integrate: $f(x) = \int f'(x) dx = F(x) + C$.
- (2) Apply the condition (e.g. $f(x_0) = y_0$) to solve for C .
- (3) If f'' is given, integrate twice: $f'(x) = \int f''(x) dx + C_1$, then $f(x) = \int f'(x) dx + C_2$.

1.2 Area Problem and Riemann Sums

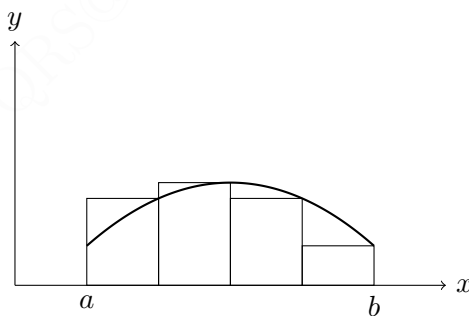
A key motivation for definite integrals is **area/accumulation**. For $f(x) \geq 0$ on $[a, b]$, the area under $y = f(x)$ from a to b can be approximated by rectangles.

Let $[a, b]$ be partitioned into n equal subintervals with

$$\Delta x = \frac{b-a}{n}, \quad x_i = a + i\Delta x \quad (i = 0, 1, \dots, n).$$

Choosing sample points $x_i^* \in [x_{i-1}, x_i]$, the associated **Riemann sum** is

$$S_n = \sum_{i=1}^n f(x_i^*) \Delta x.$$



1.3 Definite Integrals and Basic Properties

Definition 1.2: Definite integral

Suppose f is defined on $[a, b]$. Using the equal-width partition with $\Delta x = (b - a)/n$ and sample points $x_i^* \in [x_{i-1}, x_i]$, the **definite integral** of f on $[a, b]$ is

$$\int_a^b f(x) dx := \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x,$$

provided the limit exists and is independent of the choice of sample points. In that case, f is **integrable** on $[a, b]$.

If f changes sign, $\int_a^b f(x) dx$ is interpreted as **net area**: (area above x -axis) minus (area below x -axis).

Key Formula: Core properties of definite integrals

Assume f, g are integrable on $[a, b]$ and c is constant.

$$(P1) \quad \int_a^b c dx = c(b - a).$$

$$(P2) \quad \int_a^b (f(x) \pm g(x)) dx = \int_a^b f(x) dx \pm \int_a^b g(x) dx.$$

$$(P3) \quad \int_a^b c f(x) dx = c \int_a^b f(x) dx.$$

$$(P4) \quad (\text{Additivity}) \text{ If } a < c < b, \text{ then } \int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx.$$

$$(P5) \quad (\text{Comparison}) \text{ If } f(x) \geq 0 \text{ on } [a, b], \text{ then } \int_a^b f(x) dx \geq 0.$$

$$(P6) \quad (\text{Comparison}) \text{ If } f(x) \geq g(x) \text{ on } [a, b], \text{ then } \int_a^b f(x) dx \geq \int_a^b g(x) dx.$$

$$(P7) \quad (\text{Bounds}) \text{ If } m \leq f(x) \leq M \text{ on } [a, b], \text{ then } m(b - a) \leq \int_a^b f(x) dx \leq M(b - a).$$

Worked Example: Average value satisfies its own bounds

Show that if f is continuous on $[a, b]$ with $m \leq f(x) \leq M$, then the average value $f_{\text{ave}} = \frac{1}{b-a} \int_a^b f(x) dx$ satisfies $m \leq f_{\text{ave}} \leq M$.

Solution. By Property P7,

$$m(b-a) \leq \int_a^b f(x) dx \leq M(b-a).$$

Divide by $b-a > 0$ to obtain

$$m \leq \frac{1}{b-a} \int_a^b f(x) dx \leq M,$$

i.e. $m \leq f_{\text{ave}} \leq M$.

Exam Focus: Fast integral comparisons

If you can compare integrands pointwise (e.g. $f \geq g \geq 0$), then you can compare integrals *without computing antiderivatives*. This is often faster and avoids algebra mistakes.

1.4 The Fundamental Theorem of Calculus

Motivation: Why do we need the FTC?

At this point in the course, you know two distinct operations:

- **Differentiation:** Given a function f , compute its instantaneous rate of change $f'(x)$.
- **Integration (as area):** Given a function f on $[a, b]$, compute the accumulated area $\int_a^b f(x) dx$ by taking the limit of Riemann sums.

These look completely unrelated—one is about *rates*, the other is about *totals*. Computing $\int_0^1 x^2 dx$ from the definition means evaluating $\lim_{n \rightarrow \infty} \sum_{i=1}^n \left(\frac{i}{n}\right)^2 \cdot \frac{1}{n}$, which requires summation formulas and careful limit-taking.

The Fundamental Theorem of Calculus reveals that differentiation and integration are **inverse processes**. This has two immediate practical consequences:

- If you *integrate then differentiate*, you recover the original function (FTC Part I).
- If you know an *antiderivative*, you can compute a definite integral instantly without Riemann sums (FTC Part II).

In effect, the FTC transforms integration from a *limit-of-sums calculation* (slow, algebraically tedious) into an *antiderivative evaluation* (fast, routine). This is why we can evaluate $\int_0^1 x^2 dx$ by writing $\left[\frac{x^3}{3}\right]_0^1 = \frac{1}{3}$ instead of summing rectangles.

We now state and prove both parts of the theorem carefully, emphasizing *why* each formula takes the form it does.

Theorem 1.2: Fundamental Theorem of Calculus (Part I)

If f is continuous on $[a, b]$ and

$$F(x) = \int_a^x f(t) dt \quad (a \leq x \leq b),$$

then F is differentiable on (a, b) and

$$F'(x) = f(x) \quad (a < x < b).$$

Intuition behind Part I: Why does the derivative of an integral give back the original function?

Think of $F(x) = \int_a^x f(t) dt$ as the **accumulated area** under $y = f(t)$ from a up to x . When you increase x by a small amount h , the new area is approximately the area of a thin rectangle of width h and height $f(x)$, so

$$F(x+h) - F(x) \approx f(x) \cdot h.$$

Dividing by h and taking $h \rightarrow 0$ gives $F'(x) = f(x)$. The proof below makes this intuition rigorous using the Mean Value Theorem for Integrals.

Proof. Strategy: Use the definition of the derivative and then apply the Mean Value Theorem for Integrals to evaluate the resulting integral increment.

Step 1: Expand $F'(x)$ using the definition.

Let $x \in (a, b)$. By definition,

$$F'(x) = \lim_{h \rightarrow 0} \frac{F(x+h) - F(x)}{h}.$$

Since $F(x) = \int_a^x f(t) dt$ and $F(x+h) = \int_a^{x+h} f(t) dt$, we have

$$F(x+h) - F(x) = \int_a^{x+h} f(t) dt - \int_a^x f(t) dt.$$

By the additivity property (P4), the interval $[a, x+h]$ splits as $[a, x] \cup [x, x+h]$, giving

$$F(x+h) - F(x) = \int_x^{x+h} f(t) dt.$$

Therefore

$$F'(x) = \lim_{h \rightarrow 0} \frac{1}{h} \int_x^{x+h} f(t) dt.$$

Step 2: Apply the Mean Value Theorem for Integrals.

Since f is continuous on $[x, x+h]$ (for small $h > 0$), the Mean Value Theorem for Integrals guarantees a point $c \in [x, x+h]$ such that

$$\int_x^{x+h} f(t) dt = f(c) \cdot h.$$

Substituting into our expression for $F'(x)$,

$$F'(x) = \lim_{h \rightarrow 0} \frac{f(c) \cdot h}{h} = \lim_{h \rightarrow 0} f(c).$$

Step 3: Take the limit as $h \rightarrow 0$.

Because $c \in [x, x + h]$, as $h \rightarrow 0$ we have $c \rightarrow x$ (the point c is squeezed between x and $x + h$). Since f is continuous at x ,

$$\lim_{h \rightarrow 0} f(c) = f(x).$$

Hence $F'(x) = f(x)$, as desired. □

Why is continuity of f required?

The proof relies on two facts:

- The Mean Value Theorem for Integrals, which requires f to be continuous.
- The limit $\lim_{c \rightarrow x} f(c) = f(x)$, which is the definition of continuity at x .

If f has a jump discontinuity, the accumulated area function $F(x)$ may still be continuous, but it will fail to be differentiable at the jump.

Summary of Part I in words: If $F(x)$ accumulates the values of f from a to x , then the *rate of change* of this accumulation at x is precisely $f(x)$ itself.

Worked Example: FTC Part I (direct application)

Find $\frac{d}{dx} \int_1^x \frac{1}{1+t^2} dt$.

Solution. The integrand $f(t) = \frac{1}{1+t^2}$ is continuous on $\mathbb{R}$. By FTC Part I, with $a = 1$,

$$\frac{d}{dx} \int_1^x \frac{1}{1+t^2} dt = f(x) = \frac{1}{1+x^2}.$$

Worked Example: FTC Part I with Chain Rule

Find $\frac{d}{dx} \int_1^{x^4} \sec t \, dt$.

Solution. Let $u = x^4$. Then

$$\frac{d}{dx} \int_1^{x^4} \sec t \, dt = \frac{d}{dx} \int_1^u \sec t \, dt = \frac{d}{du} \left(\int_1^u \sec t \, dt \right) \cdot \frac{du}{dx}.$$

By FTC Part I, $\frac{d}{du} \int_1^u \sec t \, dt = \sec u$. Since $\frac{du}{dx} = 4x^3$, we obtain

$$\frac{d}{dx} \int_1^{x^4} \sec t \, dt = \sec(x^4) \cdot 4x^3 = 4x^3 \sec(x^4).$$

Worked Example: FTC Part I with two variable limits (Leibniz rule)

Find $\frac{d}{dx} \int_x^{x^2} \sin t \, dt$.

Solution. Method: Split the integral at a fixed point (say, 0) and apply FTC Part I to each piece. Write

$$\int_x^{x^2} \sin t \, dt = \int_x^0 \sin t \, dt + \int_0^{x^2} \sin t \, dt = - \int_0^x \sin t \, dt + \int_0^{x^2} \sin t \, dt.$$

Differentiate term-by-term using FTC Part I and the Chain Rule:

$$\frac{d}{dx} \left(- \int_0^x \sin t \, dt \right) = - \sin x,$$

$$\frac{d}{dx} \int_0^{x^2} \sin t \, dt = \sin(x^2) \cdot 2x.$$

Hence

$$\frac{d}{dx} \int_x^{x^2} \sin t \, dt = - \sin x + 2x \sin(x^2).$$

General form (Leibniz rule):

$$\frac{d}{dx} \int_{a(x)}^{b(x)} f(t) \, dt = f(b(x)) \cdot b'(x) - f(a(x)) \cdot a'(x).$$

Exam Focus: Variable limits and Chain Rule

When the upper limit is not just x but a composite function $g(x)$, you *must* apply the Chain Rule:

$$\frac{d}{dx} \int_a^{g(x)} f(t) \, dt = f(g(x)) \cdot g'(x).$$

When both limits are variable, split at a convenient constant and differentiate each piece separately.

Theorem 1.3: Fundamental Theorem of Calculus (Part II)

If f is continuous on $[a, b]$ and F is *any* antiderivative of f (i.e., $F'(x) = f(x)$ on $[a, b]$), then

$$\int_a^b f(x) \, dx = F(b) - F(a).$$

Intuition behind Part II: Why does an antiderivative let us evaluate definite integrals?

Part I showed that the accumulation function $G(x) = \int_a^x f(t) \, dt$ is an antiderivative of f . If F is *any other* antiderivative of f , then F and G differ by a constant (Theorem ??):

$$F(x) = G(x) + C.$$

Evaluating at $x = b$,

$$F(b) = G(b) + C = \int_a^b f(t) dt + C.$$

Evaluating at $x = a$,

$$F(a) = G(a) + C = \int_a^a f(t) dt + C = 0 + C = C.$$

Subtracting,

$$F(b) - F(a) = \int_a^b f(t) dt.$$

The constant C cancels in the subtraction, which is why you can use *any* antiderivative F (with or without $+C$) to compute the definite integral.

Proof. Strategy: Define $G(x) = \int_a^x f(t) dt$. Both G and F are antiderivatives of f , so they differ by a constant. Exploit this to relate $\int_a^b f$ to $F(b) - F(a)$.

By FTC Part I, $G'(x) = f(x)$ on (a, b) . Since F is also an antiderivative of f , we have $F'(x) = f(x) = G'(x)$ on (a, b) . By Theorem ??, there exists a constant C such that

$$F(x) = G(x) + C \quad \text{for all } x \in [a, b].$$

Evaluate at $x = b$:

$$F(b) = G(b) + C = \int_a^b f(t) dt + C.$$

Evaluate at $x = a$:

$$F(a) = G(a) + C = \int_a^a f(t) dt + C = 0 + C = C.$$

Subtract the second equation from the first:

$$F(b) - F(a) = \int_a^b f(t) dt.$$

□

Why is the dummy variable irrelevant?

In the statement of the theorem, we wrote $\int_a^b f(x) dx$ instead of $\int_a^b f(t) dt$. The choice of variable inside the integral doesn't matter:

$$\int_a^b f(x) dx = \int_a^b f(t) dt = \int_a^b f(u) du.$$

All represent the same number (the accumulated area from a to b). The variable is a **dummy variable**—it is local to the integral and does not appear in the final answer $F(b) - F(a)$.

Notation: We often write $F(b) - F(a)$ as $[F(x)]_a^b$ or $F(x) \Big|_a^b$.

Worked Example: FTC Part II (direct evaluation)

Evaluate $\int_{-1}^2 x^3 dx$.

Solution. An antiderivative of $f(x) = x^3$ is $F(x) = \frac{x^4}{4}$ (we omit $+C$ because it will cancel). By FTC Part II,

$$\int_{-1}^2 x^3 dx = \left[\frac{x^4}{4} \right]_{-1}^2 = \frac{2^4}{4} - \frac{(-1)^4}{4} = \frac{16}{4} - \frac{1}{4} = \frac{15}{4}.$$

Worked Example: FTC Part II with endpoint checks

Evaluate $\int_0^\pi \cos x \, dx$.

Solution. An antiderivative of $\cos x$ is $\sin x$. Hence

$$\int_0^\pi \cos x \, dx = [\sin x]_0^\pi = \sin \pi - \sin 0 = 0 - 0 = 0.$$

Interpretation: The area above the x -axis on $[0, \frac{\pi}{2}]$ exactly cancels the area below the x -axis on $[\frac{\pi}{2}, \pi]$, giving net area zero.

Worked Example: FTC Part II with absolute value (sign change)

Evaluate $\int_{-2}^2 |x| \, dx$.

Solution. The function $|x|$ is not differentiable at $x = 0$, so we must split the integral at the point where the formula changes:

$$\int_{-2}^2 |x| \, dx = \int_{-2}^0 (-x) \, dx + \int_0^2 x \, dx.$$

Compute each piece:

$$\int_{-2}^0 (-x) \, dx = \left[-\frac{x^2}{2} \right]_{-2}^0 = 0 - \left(-\frac{4}{2} \right) = 2,$$

$$\int_0^2 x \, dx = \left[\frac{x^2}{2} \right]_0^2 = 2 - 0 = 2.$$

Hence $\int_{-2}^2 |x| \, dx = 2 + 2 = 4$.

Sanity check: By symmetry, the two halves contribute equally; each is a triangle of base 2 and height 2, so area $\frac{1}{2}(2)(2) = 2$.

Common Mistake: Ignoring sign changes in the integrand

If f changes sign or formula on $[a, b]$, you *cannot* apply FTC Part II blindly to $\int_a^b f(x) \, dx$. You must:

- (1) Identify points c where f changes sign or formula.
- (2) Split the integral: $\int_a^b f = \int_a^c f + \int_c^b f$.
- (3) Evaluate each piece separately using the correct formula for f on each subinterval.

Example: $\int_{-1}^1 \frac{1}{x} \, dx$ is **not** $\ln|x| \Big|_{-1}^1 = 0$. The integrand has a singularity at $x = 0$; the integral is improper and divergent.

Exam Focus: FTC Part II workflow

When evaluating $\int_a^b f(x) dx$:

- (1) Find *any* antiderivative $F(x)$ of $f(x)$ (omit $+C$).
- (2) Compute $F(b) - F(a)$.
- (3) Sanity check:
 - If $f \geq 0$ on $[a, b]$, the answer should be ≥ 0 .
 - If f is odd and the interval is symmetric, the answer may be zero.
 - If f changes sign, check that you've split correctly.

Exam Focus: When to use Part I vs. Part II

Use **FTC Part I** when:

- You see $\frac{d}{dx} \int_a^{g(x)} f(t) dt$ (differentiate an integral).
- The question asks for the derivative of an accumulation function.

Use **FTC Part II** when:

- You need to evaluate $\int_a^b f(x) dx$ numerically.
- You are asked to compute a definite integral using an antiderivative.

1.5 The Substitution Rule

Integration by substitution (also called u -substitution) is the reverse of the Chain Rule for differentiation. It allows us to evaluate integrals by transforming them into simpler integrals in a new variable.

Theorem 1.4: Substitution Rule (Indefinite Integrals)

If $u = g(x)$ is differentiable and F is an antiderivative of f , then

$$\int f(g(x)) g'(x) dx = F(g(x)) + C.$$

Equivalently, writing $du = g'(x) dx$,

$$\int f(u) du = F(u) + C.$$

Practical recipe:

- (1) Choose a substitution $u = g(x)$.
- (2) Compute $du = g'(x) dx$.
- (3) Rewrite the integrand and differential in terms of u and du .

- (4) Integrate with respect to u .
- (5) Substitute back $u = g(x)$ to express the answer in terms of x .

Worked Example: u -substitution for indefinite integrals

Evaluate $\int x \cos(x^2) dx$.

Solution. Let $u = x^2$. Then $du = 2x dx$, so $x dx = \frac{1}{2} du$. Substitute:

$$\int x \cos(x^2) dx = \int \cos u \cdot \frac{1}{2} du = \frac{1}{2} \sin u + C = \frac{1}{2} \sin(x^2) + C.$$

Theorem 1.5: Substitution Rule (Definite Integrals)

If g is continuously differentiable on $[a, b]$ and f is continuous on the range of g , then

$$\int_a^b f(g(x)) g'(x) dx = \int_{g(a)}^{g(b)} f(u) du.$$

Key observation: When you substitute $u = g(x)$ in a definite integral, you must also transform the limits of integration: $x = a$ becomes $u = g(a)$, and $x = b$ becomes $u = g(b)$.

Worked Example: u -substitution for definite integrals

Evaluate $\int_2^5 \frac{1}{3-5x} dx$.

Solution. Let $u = 3 - 5x$. Then $du = -5 dx$, so $dx = -\frac{1}{5} du$. Transform the limits:

$$x = 2 \Rightarrow u = 3 - 5(2) = -7, \quad x = 5 \Rightarrow u = 3 - 5(5) = -22.$$

Substitute:

$$\int_2^5 \frac{1}{3-5x} dx = \int_{-7}^{-22} \frac{1}{u} \cdot \left(-\frac{1}{5}\right) du = -\frac{1}{5} \int_{-7}^{-22} \frac{1}{u} du.$$

Reverse the limits (introduces a minus sign):

$$= \frac{1}{5} \int_{-22}^{-7} \frac{1}{u} du = \frac{1}{5} [\ln |u|]_{-22}^{-7} = \frac{1}{5} (\ln 7 - \ln 22) = \frac{1}{5} \ln \left(\frac{7}{22}\right).$$

Exam Focus: Substitution workflow for definite integrals

When using substitution on $\int_a^b f(x) dx$:

- (1) Choose $u = g(x)$ and compute $du = g'(x) dx$.
- (2) Transform limits: $u_1 = g(a)$, $u_2 = g(b)$.
- (3) Rewrite the integral entirely in terms of u (no x should remain).
- (4) Integrate with respect to u from u_1 to u_2 .
- (5) **Do not** substitute back to x —evaluate directly in u .

1.6 Improper Integrals

So far, we have defined $\int_a^b f(x) dx$ only when $[a, b]$ is a finite interval and f is continuous (or at least bounded and integrable) on $[a, b]$. An **improper integral** extends the concept of integration to:

- **Type 1:** Infinite intervals (e.g., $\int_1^\infty \frac{1}{x^2} dx$).
- **Type 2:** Unbounded integrands (e.g., $\int_0^1 \frac{1}{\sqrt{x}} dx$, where $f(x) \rightarrow \infty$ as $x \rightarrow 0^+$).

Definition 1.3: Improper Integral of Type 1 (Infinite Interval)

(a) If $\int_a^t f(x) dx$ exists for all $t \geq a$, then

$$\int_a^\infty f(x) dx := \lim_{t \rightarrow \infty} \int_a^t f(x) dx.$$

(b) If $\int_t^b f(x) dx$ exists for all $t \leq b$, then

$$\int_{-\infty}^b f(x) dx := \lim_{t \rightarrow -\infty} \int_t^b f(x) dx.$$

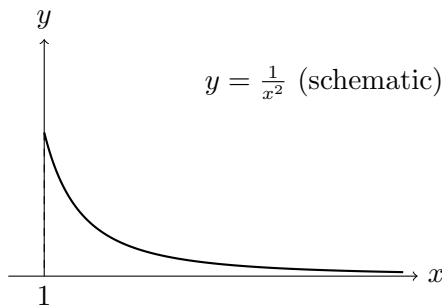
(c) If both $\int_{-\infty}^a f$ and $\int_a^\infty f$ converge, then

$$\int_{-\infty}^\infty f(x) dx := \int_{-\infty}^a f(x) dx + \int_a^\infty f(x) dx.$$

In each case, the integral **converges** if the limit exists (and is finite); otherwise it **diverges**.

Key Formula: p -integrals (Type 1 and Type 2)

$$\int_1^\infty \frac{1}{x^p} dx \text{ converges} \iff p > 1, \quad \int_0^1 \frac{1}{x^p} dx \text{ converges} \iff p < 1.$$



Worked Example: Type 1 divergence

Determine whether $\int_1^\infty \frac{1}{x} dx$ converges.

Solution. Write the definition:

$$\int_1^\infty \frac{1}{x} dx = \lim_{t \rightarrow \infty} \int_1^t \frac{1}{x} dx = \lim_{t \rightarrow \infty} [\ln |x|]_1^t = \lim_{t \rightarrow \infty} \ln t = \infty.$$

Hence the improper integral **diverges**.

Worked Example: Type 1 evaluation (split correctly)

Evaluate $\int_{-\infty}^\infty \frac{1}{1+x^2} dx$.

Solution. Split at 0 and take one-sided limits:

$$\int_{-\infty}^\infty \frac{1}{1+x^2} dx = \int_{-\infty}^0 \frac{1}{1+x^2} dx + \int_0^\infty \frac{1}{1+x^2} dx,$$

provided both sides converge.

Since $\int \frac{1}{1+x^2} dx = \tan^{-1} x + C$,

$$\int_0^\infty \frac{1}{1+x^2} dx = \lim_{t \rightarrow \infty} [\tan^{-1} x]_0^t = \frac{\pi}{2}, \quad \int_{-\infty}^0 \frac{1}{1+x^2} dx = \lim_{t \rightarrow -\infty} [\tan^{-1} x]_t^0 = \frac{\pi}{2}.$$

Therefore $\int_{-\infty}^\infty \frac{1}{1+x^2} dx = \pi$.

Definition 1.4: Improper Integral of Type 2 (Unbounded Integrand)

(a) If f is continuous on $[a, b)$ and $\lim_{x \rightarrow b^-} f(x) = \pm\infty$, then

$$\int_a^b f(x) dx := \lim_{t \uparrow b} \int_a^t f(x) dx.$$

(b) If f is continuous on $(a, b]$ and $\lim_{x \rightarrow a^+} f(x) = \pm\infty$, then

$$\int_a^b f(x) dx := \lim_{t \downarrow a} \int_t^b f(x) dx.$$

(c) If f has a discontinuity at $c \in (a, b)$ and both $\int_a^c f$ and $\int_c^b f$ (as improper integrals) converge, then

$$\int_a^b f(x) dx := \int_a^c f(x) dx + \int_c^b f(x) dx.$$

The integral converges if the limit(s) exist and are finite; otherwise it diverges.

Worked Example: Type 2 at an endpoint

Evaluate $\int_2^5 \frac{1}{\sqrt{x-2}} dx$.

Solution. This is improper at $x = 2$. By definition,

$$\int_2^5 \frac{1}{\sqrt{x-2}} dx = \lim_{t \downarrow 2} \int_t^5 \frac{1}{\sqrt{x-2}} dx.$$

Let $u = x - 2$ so $du = dx$. Then $\int u^{-1/2} du = 2u^{1/2}$, hence

$$\int_t^5 \frac{1}{\sqrt{x-2}} dx = \left[2\sqrt{x-2} \right]_t^5 = 2\sqrt{3} - 2\sqrt{t-2}.$$

Taking $t \downarrow 2$ gives $2\sqrt{t-2} \rightarrow 0$, so the integral converges and equals $2\sqrt{3}$.

Common Mistake: Treating $\int_{-\infty}^{\infty}$ as a symmetric single limit

The definition is *not* $\lim_{A \rightarrow \infty} \int_{-A}^A f(x) dx$.

A two-sided improper integral converges only if **both** one-sided integrals converge:

$$\int_{-\infty}^{\infty} f := \int_{-\infty}^a f + \int_a^{\infty} f, \quad \text{with both terms finite.}$$

Symmetric cancellation can produce a finite number even when the improper integral diverges (see extension below).

Exam Focus: Improper integrals (non-negotiable workflow)

- (1) Identify why it is improper (infinite interval / unbounded integrand / interior singularity).
- (2) Rewrite using the correct one-sided limit(s) *before* manipulating.
- (3) Decide convergence; only then compute the finite value (if it converges).

Extension (Tutorial 3, Question 2, Method 1)

Syllabus link: Chapter 1 — §1.6 *Improper Integrals*.

Proposition (independence of split point).

Assume that for every real c , both $\int_{-\infty}^c f(x) dx$ and $\int_c^{\infty} f(x) dx$ converge. Then for any real a, b ,

$$\int_{-\infty}^a f(x) dx + \int_a^{\infty} f(x) dx = \int_{-\infty}^b f(x) dx + \int_b^{\infty} f(x) dx.$$

Proof. WLOG let $a < b$. For $M > 0$, finite additivity gives

$$\int_{-M}^b f = \int_{-M}^a f + \int_a^b f.$$

Take $M \rightarrow \infty$ (the limits exist by hypothesis) to obtain

$$\int_{-\infty}^b f = \int_{-\infty}^a f + \int_a^b f. \quad (1)$$

Similarly, for $N > 0$, finite additivity gives $\int_a^N f = \int_a^b f + \int_b^N f$; letting $N \rightarrow \infty$ yields

$$\int_a^{\infty} f = \int_a^b f + \int_b^{\infty} f. \quad (2)$$

Rearrange (2) as $\int_b^{\infty} f = \int_a^{\infty} f - \int_a^b f$ and add to (1):

$$\int_{-\infty}^b f + \int_b^{\infty} f = \left(\int_{-\infty}^a f + \int_a^b f \right) + \left(\int_a^{\infty} f - \int_a^b f \right) = \int_{-\infty}^a f + \int_a^{\infty} f.$$

Extension (Beyond Syllabus: Cauchy principal value vs. improper integral)

Syllabus link: Chapter 1 — §1.6 *Improper Integrals*.

Definition. For integrable f on every finite interval, the *Cauchy principal value* is

$$\text{PV} \int_{-\infty}^{\infty} f(x) dx := \lim_{A \rightarrow \infty} \int_{-A}^A f(x) dx \quad (\text{if the limit exists}).$$

This is *not* the same as the improper integral, which requires both $\int_{-\infty}^0 f$ and $\int_0^{\infty} f$ to converge (finite).

Example (odd integrand with finite PV but divergent improper integral).

Let $f(x) = \frac{x}{1+x^2}$ (odd, continuous on $\mathbb{R}$). Compute one-sided integrals:

$$\int_0^A \frac{x}{1+x^2} dx = \left[\frac{1}{2} \ln(1+x^2) \right]_0^A = \frac{1}{2} \ln(1+A^2) \xrightarrow{A \rightarrow \infty} \infty,$$

$$\int_{-A}^0 \frac{x}{1+x^2} dx = \left[\frac{1}{2} \ln(1+x^2) \right]_{-A}^0 = -\frac{1}{2} \ln(1+A^2) \xrightarrow{A \rightarrow \infty} -\infty.$$

Hence the improper integral $\int_{-\infty}^{\infty} f$ *does not converge* (one-sided integrals are not both finite).

But for every $A > 0$, oddness gives $\int_{-A}^A f(x) dx = 0$, so

$$\text{PV} \int_{-\infty}^{\infty} \frac{x}{1+x^2} dx = \lim_{A \rightarrow \infty} 0 = 0.$$

2 Applications of Integrals

2.1 Area Between Curves

Many geometry questions reduce to (i) identifying the correct bounding curves/lines, (ii) locating intersection points, and (iii) setting up a *nonnegative* integrand (area is never negative).

Motivation: From “area under a curve” to “area between curves.”

In Chapter 1 we saw that $\int_a^b f(x) dx$ gives the *signed* area between $y = f(x)$ and the x -axis. What if we want the area of a region enclosed between *two* curves $y = f(x)$ and $y = g(x)$? The idea is simple: at each x , the vertical “height” of the region is $(f(x) - g(x))$ (assuming f is on top). Summing these infinitesimal vertical strips gives the total area. This is exactly how the Riemann-sum derivation works:

- Divide $[a, b]$ into n strips of width Δx .
- Each strip is approximately a rectangle of width Δx and height $f(x_i) - g(x_i)$.
- Summing and taking $n \rightarrow \infty$ gives $A = \int_a^b (f(x) - g(x)) dx$.

The requirement $f(x) \geq g(x)$ ensures every strip has nonnegative height, so we get geometric (unsigned) area.

Theorem 2.1: Area between two curves (w.r.t. x)

Let f, g be continuous on $[a, b]$ and suppose $f(x) \geq g(x)$ for all $x \in [a, b]$. The area of the region bounded by $y = f(x)$, $y = g(x)$, $x = a$, $x = b$ is

$$A = \int_a^b (f(x) - g(x)) dx.$$

Why “top minus bottom”? The integrand $(f - g)$ measures the *vertical distance* between the two curves at position x . If $f(x) \geq g(x)$ on the whole interval, this distance is nonnegative and we get a single integral. If the ordering swaps (i.e., g is on top for part of the interval), the integrand $(f - g)$ becomes negative there, yielding signed area rather than geometric area. To get geometric area, you must **split** at intersection points and always write $(\text{top} - \text{bottom}) \geq 0$.

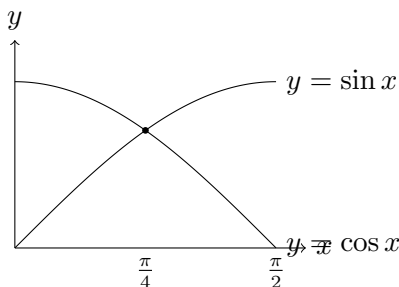
Key Formula: Area Between Curves (Choose dx or dy)

- **Vertical slices (dx):** $A = \int_{x=a}^{x=b} (\text{top} - \text{bottom}) dx$.
- **Horizontal slices (dy):** $A = \int_{y=c}^{y=d} (\text{right} - \text{left}) dy$.
- **Workflow:** find intersection points $\Rightarrow$ split the interval whenever the ordering (top/bottom or right/left) changes.

When to use dx vs. dy :

The choice is a matter of convenience, not correctness. Ask yourself:

- Can I express both boundaries as functions of a *single* variable with *one* formula each? If yes, integrate with respect to that variable.
- Does one choice avoid splitting the integral? For example, a parabola $x = y^2$ and a line $x = y + 2$ are simpler to integrate with dy (one integral from $y = -1$ to $y = 2$), whereas using dx would require two separate integrals.


Worked Example (Lecture Example 2.2): $y = \sin x$ vs. $y = \cos x$ on $[0, \frac{\pi}{2}]$

Find the area bounded by $y = \sin x$, $y = \cos x$, $x = 0$, $x = \frac{\pi}{2}$.

Solution. Solve $\sin x = \cos x \Rightarrow x = \frac{\pi}{4}$ on $[0, \frac{\pi}{2}]$. On $[0, \frac{\pi}{4}]$, $\cos x \geq \sin x$; on $[\frac{\pi}{4}, \frac{\pi}{2}]$, $\sin x \geq \cos x$. Hence

$$\begin{aligned} A &= \int_0^{\pi/4} (\cos x - \sin x) dx + \int_{\pi/4}^{\pi/2} (\sin x - \cos x) dx \\ &= [\sin x + \cos x]_0^{\pi/4} + [-\cos x - \sin x]_{\pi/4}^{\pi/2} = (\sqrt{2} - 1) + (\sqrt{2} - 1) \\ &= 2\sqrt{2} - 2. \end{aligned}$$

Worked Example (Lecture Example 2.1): $y = x^2 + 1$ vs. $y = x$ on $[0, 1]$

Find the area bounded above by $y = x^2 + 1$, below by $y = x$, and on the sides by $x = 0$ and $x = 1$.

Solution. First check ordering: for $x \in [0, 1]$, $x^2 + 1 \geq 1 > x$ (since $x \leq 1$), so $f(x) = x^2 + 1$ is on top throughout. No splitting needed.

$$A = \int_0^1 ((x^2 + 1) - x) dx = \int_0^1 (x^2 - x + 1) dx = \left[\frac{x^3}{3} - \frac{x^2}{2} + x \right]_0^1 = \frac{1}{3} - \frac{1}{2} + 1 = \frac{5}{6}.$$

Sanity check: The region lies inside the unit square $[0, 1] \times [0, 2]$, so $A < 2$. Also $A > 0$ since the integrand is positive. ✓

Worked Example: Horizontal slices (dy) are simpler (Tutorial 3, Q3(b) style)

Find the area enclosed by the parabola $4x + y^2 = 12$ and the line $x = y$.

Solution. Rewrite the parabola as $x = 3 - \frac{y^2}{4}$, and the line as $x = y$.

Intersections: Set $y = 3 - \frac{y^2}{4}$, i.e. $y^2 + 4y - 12 = 0$, so $(y + 6)(y - 2) = 0$, giving $y = -6$ and $y = 2$. For $y \in [-6, 2]$, the parabola lies to the *right* of the line (check: at $y = 0$, parabola gives $x = 3$, line gives $x = 0$). Hence

$$A = \int_{-6}^2 \left[\left(3 - \frac{y^2}{4} \right) - y \right] dy = \int_{-6}^2 \left(3 - y - \frac{y^2}{4} \right) dy.$$

Evaluate:

$$= \left[3y - \frac{y^2}{2} - \frac{y^3}{12} \right]_{-6}^2 = \left(6 - 2 - \frac{8}{12} \right) - (-18 - 18 + 18) = \frac{10}{3} - (-18) = \frac{64}{3}.$$

Using dx would require solving for y (two branches $y = \pm\sqrt{12 - 4x}$) and splitting the integral at $x = -6$ and $x = 2$ —much messier. This illustrates why choosing the right variable matters.

Common Mistake: Not Splitting Where the Curves Cross

If the “top” and “bottom” curves swap order inside $[a, b]$, then $\int_a^b (f - g) dx$ gives *signed area*, not geometric area. Always find intersection points first and split accordingly.

Common Mistake: Forgetting to check the ordering

Even if you correctly find intersection points, you still need to determine *which curve is on top* in each sub-interval. A quick way: plug in a test value of x (or y) from each sub-interval and compare.

For instance, between $y = x^3$ and $y = x$ on $[-1, 1]$: the intersections are $x = -1, 0, 1$. On $(-1, 0)$, test $x = -0.5$: $(-0.5)^3 = -0.125$ vs. -0.5 , so $x^3 > x$ there. On $(0, 1)$, test $x = 0.5$: $0.125 < 0.5$, so $x > x^3$. The ordering reverses.

Exam Focus: Fast Setup Checklist for Area Problems

- Sketch *roughly* and label intersections (solve algebraically).
- Decide dx vs. dy by whichever gives **simpler bounds and simpler integrand**.
- Write the integrand explicitly as **(top–bottom)** or **(right–left)** and ensure it is ≥ 0 on each sub-interval.

Extension (Beyond Syllabus: absolute-value area formula)

Syllabus link: Chapter 2 — 2.1 Area Between Curves.

Proposition. Let f, g be continuous on $[a, b]$. Assume there exists a partition $a = x_0 < x_1 < \cdots < x_n = b$ such that $f - g$ does not change sign on each (x_{k-1}, x_k) . Then the area between the two curves is

$$A = \int_a^b |f(x) - g(x)| dx = \sum_{k=1}^n \left| \int_{x_{k-1}}^{x_k} (f(x) - g(x)) dx \right|.$$

This is just the rigorous statement of “split and take absolute values.” It can simplify setup when $f - g$ factors nicely.

2.2 Volumes by Disk/Washer Method

Motivation: From area to volume via slicing.

We already know how to compute areas of planar regions using integrals. Volumes of three-dimensional solids can be computed by a natural extension of the same idea: instead of summing the “heights” of thin rectangles, we sum the *areas* of thin cross-sectional slices.

Consider a solid S lying along the x -axis between $x = a$ and $x = b$. If you cut the solid with a plane perpendicular to the x -axis at position x , you obtain a cross-section whose area we call $A(x)$. The volume of a thin slab of thickness Δx is approximately $A(x) \Delta x$. Summing all such slabs and taking the limit gives:

$$V = \int_a^b A(x) dx.$$

This is the **general slicing formula**—it works for *any* solid whose cross-sectional area $A(x)$ you can express as a function of x . Similarly, if slicing along the y -axis, $V = \int_c^d A(y) dy$.

Why disks and washers?

When a solid is formed by *revolving* a plane region about an axis, every cross-section perpendicular to that axis is either:

- a **disk** (solid circle) if the region touches the axis — area = πR^2 , or
- a **washer** (annulus = disk with a hole) if the region has a gap from the axis — area = $\pi(R_{\text{out}}^2 - R_{\text{in}}^2)$.

The **radii** are *distances* from the axis of rotation to the boundary curves. This distance interpretation is critical, especially when the axis is shifted.

Key Formula: General Slicing and Disk/Washer Method

General slicing:

$$V = \int_a^b A(x) dx \quad (\text{slicing perpendicular to the } x\text{-axis}).$$

Disk method (cross-section is a full circle):

$$V = \int_a^b \pi [R(x)]^2 dx, \quad R(x) = \text{distance from axis to curve.}$$

Washer method (cross-section is an annulus):

$$V = \int_a^b \pi [(R_{\text{out}}(x))^2 - (R_{\text{in}}(x))^2] dx.$$

(Replace x with y and $[a, b]$ with $[c, d]$ when slicing perpendicular to the y -axis.)

Step-by-step workflow for disk/washer problems:

- (1) **Sketch** the region and identify the axis of rotation.
- (2) **Decide the slicing variable:** slice *perpendicular* to the axis of rotation. If the axis is horizontal, you typically slice with dx ; if vertical, with dy .
- (3) **Identify the radii:** For each slice at position x (or y), find the outer radius R_{out} and inner radius R_{in} as *distances* from the axis to the boundary curves.
- (4) **Set up the integral:** $V = \pi \int (R_{\text{out}}^2 - R_{\text{in}}^2) d(\text{slicing variable})$.

(5) **Evaluate** using FTC Part II.

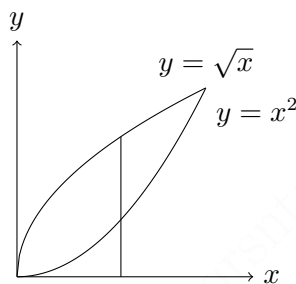
Worked Example (Lecture Example 2.3): Volume of a sphere

Show that the volume of a sphere of radius r is $V = \frac{4}{3}\pi r^3$.

Solution. Place the sphere centered at the origin. A cross-section at position x (for $-r \leq x \leq r$) is a disk of radius $R(x) = \sqrt{r^2 - x^2}$ (by the Pythagorean theorem). Hence

$$V = \int_{-r}^r \pi(r^2 - x^2) dx = \pi \left[r^2 x - \frac{x^3}{3} \right]_{-r}^r = \pi \left[\left(r^3 - \frac{r^3}{3} \right) - \left(-r^3 + \frac{r^3}{3} \right) \right] = \frac{4}{3}\pi r^3.$$

Sanity check: The sphere fits inside a cylinder of radius r and height $2r$ (volume $2\pi r^3$), and we expect $V_{\text{sphere}} < V_{\text{cylinder}}$. Indeed $\frac{4}{3}\pi r^3 < 2\pi r^3$. ✓



Worked Example (Lecture Example 2.6): Rotate region between $y = x^2$ and $x = y^2$ about $y = 0$

In the first quadrant, the curves $y = x^2$ and $x = y^2$ intersect at $(0,0)$ and $(1,1)$. Rotate the region between them about $y = 0$ (the x -axis).

Solution (washer method w.r.t. x). Write both as y -functions: $y = x^2$ and $y = \sqrt{x}$ on $x \in [0, 1]$. Outer radius $R(x) = \sqrt{x}$, inner radius $r(x) = x^2$. Then

$$V = \pi \int_0^1 (R(x)^2 - r(x)^2) dx = \pi \int_0^1 (x - x^4) dx = \pi \left[\frac{x^2}{2} - \frac{x^5}{5} \right]_0^1 = \frac{3\pi}{10}.$$

Worked Example (Lecture Example 2.4): Disk method about the y -axis

Find the volume of the solid obtained by rotating the region bounded by $y = x^3$, $y = 8$, and $x = 0$ about the y -axis.

Solution (disk method w.r.t. y). Since the axis is vertical (y -axis), slice perpendicular to it with dy .

For a given $y \in [0, 8]$, the curve $y = x^3$ gives $x = y^{1/3}$. The cross-section is a full disk (no hole, since the region touches the y -axis at $x = 0$) with radius $R(y) = y^{1/3}$.

$$V = \int_0^8 \pi [y^{1/3}]^2 dy = \pi \int_0^8 y^{2/3} dy = \pi \left[\frac{3}{5} y^{5/3} \right]_0^8 = \pi \cdot \frac{3}{5} \cdot 32 = \frac{96\pi}{5}.$$

Worked Example (Tutorial 4, Q4(b)): Washer method with shifted axis

Find the volume of the solid obtained by rotating the region bounded by $y = \frac{1}{4}x^2$ and $y = 5 - x^2$ about the x -axis.

Solution (washer method w.r.t. x). **Step 1: Intersections.** $\frac{1}{4}x^2 = 5 - x^2 \Rightarrow \frac{5}{4}x^2 = 5 \Rightarrow x^2 = 4 \Rightarrow x = \pm 2$.

Step 2: Identify radii. For $x \in [-2, 2]$, the outer curve (farther from x -axis) is $y = 5 - x^2$, and the inner curve (closer to x -axis) is $y = \frac{1}{4}x^2$ (both are nonnegative here). So:

$$R_{\text{out}}(x) = 5 - x^2, \quad R_{\text{in}}(x) = \frac{1}{4}x^2.$$

Step 3: Set up and evaluate. The integrand is even, so

$$\begin{aligned} V &= \pi \int_{-2}^2 \left[(5 - x^2)^2 - \left(\frac{x^2}{4} \right)^2 \right] dx = 2\pi \int_0^2 \left[25 - 10x^2 + x^4 - \frac{x^4}{16} \right] dx \\ &= 2\pi \int_0^2 \left(25 - 10x^2 + \frac{15}{16}x^4 \right) dx = 2\pi \left[25x - \frac{10x^3}{3} + \frac{15x^5}{80} \right]_0^2 \\ &= 2\pi \left(50 - \frac{80}{3} + \frac{480}{80} \right) = 2\pi \left(50 - \frac{80}{3} + 6 \right) = 2\pi \cdot \frac{168 - 80}{3} = \frac{176\pi}{3}. \end{aligned}$$

Common Mistake: Wrong Radius (Distance to the Axis)

When rotating about a *shifted* axis (e.g. $x = 2$ or $y = -1$), the radius is a distance:

$$R(x) = |f(x) - c| \quad (\text{about } y = c), \quad R(y) = |g(y) - c| \quad (\text{about } x = c).$$

Forgetting the shift (using $f(x)$ instead of $f(x) - c$) is a frequent setup error.

Common Mistake: Squaring a difference vs. difference of squares

The washer formula is $\pi(R^2 - r^2)$, which is a *difference of squares*. A very common error is writing $\pi(R - r)^2$ instead. These are not the same:

$$R^2 - r^2 = (R + r)(R - r) \neq (R - r)^2.$$

Always square *each radius separately*, then subtract.

Exam Focus: Disk/Washer Method Selection

- If the axis is **horizontal**, slicing with dx is often natural; if the axis is **vertical**, slicing with dy is often natural.
- If the setup forces you to solve a hard inverse (e.g. cubic $y = 2x^2 - x^3$ for x), consider **cylindrical shells** instead.
- Always state **outer radius** and **inner radius** explicitly before integrating.

Exam Focus: Quick dimensional/sign sanity checks for volumes

- Volume must be **positive**. If your answer is negative, you likely swapped R_{out} and R_{in} .
- Volume should have dimensions of $(\text{length})^3$. If boundaries are pure numbers (e.g. $x \in [0, 2]$, $y \in [0, 4]$), the answer should be a number times π .
- For a sanity bound: the solid of revolution fits inside a cylinder. Compute the cylinder's volume as a quick upper bound. If your answer exceeds it, something is wrong.

2.3 Volumes by Cylindrical Shells

The shell method computes volume by integrating volumes of *thin cylindrical shells* obtained from slices *parallel* to the axis of rotation.

Motivation: When washers fail.

Consider rotating the region under $y = 2x^2 - x^3$ (for $x \in [0, 2]$) about the y -axis. If you try the washer method, you need to slice perpendicular to the y -axis (i.e., horizontally), which requires writing x as a function of y . But solving $y = 2x^2 - x^3$ for x means solving a *cubic*—messy and impractical. The shell method avoids this by slicing *parallel* to the axis (vertically, using dx), so the curve stays in its natural form $y = f(x)$.

How a shell is formed:

Take a thin vertical strip at position x with width Δx and height $h(x)$. When this strip is revolved about a vertical axis, it sweeps out a thin cylindrical shell (like a hollow pipe). The volume of this shell is approximately:

$$\Delta V \approx (\text{circumference}) \times (\text{height}) \times (\text{thickness}) = 2\pi r(x) \cdot h(x) \cdot \Delta x,$$

where $r(x)$ is the distance from the strip to the axis (the shell radius). Summing all shells and taking $\Delta x \rightarrow 0$ gives the integral formula.

Theorem 2.2: Volume by cylindrical shells

- (a) (Rotation about a **vertical** line.) If a region between $x = a$ and $x = b$ is revolved about a vertical axis, and a typical *vertical* slice at x produces a shell of radius $r(x)$ and height $h(x)$, then

$$V = \int_a^b 2\pi r(x) h(x) dx.$$

- (b) (Rotation about a **horizontal** line.) If a region between $y = c$ and $y = d$ is revolved about a horizontal axis, and a typical *horizontal* slice at y produces a shell of radius $r(y)$ and height $h(y)$, then

$$V = \int_c^d 2\pi r(y) h(y) dy.$$

Why $2\pi r \cdot h$?

A cylindrical shell with inner radius r_1 , outer radius r_2 , and height h has exact volume

$$V_{\text{shell}} = \pi r_2^2 h - \pi r_1^2 h = \pi h(r_2^2 - r_1^2) = \pi h(r_2 + r_1)(r_2 - r_1).$$

Let $r = \frac{r_1 + r_2}{2}$ (average radius) and $\Delta r = r_2 - r_1$ (thickness). Then $V_{\text{shell}} = 2\pi r h \Delta r$. In the limit $\Delta r \rightarrow 0$, this becomes $2\pi r h dr$, which is the integrand.

Key Formula: Shell Geometry

For a shell created by a slice parallel to the axis:

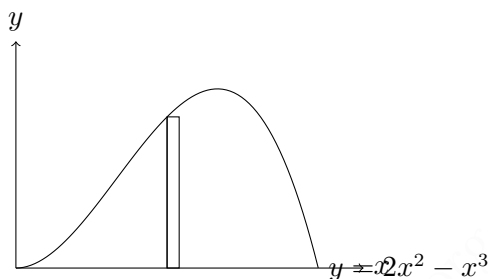
$$\text{shell volume} \approx (\text{circumference}) \cdot (\text{height}) \cdot (\text{thickness}) = (2\pi r) h (\Delta r).$$

So the integral is always of the form $V = \int 2\pi(\text{radius})(\text{height}) d(\text{slice variable})$.

Step-by-step workflow for shell problems:

- (1) **Sketch** the region and identify the axis of rotation.

- (2) **Decide the slicing variable:** slice *parallel* to the axis of rotation. If the axis is vertical, use vertical strips (dx); if horizontal, use horizontal strips (dy).
- (3) **Identify radius and height:**
- Radius r = distance from the strip to the axis.
 - Height h = length of the strip = (top curve) – (bottom curve) for vertical strips.
- (4) **Set up the integral:** $V = \int_a^b 2\pi r \cdot h d(\text{slice variable})$.
- (5) **Evaluate.**



Worked Example (Lecture Example 2.7): Rotate $y = 2x^2 - x^3$ about the y -axis

Find the volume obtained by rotating the region bounded by $y = 2x^2 - x^3$ and $y = 0$ about the y -axis.

Solution (shell method w.r.t. x). The curve meets $y = 0$ at $x = 0$ and $x = 2$ (since $2x^2 - x^3 = x^2(2 - x)$). A shell at position $x \in [0, 2]$ has radius $r(x) = x$ and height $h(x) = 2x^2 - x^3$. Hence

$$\begin{aligned} V &= \int_0^2 2\pi x (2x^2 - x^3) dx = 2\pi \int_0^2 (2x^3 - x^4) dx \\ &= 2\pi \left[\frac{x^4}{2} - \frac{x^5}{5} \right]_0^2 = 2\pi \left(8 - \frac{32}{5} \right) = \frac{16\pi}{5}. \end{aligned}$$

Worked Example (Tutorial 4, Question 1(a)): Shells about $x = 0$

Use cylindrical shells to find the volume generated by rotating the region bounded by $y = x^2$ and $y = 6x - 2x^2$ about $x = 0$.

Solution (shell method w.r.t. x). Intersections: $x^2 = 6x - 2x^2 \Rightarrow 3x^2 - 6x = 0 \Rightarrow x = 0, 2$. On $[0, 2]$, the shell radius is $r(x) = x$ and height is $h(x) = (6x - 2x^2) - x^2 = 6x - 3x^2$. Thus

$$\begin{aligned} V &= \int_0^2 2\pi x (6x - 3x^2) dx = 2\pi \int_0^2 (6x^2 - 3x^3) dx \\ &= 2\pi \left[2x^3 - \frac{3}{4}x^4 \right]_0^2 = 2\pi(16 - 12) = 8\pi. \end{aligned}$$

Worked Example (Tutorial 4, Q1(b)): Shells about a shifted vertical axis $x = 3$

Find the volume of the solid generated by rotating the region bounded by $y = x^3$, $y = 8$, and $x = 0$ about the line $x = 3$.

Solution (shell method w.r.t. x). The curves $y = x^3$ and $y = 8$ intersect at $x = 2$. The region spans $x \in [0, 2]$.

A vertical shell at position x has:

- Radius: $r(x) = 3 - x$ (distance from x to the axis $x = 3$; note $3 - x > 0$ on $[0, 2]$).
- Height: $h(x) = 8 - x^3$ (from $y = x^3$ up to $y = 8$).

$$\begin{aligned} V &= \int_0^2 2\pi (3 - x)(8 - x^3) dx = 2\pi \int_0^2 (24 - 3x^3 - 8x + x^4) dx \\ &= 2\pi \left[24x - \frac{3x^4}{4} - 4x^2 + \frac{x^5}{5} \right]_0^2 \\ &= 2\pi \left(48 - 12 - 16 + \frac{32}{5} \right) = 2\pi \cdot \frac{132}{5} = \frac{264\pi}{5}. \end{aligned}$$

Sanity check: Compare with the disk method answer for the same region about the y -axis ($V = \frac{96\pi}{5}$). Rotating about $x = 3$ (farther away) should give a *larger* volume, and indeed $\frac{264\pi}{5} > \frac{96\pi}{5}$. ✓

Worked Example (Tutorial 4, Q1(c)): Shells about a shifted horizontal axis $y = -2$

Find the volume of the solid generated by rotating the region enclosed by $x = 2y^2$ and $x = y^2 + 1$ about the line $y = -2$.

Solution (shell method w.r.t. y). Intersections: $2y^2 = y^2 + 1 \Rightarrow y^2 = 1 \Rightarrow y = \pm 1$.

For $y \in [-1, 1]$, horizontal shells are used. A shell at height y has:

- Radius: $r(y) = y - (-2) = y + 2$ (distance from y to axis $y = -2$; note $y + 2 > 0$ for $y \geq -1$).
- Height: $h(y) = (y^2 + 1) - 2y^2 = 1 - y^2$ (right curve minus left curve).

$$V = \int_{-1}^1 2\pi (y + 2)(1 - y^2) dy = 2\pi \int_{-1}^1 (y + 2 - y^3 - 2y^2) dy.$$

Split into odd and even parts. The integrand is $(y - y^3) + (2 - 2y^2)$. On $[-1, 1]$, the odd part $(y - y^3)$ integrates to 0 by symmetry. The even part is:

$$2\pi \int_{-1}^1 (2 - 2y^2) dy = 2\pi \cdot 2 \int_0^1 (2 - 2y^2) dy = 4\pi \left[2y - \frac{2y^3}{3} \right]_0^1 = 4\pi \cdot \frac{4}{3} = \frac{16\pi}{3}.$$

Common Mistake: Radius for Shells Is a Distance

For shells, the radius is measured *to the axis of rotation*. About $x = c$ (vertical axis): $r(x) = |x - c|$. About $y = c$ (horizontal axis): $r(y) = |y - c|$. Do not drop the absolute value until you have fixed the interval where the sign is known.

Common Mistake: Using the wrong slicing direction

Remember the key distinction:

- **Washers:** slice *perpendicular* to the axis $\Rightarrow$ cross-sections are circles/annuli.
- **Shells:** slice *parallel* to the axis $\Rightarrow$ strips sweep out cylindrical shells.

If you accidentally slice in the wrong direction, you will mix up the geometric setup and get incorrect radii/heights.

Exam Focus: When Shells Beat Washers

Shells are often the cleanest choice when:

- the region is naturally described as $y = f(x)$ and the axis is vertical, but washers would require solving x in terms of y ;
- the intersection analysis is simpler in the slice-variable used by shells;
- you want a quick cross-check against a washer setup (when both are manageable).

Exam Focus: Method comparison decision tree

When faced with a volume-of-revolution problem, use this decision tree:

- (1) **Identify the axis of rotation** (horizontal or vertical? standard or shifted?).
 - (2) **Try washers first** (perpendicular slicing): Can you express both boundary curves as functions of the slice variable? If yes, proceed with washers.
 - (3) **If washers are awkward** (e.g., solving a cubic for $x(y)$), switch to **shells** (parallel slicing).
 - (4) **If time permits**, set up both methods and compare answers as a cross-check.
- Scenario — Preferred Method — ——— — Curves given as $y = f(x)$, axis is horizontal ($y = c$)
 — Washers w.r.t. x — — Curves given as $y = f(x)$, axis is vertical ($x = c$) — Shells w.r.t. x — —
 Curves given as $x = g(y)$, axis is vertical ($x = c$) — Washers w.r.t. y — — Curves given as $x = g(y)$,
 axis is horizontal ($y = c$) — Shells w.r.t. y —

Extension (Beyond Syllabus: Pappus's centroid theorem for volumes)

Syllabus link: Chapter 2 — 2.3 Volumes by Cylindrical Shells.

Theorem (Pappus, Volume Theorem). Let D be a plane region of finite area A , and let ℓ be a line in the same plane that does *not* intersect D . If D is revolved about ℓ , then the resulting solid has volume

$$V = A \cdot (2\pi d),$$

where d is the perpendicular distance from the centroid of D to the axis ℓ .

Working (torus volume in one line). Let D be a disk of radius r (area $A = \pi r^2$) whose center is at distance $R > r$ from the rotation axis ℓ . The centroid of the disk is its center, so $d = R$. Hence

$$V = \pi r^2 \cdot 2\pi R = 2\pi^2 R r^2.$$

This is a fast cross-check alternative to a longer shell integral for the same solid.

Cross-check example (Tutorial 4, Q1(b)). The region bounded by $y = x^3$, $y = 8$, $x = 0$ has area $A = \int_0^2 (8 - x^3) dx = 12$ and centroid $\bar{x} = \frac{1}{12} \int_0^2 x(8 - x^3) dx = \frac{4}{5}$. Rotating about $x = 3$: $d = 3 - \frac{4}{5} = \frac{11}{5}$. Thus $V = 12 \cdot 2\pi \cdot \frac{11}{5} = \frac{264\pi}{5}$, matching the shell computation.

Extension (Tutorial 4, Q5: Why shells are awkward for $y = x(x - 1)^2$)

Syllabus link: Chapter 2 — 2.3 Volumes by Cylindrical Shells.

The region under $y = x(x - 1)^2$ on $[0, 1]$ rotated about the y -axis is a natural shell problem: $r(x) = x$, $h(x) = x(x - 1)^2$.

Why not washers? The function $f(x) = x(x - 1)^2 = x^3 - 2x^2 + x$ is *not* one-to-one on $[0, 1]$: it increases on $[0, \frac{1}{3}]$ and decreases on $[\frac{1}{3}, 1]$ (verify via $f'(x) = 3x^2 - 4x + 1 = (3x - 1)(x - 1)$). So for each $y \in (0, \frac{4}{27})$, a horizontal line meets the curve at *two* x -values, requiring you to solve a cubic for $x(y)$ —impractical.

With shells: $V = 2\pi \int_0^1 x \cdot x(x - 1)^2 dx = 2\pi \int_0^1 (x^4 - 2x^3 + x^2) dx = 2\pi \left[\frac{1}{5} - \frac{1}{2} + \frac{1}{3} \right] = \frac{\pi}{15}$.

3 Techniques of Integration

Overview. Chapter 1 gave us the Fundamental Theorem of Calculus: if we can find an antiderivative F of f , then $\int_a^b f(x) dx = F(b) - F(a)$. The practical difficulty is that most integrands do *not* yield to a single application of the power rule or a basic substitution. This chapter develops a toolkit of techniques—integration by parts, trigonometric identities, trigonometric substitution, partial fractions, and numerical approximation—that, taken together, let you evaluate (or at least approximate) virtually any integral that arises in MH1101.

Each technique answers a specific structural question about the integrand:

- **IBP:** “Is the integrand a *product* of two functions, one of which simplifies when differentiated?”
- **Trig integrals:** “Does the integrand involve powers or products of trig functions that can be reduced using identities?”
- **Trig substitution:** “Does the integrand contain a radical $\sqrt{a^2 - x^2}$, $\sqrt{a^2 + x^2}$, or $\sqrt{x^2 - a^2}$?”
- **Partial fractions:** “Is the integrand a rational function $P(x)/Q(x)$?”
- **Numerical integration:** “Is an antiderivative unavailable or impractical? Can we approximate the integral instead?”

3.1 Integration by Parts

Motivation: reversing the Product Rule.

We know the Product Rule for differentiation:

$$\frac{d}{dx}(u(x)v(x)) = u(x)v'(x) + u'(x)v(x).$$

This tells us how to *differentiate* a product. But what if we need to *integrate* a product? Rearranging the Product Rule and integrating both sides gives us a way to “trade” one integral for another—hopefully simpler—one.

Integrate both sides of the Product Rule over any interval (or indefinitely):

$$u(x)v(x) = \int u(x)v'(x) dx + \int u'(x)v(x) dx.$$

Solving for the first integral on the right:

$$\int u(x)v'(x) dx = u(x)v(x) - \int u'(x)v(x) dx.$$

This is the **integration-by-parts (IBP) formula**. Notice what it does: it replaces $\int u dv$ with $uv - \int v du$. The hope is that $\int v du$ is easier to evaluate than the original integral.

Key Formula: Integration by Parts (Indefinite and Definite)

The rule is the integration analogue of the Product Rule.

If $u = u(x)$ and $v = v(x)$ are differentiable, then

$$\frac{d}{dx}(u(x)v(x)) = u(x)v'(x) + u'(x)v(x).$$

Integrating both sides gives:

$$\int u(x)v'(x) dx = u(x)v(x) - \int u'(x)v(x) dx.$$

For definite integrals,

$$\int_a^b u(x)v'(x) dx = [u(x)v(x)]_a^b - \int_a^b u'(x)v(x) dx.$$

How to choose u and dv . The goal is to *trade* an integral that looks hard for one that looks easier:

$$\int \underbrace{u(x)}_{\text{differentiate}} \underbrace{dv}_{\text{integrate}} \longrightarrow uv - \int v du.$$

A good choice typically makes:

- du simpler than u (e.g. polynomials shrink in degree; $\ln x$ becomes $1/x$; $\arctan x$ becomes $1/(1+x^2)$),
- $v = \int dv$ computable immediately (avoid dv whose antiderivative is unknown).

Why the trade works (intuition).

Think of IBP as shifting “complexity” from one factor to the other. If u becomes simpler when differentiated (e.g. $x^n \rightarrow nx^{n-1}$), and dv can be integrated without difficulty (e.g. $\sin x dx \rightarrow -\cos x$), then $\int v du$ has a *simpler* first factor (du) multiplied by a known function (v). After one or more rounds of IBP, the integral may reduce to a standard form.

Common Mistake: Forgetting the boundary term in definite IBP

For $\int_a^b u dv$, the boundary contribution $[uv]_a^b$ is **not optional**.

A frequent error is to write $\int_a^b u dv = uv - \int_a^b v du$ but forget to evaluate uv at a, b .

Worked Example: $\int x \sin x dx$

Let $u = x$ and $dv = \sin x dx$. Then $du = dx$ and $v = -\cos x$. Hence

$$\int x \sin x dx = uv - \int v du = x(-\cos x) - \int (-\cos x) dx = -x \cos x + \sin x + C.$$

Quick check: Differentiate $-x \cos x + \sin x$: by the product rule we get $-\cos x + x \sin x + \cos x = x \sin x$.
✓

Worked Example: $\int \ln x \, dx$ (why IBP is necessary)

There is no direct “antiderivative rule” for $\ln x$ itself. Treat $\ln x$ as a product:

$$\int \ln x \, dx = \int (\ln x) \cdot 1 \, dx.$$

Choose $u = \ln x$ and $dv = dx$. Then $du = \frac{1}{x} dx$ and $v = x$. Thus

$$\int \ln x \, dx = x \ln x - \int x \cdot \frac{1}{x} dx = x \ln x - \int 1 \, dx = x \ln x - x + C.$$

Why this choice? Choosing $u = \ln x$ converts $\ln x$ (which we cannot integrate directly) into $du = \frac{1}{x} dx$ (which combines nicely with $v = x$ to give a trivial integral $\int 1 \, dx$). Choosing $u = 1$ and $dv = \ln x \, dx$ would require knowing $\int \ln x \, dx$ in advance—circular!

Key Formula: Derivatives often paired with IBP

These frequently appear when u is an inverse trig or logarithmic function:

$$\frac{d}{dx}(\arctan x) = \frac{1}{1+x^2}, \quad \frac{d}{dx}(\arcsin x) = \frac{1}{\sqrt{1-x^2}}, \quad \frac{d}{dx}(\arccos x) = -\frac{1}{\sqrt{1-x^2}}.$$

(Use the course’s standard inverse-trig derivative list when needed.)

Why inverse trig and log functions are always chosen as u :

Functions like $\ln x$, $\arctan x$, and $\arcsin x$ are hard to integrate directly but *easy to differentiate*—their derivatives are algebraic. So making them u converts them into simpler du terms. Meanwhile, the remaining factor (often algebraic or trigonometric) serves as dv and is easy to integrate.

Worked Example: Definite IBP with a secondary substitution

Evaluate $\int_0^1 \arctan x \, dx$.

Choose $u = \arctan x$ and $dv = dx$, so $du = \frac{1}{1+x^2} dx$ and $v = x$.

Then

$$\begin{aligned}\int_0^1 \arctan x \, dx &= [x \arctan x]_0^1 - \int_0^1 x \cdot \frac{1}{1+x^2} dx \\ &= \left(1 \cdot \frac{\pi}{4} - 0\right) - \int_0^1 \frac{x}{1+x^2} dx.\end{aligned}$$

For the remaining integral, set $t = 1 + x^2$, so $dt = 2x \, dx$:

$$\int_0^1 \frac{x}{1+x^2} dx = \frac{1}{2} \int_{t=1}^2 \frac{1}{t} dt = \frac{1}{2} [\ln t]_1^2 = \frac{1}{2} \ln 2.$$

Hence

$$\int_0^1 \arctan x \, dx = \frac{\pi}{4} - \frac{1}{2} \ln 2.$$

Worked Example: $\int x^2 e^x \, dx$ (two rounds of IBP)

Here the polynomial factor x^2 needs *two* differentiations to reach 0.

Round 1: Let $u = x^2$, $dv = e^x \, dx$, so $du = 2x \, dx$, $v = e^x$:

$$\int x^2 e^x \, dx = x^2 e^x - 2 \int x e^x \, dx.$$

Round 2: For $\int x e^x \, dx$, let $u = x$, $dv = e^x \, dx$, so $du = dx$, $v = e^x$:

$$\int x e^x \, dx = x e^x - \int e^x \, dx = x e^x - e^x + C.$$

Combine:

$$\int x^2 e^x \, dx = x^2 e^x - 2(x e^x - e^x) + C = e^x(x^2 - 2x + 2) + C.$$

Pattern: Each IBP round reduces the polynomial degree by 1. For $\int x^n e^x \, dx$, you need n rounds (or use the Tabular method in the Extension below).

Repeated integration by parts (“solve-for-the-integral”)

Some integrals produce the *same* integral again after applying IBP. In that case, carry the repeated term to the left and solve algebraically.

Why does the integral reappear?

When both factors in the product “cycle” under differentiation and integration (e.g. e^x stays e^x , while $\sin x \rightarrow \cos x \rightarrow -\sin x$), applying IBP twice brings back the original integrand. The trick is to **name** the original integral (e.g. I), and after two rounds you get an equation of the form $I = (\text{stuff}) - I$, which you

solve as $2I = \text{stuff}$.

Worked Example: $\int e^x \sin x \, dx$ (two rounds of IBP)

Let

$$I = \int e^x \sin x \, dx.$$

First IBP: choose $u = e^x$, $dv = \sin x \, dx$. Then $du = e^x \, dx$, $v = -\cos x$:

$$I = -e^x \cos x + \int e^x \cos x \, dx.$$

Let $J = \int e^x \cos x \, dx$. Apply IBP again with $u = e^x$, $dv = \cos x \, dx$ so $v = \sin x$:

$$J = e^x \sin x - \int e^x \sin x \, dx = e^x \sin x - I.$$

Substitute into the first relation:

$$I = -e^x \cos x + (e^x \sin x - I) \implies 2I = e^x(\sin x - \cos x).$$

Therefore,

$$\int e^x \sin x \, dx = \frac{1}{2}e^x(\sin x - \cos x) + C.$$

Exam Focus: When repeated IBP is likely

Repeated IBP is common when the integrand is a product of:

- e^{ax} with $\sin bx$ or $\cos bx$;
- a polynomial with e^{ax} , $\sin bx$, or $\cos bx$ (eventually the polynomial differentiates to 0).

When you see the same integral reappear, **name it** (e.g. I), then solve for I explicitly.

Exam Focus: Sign consistency in “solve-for-the-integral”

When applying IBP twice to $\int e^{ax} \sin bx \, dx$ or $\int e^{ax} \cos bx \, dx$, you **must** use the *same* choice pattern in both rounds (e.g. always let $u = e^{ax}$). If you switch (e.g. $u = e^{ax}$ in round 1 but $u = \cos bx$ in round 2), the original integral does not reappear, and you go in circles.

Reduction formulas

Motivation: Why reduction formulas?

Some families of integrals (like $\int \sin^n x \, dx$) do not have a closed-form antiderivative in terms of elementary functions for general n . Instead, we derive a *recursive* relationship: express $\int \sin^n x \, dx$ in terms of $\int \sin^{n-2} x \, dx$. Applying this recursion repeatedly reduces the power until you reach a base case ($n = 0$ or $n = 1$) that you can integrate directly.

The derivation itself is just a clever IBP combined with the Pythagorean identity.

Theorem 3.1: Reduction formula for $\int \sin^n x \, dx$

Let $n \geq 2$ be an integer. Then

$$\int \sin^n x \, dx = -\frac{1}{n} \cos x \sin^{n-1} x + \frac{n-1}{n} \int \sin^{n-2} x \, dx.$$

Proof idea (exam-usable). Write $\sin^n x = \sin^{n-1} x \cdot \sin x$ and apply integration by parts with $u = \sin^{n-1} x$ and $dv = \sin x \, dx$; then use $\cos^2 x = 1 - \sin^2 x$ to eliminate $\cos^2 x$ and isolate the original integral on the left.

Step-by-step derivation.

Set $u = \sin^{n-1} x$ and $dv = \sin x \, dx$. Then $du = (n-1) \sin^{n-2} x \cos x \, dx$ and $v = -\cos x$. By IBP:

$$\int \sin^n x \, dx = -\cos x \sin^{n-1} x + (n-1) \int \sin^{n-2} x \cos^2 x \, dx.$$

Now replace $\cos^2 x = 1 - \sin^2 x$:

$$\int \sin^n x \, dx = -\cos x \sin^{n-1} x + (n-1) \int \sin^{n-2} x \, dx - (n-1) \int \sin^n x \, dx.$$

Move the last term to the left side:

$$n \int \sin^n x \, dx = -\cos x \sin^{n-1} x + (n-1) \int \sin^{n-2} x \, dx.$$

Divide by n to get the stated formula.

Extension (Tutorial 4, Question 5, Method 1: Reduction for $\cos^n x$)

Syllabus link: Chapter 3 — §3.1 Integration by Parts.

Proposition. Let $n \geq 2$ be an integer. Then

$$\int \cos^n x \, dx = \frac{1}{n} \cos^{n-1} x \sin x + \frac{n-1}{n} \int \cos^{n-2} x \, dx.$$

Full working.

Start from $\int \cos^n x \, dx = \int \cos^{n-1} x \cdot \cos x \, dx$.

Choose

$$u = \cos^{n-1} x, \quad dv = \cos x \, dx.$$

Then $v = \sin x$ and

$$du = (n-1) \cos^{n-2} x (-\sin x) \, dx = -(n-1) \cos^{n-2} x \sin x \, dx.$$

By integration by parts,

$$\begin{aligned} \int \cos^n x \, dx &= uv - \int v \, du \\ &= \cos^{n-1} x \sin x - \int \sin x (-(n-1) \cos^{n-2} x \sin x) \, dx \\ &= \cos^{n-1} x \sin x + (n-1) \int \cos^{n-2} x \sin^2 x \, dx. \end{aligned}$$

Use $\sin^2 x = 1 - \cos^2 x$:

$$\begin{aligned} \int \cos^n x \, dx &= \cos^{n-1} x \sin x + (n-1) \int \cos^{n-2} x (1 - \cos^2 x) \, dx \\ &= \cos^{n-1} x \sin x + (n-1) \int \cos^{n-2} x \, dx - (n-1) \int \cos^n x \, dx. \end{aligned}$$

Bring the last term to the left:

$$n \int \cos^n x \, dx = \cos^{n-1} x \sin x + (n-1) \int \cos^{n-2} x \, dx,$$

and divide by n to obtain the claimed formula.

A warning on heuristics (LIATE)

Common Mistake: Blindly following LIATE

Heuristics like LIATE (Log, Inverse trig, Algebraic, Trig, Exponential) can suggest a choice for u , but the **actual criterion** is: does $\int v \, du$ simplify?

If a substitution is available (chain-rule structure), it often beats IBP even when LIATE “permits” IBP.

Worked Example: When substitution beats IBP

Evaluate $\int x^3 \sin(x^2) dx$.

A fast approach is to expose a $2x dx$ term.

Write $x^3 dx = x^2(x dx)$. Let $u = x^2$ so that $du = 2x dx$ and $x^2 = u$:

$$\int x^3 \sin(x^2) dx = \frac{1}{2} \int u \sin u du.$$

Now apply IBP in u :

choose $\tilde{u} = u$, $d\tilde{v} = \sin u du$, so $\tilde{v} = -\cos u$:

$$\frac{1}{2} \int u \sin u du = \frac{1}{2} \left(-u \cos u + \int \cos u du \right) = \frac{1}{2} (-u \cos u + \sin u) + C.$$

Substitute back $u = x^2$:

$$\int x^3 \sin(x^2) dx = \frac{1}{2} (-x^2 \cos(x^2) + \sin(x^2)) + C.$$

Extension (Beyond Syllabus: Tabular Integration by Parts)

Syllabus link: Chapter 3 — §3.1 Integration by Parts.

Proposition (finite-derivative case).

Let u be a function such that $u^{(m+1)}(x) \equiv 0$ for some integer $m \geq 0$ (e.g. a polynomial of degree m), and let v be differentiable.

Then repeated integration by parts yields

$$\int u(x)v(x) dx = \sum_{k=0}^m (-1)^k u^{(k)}(x) V^{(k)}(x) + C,$$

where $V^{(0)} = \int v dx$ and $V^{(k)} = \int V^{(k-1)} dx$ (i.e. $V^{(k)}$ is the k -fold antiderivative of v).

Full working (derivation).

Apply IBP once to $\int u v dx$ with $dv = v dx$:

$$\int uv dx = uV^{(0)} - \int u'V^{(0)} dx.$$

Apply IBP again to the remaining integral:

$$\int u'V^{(0)} dx = u'V^{(1)} - \int u''V^{(1)} dx,$$

so

$$\int uv dx = uV^{(0)} - u'V^{(1)} + \int u''V^{(1)} dx.$$

Continuing this process m times produces alternating signs and ends because $u^{(m+1)} \equiv 0$. This yields the stated finite sum.

Micro-example.

For $\int x \cos(5x) dx$, take $u = x$ (so $u'' = 0$) and $v = \cos(5x)$.

Compute $V^{(0)} = \int \cos(5x) dx = \frac{1}{5} \sin(5x)$ and

$V^{(1)} = \int V^{(0)} dx = \int \frac{1}{5} \sin(5x) dx = -\frac{1}{25} \cos(5x)$.

Thus

$$\int x \cos(5x) dx = x \cdot \frac{1}{5} \sin(5x) - 1 \cdot \left(-\frac{1}{25} \cos(5x) \right) + C = \frac{x}{5} \sin(5x) + \frac{1}{25} \cos(5x) + C.$$

3.2 Trigonometric Integrals

Motivation.

Integrals involving powers and products of trigonometric functions arise naturally after trigonometric substitution (Section 3.3) and in Fourier analysis. The key idea: use trig identities to rewrite the integrand into a form amenable to a simple u -substitution or direct integration. The choice of identity depends on the *parity* (odd/even) of the powers involved.

Powers of $\sin x$ and $\cos x$

Key Formula: Power-reduction and Pythagorean identities

$$\sin^2 x + \cos^2 x = 1, \quad \sin^2 x = \frac{1 - \cos 2x}{2}, \quad \cos^2 x = \frac{1 + \cos 2x}{2}, \quad \sin x \cos x = \frac{1}{2} \sin 2x.$$

Why these identities? The Pythagorean identity $\sin^2 x + \cos^2 x = 1$ lets you convert between sin and cos powers, which is essential when one power is odd (you “peel off” one factor and convert the rest). The half-angle formulas are needed when *both* powers are even, because there is no odd factor to peel off; instead, you reduce the degree by expressing $\sin^2 x$ or $\cos^2 x$ in terms of $\cos 2x$.

Method-selection logic for $\int \sin^m x \cos^n x dx$.

- If n is odd: save one $\cos x$, convert the rest using $\cos^2 x = 1 - \sin^2 x$, then substitute $u = \sin x$.
- If m is odd: save one $\sin x$, convert the rest using $\sin^2 x = 1 - \cos^2 x$, then substitute $u = \cos x$.
- If both m, n are even: use power-reduction (half-angle) formulas until integrable.

Why “save one factor”?

The substitution $u = \sin x$ requires $du = \cos x dx$, so you need exactly one leftover $\cos x$ to pair with dx . All remaining cos factors must be converted to sin via $\cos^2 x = 1 - \sin^2 x$, so the entire integrand becomes a polynomial in $u = \sin x$. This only works if the number of leftover cos factors after saving one is *even*—which is guaranteed when n is odd.

Worked Example: $\int \cos^3 x dx$ (odd power)

Write $\cos^3 x = \cos x \cos^2 x = \cos x(1 - \sin^2 x)$.

Let $u = \sin x$, $du = \cos x dx$:

$$\int \cos^3 x dx = \int (1 - u^2) du = u - \frac{u^3}{3} + C = \sin x - \frac{\sin^3 x}{3} + C.$$

Worked Example: $\int \sin^5 x \cos^2 x dx$ (Lecture Example 3.9)

Since $m = 5$ is odd, save one $\sin x$ and convert $\sin^4 x = (\sin^2 x)^2 = (1 - \cos^2 x)^2$:

$$\int \sin^5 x \cos^2 x dx = \int (1 - \cos^2 x)^2 \cos^2 x \sin x dx.$$

Let $u = \cos x$, $du = -\sin x dx$:

$$\begin{aligned} &= - \int (1 - u^2)^2 u^2 du = - \int (u^2 - 2u^4 + u^6) du \\ &= -\frac{u^3}{3} + \frac{2u^5}{5} - \frac{u^7}{7} + C = -\frac{\cos^3 x}{3} + \frac{2\cos^5 x}{5} - \frac{\cos^7 x}{7} + C. \end{aligned}$$

Worked Example: $\int \sin^4 x \, dx$ (even power)

Use $\sin^2 x = \frac{1 - \cos 2x}{2}$:

$$\sin^4 x = \left(\frac{1 - \cos 2x}{2} \right)^2 = \frac{1}{4} (1 - 2 \cos 2x + \cos^2 2x).$$

Apply power-reduction again to $\cos^2 2x = \frac{1}{2}(1 + \cos 4x)$:

$$\sin^4 x = \frac{1}{4} \left(1 - 2 \cos 2x + \frac{1}{2}(1 + \cos 4x) \right) = \frac{1}{4} \left(\frac{3}{2} - 2 \cos 2x + \frac{1}{2} \cos 4x \right).$$

Integrate term-by-term:

$$\int \sin^4 x \, dx = \frac{1}{4} \left(\frac{3}{2}x - \sin 2x + \frac{1}{8} \sin 4x \right) + C.$$

Worked Example (Tutorial 5, Q1(a)): $\int \sin^6 x \cos^3 x \, dx$

Here $n = 3$ is odd. Save one $\cos x$ and convert $\cos^2 x = 1 - \sin^2 x$:

$$\int \sin^6 x \cos^3 x \, dx = \int \sin^6 x (1 - \sin^2 x) \cos x \, dx.$$

Let $u = \sin x$, $du = \cos x \, dx$:

$$= \int u^6 (1 - u^2) \, du = \int (u^6 - u^8) \, du = \frac{u^7}{7} - \frac{u^9}{9} + C = \frac{\sin^7 x}{7} - \frac{\sin^9 x}{9} + C.$$

Powers of $\tan x$ and $\sec x$ **Key Formula: Core identity for $\tan/\sec$**

$$1 + \tan^2 x = \sec^2 x, \quad \int \tan x \, dx = \ln |\sec x| + C, \quad \int \sec x \, dx = \ln |\sec x + \tan x| + C.$$

Why the parallel with $\sin/\cos$?

The identity $1 + \tan^2 x = \sec^2 x$ plays the same role as $\sin^2 x + \cos^2 x = 1$: it lets you convert between $\tan$ and $\sec$ powers. The key derivatives are $\frac{d}{dx} \tan x = \sec^2 x$ and $\frac{d}{dx} \sec x = \sec x \tan x$, which dictate what you “save” for the substitution:

- Save $\sec^2 x \, dx$ when you want $u = \tan x$ (requires the remaining $\sec$ power to be even).
- Save $\sec x \tan x \, dx$ when you want $u = \sec x$ (requires the remaining $\tan$ power to be even, i.e. original $\tan$ power is odd).

Method-selection logic for $\int \tan^m x \sec^n x \, dx$.

- If n is even: save a $\sec^2 x$ factor, convert remaining $\sec^{n-2} x$ into $(1 + \tan^2 x)^{(n-2)/2}$, then substitute $u = \tan x$.

- If m is odd: save a $\sec x \tan x$ factor, convert remaining $\tan^{m-1} x$ into $(\sec^2 x - 1)^{(m-1)/2}$, then substitute $u = \sec x$.

Worked Example: $\int \tan^3 x \, dx$ (reduce power)

Use $\tan^3 x = \tan x \cdot \tan^2 x = \tan x(\sec^2 x - 1)$:

$$\int \tan^3 x \, dx = \int \tan x \sec^2 x \, dx - \int \tan x \, dx.$$

For $\int \tan x \sec^2 x \, dx$, set $u = \tan x$ so $du = \sec^2 x \, dx$:

$$\int \tan x \sec^2 x \, dx = \int u \, du = \frac{u^2}{2} + C = \frac{\tan^2 x}{2} + C.$$

Hence

$$\int \tan^3 x \, dx = \frac{\tan^2 x}{2} - \ln |\sec x| + C.$$

Worked Example (Lecture Example 3.10): $\int \tan^6 x \sec^4 x \, dx$

Since $n = 4$ is even, save $\sec^2 x$ and convert $\sec^2 x = 1 + \tan^2 x$:

$$\int \tan^6 x \sec^4 x \, dx = \int \tan^6 x (1 + \tan^2 x) \sec^2 x \, dx.$$

Let $u = \tan x$, $du = \sec^2 x \, dx$:

$$= \int u^6 (1 + u^2) \, du = \int (u^6 + u^8) \, du = \frac{u^7}{7} + \frac{u^9}{9} + C = \frac{\tan^7 x}{7} + \frac{\tan^9 x}{9} + C.$$

Exam Focus: What if neither m is odd nor n is even?

When m is even and n is odd (e.g. $\int \sec^3 x \, dx$), the standard strategies above do not directly apply. In such cases:

- Try writing $\tan^m x = (\sec^2 x - 1)^{m/2}$ and expanding.
- Use IBP (e.g. $\int \sec^3 x \, dx$ is a classic IBP + solve-for-the-integral problem).
- Use reduction formulas if available.

Product-to-sum identities

Key Formula: Product-to-sum identities

$$\sin A \cos B = \frac{1}{2}(\sin(A - B) + \sin(A + B)),$$

$$\sin A \sin B = \frac{1}{2}(\cos(A - B) - \cos(A + B)),$$

$$\cos A \cos B = \frac{1}{2}(\cos(A - B) + \cos(A + B)).$$

Also: $\cos(-\theta) = \cos \theta$ and $\sin(-\theta) = -\sin \theta$.

When to use these:

These identities convert a *product* of trig functions with *different arguments* (e.g. $\sin 4x \cos 5x$) into a *sum* of single trig functions, each of which is trivially integrable. They are **not** needed when both factors share the same argument—in that case, the power-reduction strategies above apply.

Worked Example: $\int \sin(4x) \cos(5x) dx$

Apply $\sin A \cos B = \frac{1}{2}(\sin(A - B) + \sin(A + B))$:

$$\sin(4x) \cos(5x) = \frac{1}{2}(\sin(-x) + \sin(9x)) = \frac{1}{2}(-\sin x + \sin 9x).$$

Thus

$$\int \sin(4x) \cos(5x) dx = \frac{1}{2} \left(\int -\sin x dx + \int \sin 9x dx \right) = \frac{1}{2} \left(\cos x - \frac{1}{9} \cos 9x \right) + C.$$

Extension (Beyond Syllabus: Complex-exponential shortcut for product-to-sum)

Syllabus link: Chapter 3 — §3.2 Trigonometric Integrals.

Lemma (Euler). For all real θ ,

$$e^{i\theta} = \cos \theta + i \sin \theta, \quad \text{hence} \quad \cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}, \quad \sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}.$$

Full working (derive $\sin A \cos B$ identity).

Using Euler's formulas,

$$\sin A \cos B = \left(\frac{e^{iA} - e^{-iA}}{2i} \right) \left(\frac{e^{iB} + e^{-iB}}{2} \right) = \frac{1}{4i} \left(e^{i(A+B)} + e^{i(A-B)} - e^{-i(A-B)} - e^{-i(A+B)} \right).$$

Group the $(A \pm B)$ terms:

$$\sin A \cos B = \frac{1}{2} \left(\frac{e^{i(A+B)} - e^{-i(A+B)}}{2i} + \frac{e^{i(A-B)} - e^{-i(A-B)}}{2i} \right) = \frac{1}{2} (\sin(A + B) + \sin(A - B)).$$

This reproduces the product-to-sum identity and provides a systematic way to derive others.

3.3 Trigonometric Substitution

Motivation: eliminating radicals.

Expressions like $\sqrt{a^2 - x^2}$, $\sqrt{a^2 + x^2}$, and $\sqrt{x^2 - a^2}$ arise frequently in geometric and physics problems (arc length, surface area, gravitational potentials). These radicals resist ordinary substitution because there is no obvious “inner derivative” to exploit. The idea of trigonometric substitution is to *choose* a substitution that converts the radical into a pure trig expression via a Pythagorean identity, thereby eliminating the square root entirely.

For example, if we set $x = a \sin \theta$, then $a^2 - x^2 = a^2 - a^2 \sin^2 \theta = a^2 \cos^2 \theta$, and so $\sqrt{a^2 - x^2} = a \cos \theta$ (no more radical!). The integral becomes a trigonometric integral in θ , which we can handle using the techniques of Section 3.2.

Key Formula: Trigonometric substitution table

Trigonometric substitution is designed for radicals:

$$\sqrt{a^2 - x^2}, \quad \sqrt{a^2 + x^2}, \quad \sqrt{x^2 - a^2}.$$

A standard choice is:

$$\sqrt{a^2 - x^2} : x = a \sin \theta \quad \left(-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}\right),$$

$$\sqrt{a^2 + x^2} : x = a \tan \theta \quad \left(-\frac{\pi}{2} < \theta < \frac{\pi}{2}\right),$$

$$\sqrt{x^2 - a^2} : x = a \sec \theta \quad \left(0 \leq \theta < \frac{\pi}{2} \text{ or } \pi \leq \theta < \frac{3\pi}{2}\right).$$

The angle restrictions make the substitution one-to-one so that θ can be expressed in terms of x .

Why these particular substitutions?

Each substitution is chosen so that the expression under the radical becomes a perfect square of a trig function:

Radical	Sub.	Identity used
$a^2 - x^2$	$x = a \sin \theta$	$1 - \sin^2 \theta = \cos^2 \theta$
$a^2 + x^2$	$x = a \tan \theta$	$1 + \tan^2 \theta = \sec^2 \theta$
$x^2 - a^2$	$x = a \sec \theta$	$\sec^2 \theta - 1 = \tan^2 \theta$

After the substitution, $\sqrt{\cdot}$ simplifies to $a|\cos \theta|$, $a|\sec \theta|$, or $a|\tan \theta|$ respectively, and the angle restriction ensures the expression inside the absolute value is nonnegative, so you can drop $|\cdot|$.

Why the angle restrictions?

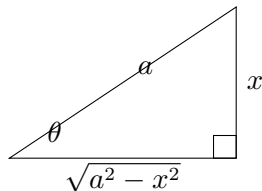
The restriction on θ serves two purposes: (1) it makes the substitution invertible (so you can write θ back in terms of x at the end), and (2) it ensures the trig function inside $\sqrt{\cdot}$ is nonnegative, so you can write $\sqrt{a^2 \cos^2 \theta} = a \cos \theta$ (not $a|\cos \theta|$).

Workflow (exam-ready).

- (1) Identify the radical form and choose the matching substitution $x = g(\theta)$.
- (2) Compute $dx = g'(\theta) d\theta$ and rewrite the entire integrand in θ .
- (3) Simplify using the identity behind the substitution (e.g. $1 - \sin^2 \theta = \cos^2 \theta$).
- (4) Integrate in θ .
- (5) Convert back to x via a triangle (or an explicit inverse relation such as $\theta = \arcsin(x/a)$).

Completing the square first.

If the expression under the radical is not already in one of the three standard forms (e.g. $\sqrt{5 + 4x - x^2}$), complete the square first: $5 + 4x - x^2 = 9 - (x - 2)^2$. Then set $u = x - 2$ so the radical becomes $\sqrt{9 - u^2}$, and proceed with $u = 3 \sin \theta$.

**Worked Example:** $\int \frac{\sqrt{9 - x^2}}{x^2} dx$

Let $x = 3 \sin \theta$ so that $dx = 3 \cos \theta d\theta$ and

$$\sqrt{9 - x^2} = \sqrt{9 - 9 \sin^2 \theta} = 3 \cos \theta.$$

Then

$$\int \frac{\sqrt{9 - x^2}}{x^2} dx = \int \frac{3 \cos \theta}{9 \sin^2 \theta} \cdot 3 \cos \theta d\theta = \int \frac{\cos^2 \theta}{\sin^2 \theta} d\theta = \int \cot^2 \theta d\theta.$$

Use $\cot^2 \theta = \csc^2 \theta - 1$:

$$\int \cot^2 \theta d\theta = \int (\csc^2 \theta - 1) d\theta = -\cot \theta - \theta + C.$$

Convert back. Since $x = 3 \sin \theta$, we have $\theta = \arcsin(x/3)$.

From the triangle, $\cot \theta = \frac{\sqrt{9 - x^2}}{x}$.

Hence

$$\int \frac{\sqrt{9 - x^2}}{x^2} dx = -\frac{\sqrt{9 - x^2}}{x} - \arcsin\left(\frac{x}{3}\right) + C.$$

Domain check: This answer is valid for $|x| < 3$ (i.e. $9 - x^2 > 0$) and $x \neq 0$.

Worked Example (Lecture Example 3.15): Preliminary substitution + trig sub

Evaluate $\int_0^{3\sqrt{3}/2} \frac{x^3}{(4x^2 + 9)^{3/2}} dx$.

Step 1: Note $4x^2 + 9 \neq a^2 + x^2$ directly. Set $x = \frac{3}{2} \tan \theta$, so $4x^2 + 9 = 9 \tan^2 \theta + 9 = 9 \sec^2 \theta$ and $dx = \frac{3}{2} \sec^2 \theta d\theta$.

Limits: $x = 0 \Rightarrow \theta = 0$; $x = \frac{3\sqrt{3}}{2} \Rightarrow \tan \theta = \sqrt{3} \Rightarrow \theta = \frac{\pi}{3}$.

Step 2:

$$\begin{aligned} \int_0^{\pi/3} \frac{(\frac{3}{2} \tan \theta)^3}{(9 \sec^2 \theta)^{3/2}} \cdot \frac{3}{2} \sec^2 \theta d\theta &= \int_0^{\pi/3} \frac{\frac{27}{8} \tan^3 \theta}{27 \sec^3 \theta} \cdot \frac{3}{2} \sec^2 \theta d\theta \\ &= \frac{3}{16} \int_0^{\pi/3} \frac{\tan^3 \theta}{\sec \theta} d\theta = \frac{3}{16} \int_0^{\pi/3} \frac{\sin^3 \theta}{\cos^2 \theta} d\theta. \end{aligned}$$

Step 3: Write $\sin^3 \theta = (1 - \cos^2 \theta) \sin \theta$ and let $u = \cos \theta$:

$$\frac{3}{16} \int_1^{1/2} \frac{-(1 - u^2)}{u^2} du = \frac{3}{16} \int_{1/2}^1 \left(\frac{1}{u^2} - 1 \right) du = \frac{3}{16} \left[-\frac{1}{u} - u \right]_{1/2}^1 = \frac{3}{16} \cdot \frac{1}{2} = \frac{3}{32}.$$

Worked Example (Tutorial 5, Q2(c)): Completing the square + trig sub

Evaluate $\int \frac{1}{\sqrt{5 + 4x - x^2}} dx$.

Step 1: Complete the square: $5 + 4x - x^2 = 9 - (x - 2)^2$. Let $u = x - 2$, $du = dx$:

$$\int \frac{1}{\sqrt{9 - u^2}} du.$$

Step 2: This is a standard form: $\int \frac{du}{\sqrt{a^2 - u^2}} = \arcsin\left(\frac{u}{a}\right) + C$ with $a = 3$.

$$= \arcsin\left(\frac{u}{3}\right) + C = \arcsin\left(\frac{x - 2}{3}\right) + C.$$

Domain: Valid for $|x - 2| < 3$, i.e. $-1 < x < 5$.

Common Mistake: Losing constants in the substitution

If the radical is $\sqrt{\alpha x^2 + \beta}$, do **not** force it into $\sqrt{a^2 + x^2}$ without scaling.

A standard fix is a preliminary substitution $u = \sqrt{\alpha} x$ so that $\alpha x^2 = u^2$, then apply the trig substitution to $\sqrt{u^2 + a^2}$.

Common Mistake: Forgetting to convert back to x

After integrating in θ , you must express the answer in terms of the original variable x . Draw the reference triangle:

- For $x = a \sin \theta$: opposite = x , hypotenuse = a , adjacent = $\sqrt{a^2 - x^2}$.
- For $x = a \tan \theta$: opposite = x , adjacent = a , hypotenuse = $\sqrt{a^2 + x^2}$.
- For $x = a \sec \theta$: hypotenuse = x , adjacent = a , opposite = $\sqrt{x^2 - a^2}$.

Read off $\sin \theta$, $\cos \theta$, $\tan \theta$, etc. from the triangle to complete the back-substitution.

Exam Focus: Back-substitution sanity checks

After converting back to x :

- Check domain restrictions (e.g. $|x| \leq a$ when using $x = a \sin \theta$).
- Differentiate your final answer quickly if the integrand is algebraically delicate.
- For definite integrals, consider switching limits to θ -limits to avoid inverse trig at the end.

3.4 Rational Functions by Partial Fractions

Motivation.

A rational function $P(x)/Q(x)$ can always be integrated if we can decompose it into simpler pieces. The idea is analogous to prime factorisation for integers: just as every integer factors uniquely into primes, every polynomial over $\mathbb{R}$ factors into linear and irreducible quadratic factors. We then “un-add” the fraction into a sum of simpler fractions (partial fractions), each of which has a known antiderivative.

Why must the fraction be proper first?

A proper fraction ($\deg P < \deg Q$) decomposes into partial fractions that are genuinely “simpler” (each has degree of numerator strictly less than degree of denominator factor). If the fraction is *improper*, polynomial long division extracts the polynomial part first; the remainder is then proper and can be decomposed.

Definition 3.1: Rational function and properness

A **rational function** is a ratio of polynomials:

$$\frac{P(x)}{Q(x)} \quad (Q(x) \neq 0).$$

It is **proper** if $\deg P < \deg Q$. If not proper, perform polynomial long division first to rewrite it as

$$\frac{P(x)}{Q(x)} = S(x) + \frac{R(x)}{Q(x)}, \quad \deg R < \deg Q.$$

Key Formula: Partial fraction templates

Factor $Q(x)$ over $\mathbb{R}$ into linear and irreducible quadratic factors.

- Linear factor $(ax + b)^r$ contributes

$$\frac{A_1}{ax + b} + \frac{A_2}{(ax + b)^2} + \cdots + \frac{A_r}{(ax + b)^r}.$$

- Irreducible quadratic factor $(ax^2 + bx + c)^r$ contributes

$$\frac{A_1x + B_1}{ax^2 + bx + c} + \frac{A_2x + B_2}{(ax^2 + bx + c)^2} + \cdots + \frac{A_rx + B_r}{(ax^2 + bx + c)^r}.$$

Why the different numerator forms?

Each partial fraction must be proper with respect to its own denominator factor:

- Over a *linear* factor $(ax + b)^k$ (degree 1), a proper numerator has degree < 1 , so it is a constant A .
- Over an *irreducible quadratic* $(ax^2 + bx + c)^k$ (degree 2), a proper numerator has degree < 2 , so it is linear: $Ax + B$.

Using a constant over an irreducible quadratic would leave you with too few unknowns to match all coefficients of $P(x)$ —the system would be inconsistent.

Integration after decomposition.

- Terms $\frac{1}{ax + b}$ integrate to $\ln|ax + b|$.
- Terms $\frac{1}{(ax + b)^k}$ integrate by power rule.
- For $\frac{Ax + B}{ax^2 + bx + c}$, split into a derivative-of-denominator part plus a remainder:

$$Ax + B = \lambda(2ax + b) + \mu,$$

so that $\int \frac{\lambda(2ax + b)}{ax^2 + bx + c} dx$ is logarithmic, while

$\int \frac{\mu}{ax^2 + bx + c} dx$ uses completing the square and arctan.

Why the “derivative of denominator” trick?

If the numerator is a constant multiple of $Q'(x) = 2ax + b$, then the integral is simply $\lambda \ln|Q(x)| + C$ (by substitution $u = Q(x)$). So we deliberately decompose $Ax + B$ into a multiple of $Q'(x)$ plus a leftover constant. The leftover gives an arctan after completing the square.

Worked Example: Proper rational function (linear factors)

Evaluate $\int \frac{x^2 + 2x - 1}{2x^3 + 3x^2 - 2x} dx$.

First factor the denominator:

$$2x^3 + 3x^2 - 2x = x(2x^2 + 3x - 2) = x(2x - 1)(x + 2).$$

So write

$$\frac{x^2 + 2x - 1}{x(2x - 1)(x + 2)} = \frac{A}{x} + \frac{B}{2x - 1} + \frac{C}{x + 2}.$$

Multiply through by $x(2x - 1)(x + 2)$:

$$x^2 + 2x - 1 = A(2x - 1)(x + 2) + Bx(x + 2) + Cx(2x - 1).$$

Expanding and matching coefficients yields $A = \frac{1}{2}$, $B = \frac{1}{5}$, $C = -\frac{1}{10}$, hence

$$\int \frac{x^2 + 2x - 1}{2x^3 + 3x^2 - 2x} dx = \int \left(\frac{1}{2x} + \frac{1}{5(2x - 1)} - \frac{1}{10(x + 2)} \right) dx.$$

Integrate term-by-term:

$$= \frac{1}{2} \ln |x| + \frac{1}{10} \ln |2x - 1| - \frac{1}{10} \ln |x + 2| + C.$$

Worked Example: Improper rational function (long division first)

Evaluate $\int \frac{x^4 - 2x^2 + 4x + 1}{x^3 - x^2 - x + 1} dx$.

Since $\deg \text{ numerator} > \deg \text{ denominator}$, do long division:

$$\frac{x^4 - 2x^2 + 4x + 1}{x^3 - x^2 - x + 1} = x + 1 + \frac{4x}{x^3 - x^2 - x + 1}.$$

Factor $x^3 - x^2 - x + 1$. Since $Q(1) = 0$, $(x - 1)$ is a factor, and

$$x^3 - x^2 - x + 1 = (x - 1)^2(x + 1).$$

Decompose

$$\frac{4x}{(x - 1)^2(x + 1)} = \frac{A}{x - 1} + \frac{B}{(x - 1)^2} + \frac{C}{x + 1}.$$

Solving gives $A = 1$, $B = 2$, $C = -1$, so

$$\frac{x^4 - 2x^2 + 4x + 1}{x^3 - x^2 - x + 1} = x + 1 + \frac{1}{x - 1} + \frac{2}{(x - 1)^2} - \frac{1}{x + 1}.$$

Integrate:

$$\begin{aligned} \int (x + 1) dx &= \frac{x^2}{2} + x, & \int \frac{1}{x - 1} dx &= \ln |x - 1|, \\ \int \frac{2}{(x - 1)^2} dx &= 2 \int (x - 1)^{-2} dx = -\frac{2}{x - 1}, & \int \frac{1}{x + 1} dx &= \ln |x + 1|. \end{aligned}$$

Thus

$$\int \frac{x^4 - 2x^2 + 4x + 1}{x^3 - x^2 - x + 1} dx = \frac{x^2}{2} + x - \frac{2}{x - 1} + \ln \left| \frac{x - 1}{x + 1} \right| + C.$$

Worked Example (Tutorial 6, Q1(b)): Irreducible quadratic in the denominator

Evaluate $\int \frac{4x}{x^3 + x^2 + x + 1} dx$.

Step 1: Factor. $x^3 + x^2 + x + 1 = x^2(x + 1) + (x + 1) = (x + 1)(x^2 + 1)$.

Note $x^2 + 1$ is irreducible over $\mathbb{R}$ (discriminant < 0).

Step 2: Decompose.

$$\frac{4x}{(x + 1)(x^2 + 1)} = \frac{A}{x + 1} + \frac{Bx + C}{x^2 + 1}.$$

Multiply through: $4x = A(x^2 + 1) + (Bx + C)(x + 1)$.

Set $x = -1$: $-4 = 2A$, so $A = -2$.

Expand and compare x^2 -coefficients: $0 = A + B = -2 + B$, so $B = 2$.

Compare constant terms: $0 = A + C = -2 + C$, so $C = 2$.

Step 3: Integrate.

$$\begin{aligned} \int \frac{4x}{(x + 1)(x^2 + 1)} dx &= \int \left(\frac{-2}{x + 1} + \frac{2x + 2}{x^2 + 1} \right) dx \\ &= -2 \ln |x + 1| + \int \frac{2x}{x^2 + 1} dx + 2 \int \frac{1}{x^2 + 1} dx \\ &= -2 \ln |x + 1| + \ln(x^2 + 1) + 2 \arctan x + C. \end{aligned}$$

Note: The $\frac{2x}{x^2+1}$ term integrates as $\ln(x^2 + 1)$ because $2x$ is the derivative of $x^2 + 1$ —exactly the “derivative of denominator” trick.

Worked Example (Tutorial 6, Q1(e)): Exponential rational integral

Evaluate $\int \frac{e^{2x}}{e^{2x} + 3e^x + 2} dx$.

Key idea: Let $u = e^x$ ($u > 0$), $du = e^x dx$. Then $e^{2x} = u^2$ and $dx = \frac{du}{u}$:

$$\int \frac{u^2}{u^2 + 3u + 2} \cdot \frac{du}{u} = \int \frac{u}{(u + 1)(u + 2)} du.$$

Decompose: $\frac{u}{(u+1)(u+2)} = \frac{-1}{u+1} + \frac{2}{u+2}$.

Integrate:

$$= -\ln |u + 1| + 2 \ln |u + 2| + C = -\ln(e^x + 1) + 2 \ln(e^x + 2) + C.$$

(Since $u = e^x > 0$, absolute values are unnecessary.)

Common Mistake: Missing $(Ax + B)$ over an irreducible quadratic

If $x^2 + 1$ (or any irreducible quadratic) appears in the denominator, the numerator of its partial fraction term must be linear $(Ax + B)$, not a constant.

Otherwise you will not have enough degrees of freedom to match coefficients.

Common Mistake: Forgetting long division for improper fractions

If $\deg P \geq \deg Q$, you **must** do long division before decomposing. Setting up partial fractions on an improper fraction gives an inconsistent system of equations.

Quick degree check: look at the leading terms of P and Q . If the degree of P is at least the degree of Q , divide first.

Exam Focus: Efficient coefficient-finding strategies

- **Strategic substitution:** After multiplying both sides by $Q(x)$, plug in the roots of Q to isolate one unknown at a time (this is the Heaviside cover-up idea for distinct linear factors).
- **Coefficient matching:** For repeated or quadratic factors where strategic substitution alone does not determine all unknowns, expand and compare coefficients of x^k .
- **Hybrid:** Use strategic substitution for as many unknowns as possible, then compare one or two coefficients for the rest.

Extension (Beyond Syllabus: Heaviside cover-up for distinct linear factors)

Syllabus link: Chapter 3 — §3.4 Rational Functions by Partial Fractions.

Proposition (cover-up method).

Suppose a rational function has the form

$$\frac{P(x)}{\prod_{k=1}^m (x - a_k)} = \sum_{k=1}^m \frac{A_k}{x - a_k},$$

where the a_k are *distinct* (no repeated roots) and $\deg P < m$. Then each coefficient is

$$A_j = \left. \frac{P(x)}{\prod_{k \neq j} (x - a_k)} \right|_{x=a_j}.$$

Full working (why it works).

Multiply both sides by $\prod_{k=1}^m (x - a_k)$:

$$P(x) = \sum_{k=1}^m A_k \prod_{\ell \neq k} (x - a_\ell).$$

Now set $x = a_j$. Every term with $k \neq j$ contains a factor $(a_j - a_j) = 0$ and vanishes, leaving

$$P(a_j) = A_j \prod_{\ell \neq j} (a_j - a_\ell),$$

$$\text{so } A_j = \frac{P(a_j)}{\prod_{\ell \neq j} (a_j - a_\ell)} = \left. \frac{P(x)}{\prod_{k \neq j} (x - a_k)} \right|_{x=a_j}.$$

Micro-example.

$$\text{For } \frac{x+5}{(x-1)(x+2)} = \frac{A}{x-1} + \frac{B}{x+2}:$$

$$\text{Cover up } (x-1) \text{ and set } x=1: A = \frac{1+5}{1+2} = 2.$$

$$\text{Cover up } (x+2) \text{ and set } x=-2: B = \frac{-2+5}{-2-1} = -1.$$

3.5 Numerical Integration I: Midpoint and Trapezoidal Rules

Motivation: when antiderivatives fail.

Not every function has a “nice” closed-form antiderivative. For instance, $\int e^{-x^2} dx$ cannot be expressed in terms of elementary functions. Even when an antiderivative exists, it may be so complicated that numerical approximation is more practical. The idea is to go back to the *definition* of the definite integral as a limit of Riemann sums and use finite sums as approximations.

Riemann-sum viewpoint.

When an antiderivative is hard or unavailable, we approximate $\int_a^b f(x) dx$ using Riemann sums. Partition $[a, b]$ into n equal subintervals with $\Delta x = \frac{b-a}{n}$, $x_i = a + i \Delta x$ ($i = 0, 1, \dots, n$).

A general Riemann-sum approximation is

$$\int_a^b f(x) dx \approx \sum_{i=1}^n f(x_i^*) \Delta x,$$

where $x_i^* \in [x_{i-1}, x_i]$. Different choices of x_i^* yield different rules.

Why different rules?

- **Left/Right endpoints** (L_n/R_n): simplest but least accurate. They are first-order methods (error $\sim 1/n$).
- **Midpoint rule** (M_n): uses the midpoint of each subinterval. Surprisingly, this is more accurate than endpoints because errors on the left and right halves of each subinterval partially cancel. It is second-order (error $\sim 1/n^2$).
- **Trapezoidal rule** (T_n): averages L_n and R_n , which is equivalent to approximating f by a straight line (trapezoid) on each subinterval. Also second-order, but typically about twice the error of the midpoint rule.

Key Formula: Endpoint, Midpoint, and Trapezoidal rules

Left endpoints: $L_n = \sum_{i=1}^n f(x_{i-1}) \Delta x.$

Right endpoints: $R_n = \sum_{i=1}^n f(x_i) \Delta x.$

Midpoint rule: $M_n = \sum_{i=1}^n f\left(\frac{x_{i-1} + x_i}{2}\right) \Delta x.$

Trapezoidal rule: $T_n = \frac{L_n + R_n}{2} = \frac{\Delta x}{2} \left(f(x_0) + 2f(x_1) + 2f(x_2) + \dots + 2f(x_{n-1}) + f(x_n) \right).$

Why is T_n the average of L_n and R_n ?

On each subinterval $[x_{i-1}, x_i]$, L_n uses the left height $f(x_{i-1})$ and R_n uses the right height $f(x_i)$. Their average gives $\frac{f(x_{i-1}) + f(x_i)}{2} \cdot \Delta x$, which is exactly the area of a *trapezoid* with parallel sides $f(x_{i-1})$ and $f(x_i)$ and width Δx . Summing over all subintervals gives the formula above, with the interior $f(x_i)$ values counted twice (they are shared by adjacent trapezoids) and the endpoints counted once.

Worked Example: T_5 and M_5 for $\int_1^2 \frac{1}{x} dx$

Let $f(x) = \frac{1}{x}$, $a = 1$, $b = 2$, $n = 5$, so $\Delta x = 0.2$ and $x_0 = 1$, $x_1 = 1.2$, $x_2 = 1.4$, $x_3 = 1.6$, $x_4 = 1.8$, $x_5 = 2$.

Trapezoidal rule:

$$T_5 = \frac{0.2}{2} (f(x_0) + 2f(x_1) + 2f(x_2) + 2f(x_3) + 2f(x_4) + f(x_5)).$$

Substitute values:

$$T_5 = 0.1 \left(1 + \frac{2}{1.2} + \frac{2}{1.4} + \frac{2}{1.6} + \frac{2}{1.8} + \frac{1}{2} \right) \approx 0.695635.$$

Midpoint rule uses midpoints 1.1, 1.3, 1.5, 1.7, 1.9:

$$M_5 = 0.2 \left(\frac{1}{1.1} + \frac{1}{1.3} + \frac{1}{1.5} + \frac{1}{1.7} + \frac{1}{1.9} \right) \approx 0.691908.$$

For reference, the exact value is $\int_1^2 \frac{1}{x} dx = \ln 2 \approx 0.693147$.

Observation: M_5 is closer to the exact value than T_5 . This is typical: the midpoint rule error is roughly half the trapezoidal rule error.

Error bounds

Why error bounds matter.

Computing T_n or M_n gives a number, but how close is it to the true integral? The error bounds below answer: “how large must n be to guarantee accuracy to k decimal places?” They depend on $|f''(x)|$, which measures how “curved” f is—the more curved, the worse a piecewise-linear (trapezoidal) or piecewise-constant (midpoint) approximation performs.

Definition 3.2: Approximation errors

Define the errors

$$E_T = \int_a^b f(x) dx - T_n, \quad E_M = \int_a^b f(x) dx - M_n,$$

$$\text{and similarly } E_L = \int_a^b f(x) dx - L_n, \quad E_R = \int_a^b f(x) dx - R_n.$$

Theorem 3.2: Error bounds for Trapezoidal and Midpoint rules

Assume f'' is twice differentiable on $[a, b]$ and $|f''(x)| \leq K$ for all $x \in [a, b]$. Then

$$|E_T| \leq \frac{K(b-a)^3}{12n^2}, \quad |E_M| \leq \frac{K(b-a)^3}{24n^2}.$$

Observations from the bounds:

- The midpoint bound has 24 in the denominator vs. 12 for the trapezoidal rule, so the midpoint error bound is *half* as large. This matches the empirical observation.
- Both errors decrease as $1/n^2$: doubling n reduces the error by a factor of 4.
- The bound involves $K = \max |f''|$: for nearly-linear functions ($f'' \approx 0$), both rules are very accurate.

Exam Focus: Using error bounds efficiently

To guarantee an accuracy target:

- Compute or bound $|f''(x)|$ on $[a, b]$ and choose a valid K (often the maximum occurs at an endpoint).
- Solve the inequality from the relevant bound for n .
- Round up to the next integer since n must be an integer.

Worked Example: Choose n for midpoint accuracy

Find n such that M_n approximates $\int_1^4 \sqrt{x} \, dx$ with error at most 10^{-4} .

Let $f(x) = \sqrt{x} = x^{1/2}$. Then $f'(x) = \frac{1}{2}x^{-1/2}$, $f''(x) = -\frac{1}{4}x^{-3/2}$, $|f''(x)| = \frac{1}{4}x^{-3/2}$.

On $[1, 4]$, $x^{-3/2}$ decreases, so the maximum occurs at $x = 1$: $|f''(x)| \leq \frac{1}{4} = K$.

Apply the midpoint bound:

$$|E_M| \leq \frac{K(b-a)^3}{24n^2} = \frac{\frac{1}{4} \cdot 3^3}{24n^2} = \frac{27}{96n^2} \leq 10^{-4}.$$

So $n^2 \geq \frac{27}{96} \cdot 10^4 = 2812.5$, hence $n \geq \sqrt{2812.5} \approx 53.033$.
Thus $n = 54$ suffices.

Worked Example (Tutorial 6, Q2): T_8 , M_8 and error for $\int_0^1 e^{-x^2} \, dx$

Let $f(x) = e^{-x^2}$, $a = 0$, $b = 1$, $n = 8$, $\Delta x = 1/8$.

(a) Compute T_8 and M_8 .

$$T_8 = \frac{1/8}{2} (f(0) + 2f(1/8) + \cdots + 2f(7/8) + f(1)).$$

$$M_8 = \frac{1}{8} (f(1/16) + f(3/16) + \cdots + f(15/16)).$$

(Numerical evaluation gives $T_8 \approx 0.902333$, $M_8 \approx 0.905620$ using $f(x) = e^{-x^2}$; the exact value is ≈ 0.746824 —but for the tutorial function $f(x) = e^{x^2}$ on $[0, 1]$, the exact value is different. Substitute the actual function values as specified.)

(b) Error estimation.

$$f'(x) = -2xe^{-x^2}, \quad f''(x) = (4x^2 - 2)e^{-x^2}.$$

On $[0, 1]$: $|f''(x)| = |4x^2 - 2|e^{-x^2} \leq 2e^0 = 2$ at $x = 0$. (Check boundary and critical points.) Take $K = 6$ as a safe bound (tutorial uses $K = 6$).

Then $|E_T| \leq \frac{6}{12 \cdot 64} \approx 0.0078$ and $|E_M| \leq \frac{6}{24 \cdot 64} \approx 0.0039$.

(c) For accuracy 10^{-4} : $\frac{K}{12n^2} \leq 10^{-4}$ gives $n \geq 71$ (trapezoidal); $\frac{K}{24n^2} \leq 10^{-4}$ gives $n \geq 50$ (midpoint).

Common Mistake: Choosing K incorrectly

K is a bound on $|f''(x)|$ over $[a, b]$. You need the **maximum** of $|f''|$, not just $|f''(a)|$ or $|f''(b)|$.

- If $|f''|$ is monotone on $[a, b]$, the max occurs at an endpoint.
- If $|f''|$ has interior critical points, find them (solve $f'''(x) = 0$) and compare values.
- It is fine to *overestimate* K —this gives a slightly larger n than necessary, which is conservative but safe.

3.6 Numerical Integration II: Simpson's Rule

Motivation: can we do better than $O(1/n^2)$?

The trapezoidal and midpoint rules approximate f by straight lines (degree-1 polynomials) on each subinterval, giving error $\sim 1/n^2$. Simpson's Rule approximates f by *parabolas* (degree-2 polynomials) on pairs of subintervals, and achieves error $\sim 1/n^4$ —a dramatic improvement. Doubling n now reduces the error by a factor of 16 instead of 4.

Definition and derivation as a weighted average.

Simpson's Rule combines the trapezoidal and midpoint ideas on pairs of subintervals. From our study, E_M is roughly $-E_T/2$ (opposite sign, half the magnitude). This means a *weighted* average that puts twice as much weight on M as on T should cancel the leading error term:

$$S_n = \frac{1}{3}T_{n/2} + \frac{2}{3}M_{n/2}, \quad n \text{ even.}$$

This weighted average turns out to be equivalent to fitting a parabola through three equally spaced points (x_{i-1}, x_i, x_{i+1}) on each pair of subintervals and integrating the parabola exactly.

Let n be even and $\Delta x = \frac{b-a}{n}$.

Key Formula: Simpson's Rule (n even)

$$\int_a^b f(x) dx \approx S_n = \frac{\Delta x}{3} \left(f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + \cdots + 2f(x_{n-2}) + 4f(x_{n-1}) + f(x_n) \right),$$

where $x_i = a + i \Delta x$.

Why the coefficient pattern 1, 4, 2, 4, 2, ..., 4, 1?

Simpson's Rule processes the subintervals in *pairs*. On each pair $[x_{2k}, x_{2k+2}]$, it fits a parabola through the three points $(x_{2k}, f(x_{2k}))$, $(x_{2k+1}, f(x_{2k+1}))$, $(x_{2k+2}, f(x_{2k+2}))$. The area under this parabola is $\frac{\Delta x}{3}(f(x_{2k}) + 4f(x_{2k+1}) + 2f(x_{2k+2}))$. The endpoint values $f(x_{2k})$ and $f(x_{2k+2})$ are shared between adjacent pairs, so interior even-indexed values get coefficient 2, odd-indexed values get coefficient 4, and the two endpoints get coefficient 1.

Common Mistake: Wrong coefficient pattern in Simpson's Rule

Simpson coefficients alternate as 1, 4, 2, 4, 2, ..., 4, 1 with 4 on odd indices and 2 on even indices (excluding endpoints).

Also, n must be **even**. If n is odd, Simpson's Rule does not apply directly.

Error bound

Theorem 3.3: Error bound for Simpson's Rule

Assume f is four times differentiable on $[a, b]$ and $|f^{(4)}(x)| \leq K$ for all $x \in [a, b]$. If $E_S = \int_a^b f(x) dx - S_n$, then

$$|E_S| \leq \frac{K(b-a)^5}{180n^4}.$$

Why $f^{(4)}$ and not f'' ?

The trapezoidal/midpoint bounds involve f'' because those methods approximate f by degree-1 polynomials (any degree-1 polynomial has $f'' = 0$). Simpson's Rule uses degree-2 approximations (parabolas), so it

is automatically exact for quadratics *and* cubics (by a fortunate symmetry cancellation). The first derivative that is *not* automatically killed is $f^{(4)}$, which is why it appears in the error bound. This also explains why Simpson's Rule is exact for polynomials up to degree 3 (see the Extension below).

Worked Example: Simpson approximation with $n = 8$

Approximate $\int_2^4 \frac{1}{x^3} dx$ using S_8 .

Here $a = 2$, $b = 4$, $n = 8$, so $\Delta x = \frac{4-2}{8} = 0.25$ and $x_0 = 2$, $x_1 = 2.25$, $\dots$, $x_8 = 4$.

Let $f(x) = \frac{1}{x^3}$. Then

$$S_8 = \frac{0.25}{3} \left(f(x_0) + 4f(x_1) + 2f(x_2) + \dots + 4f(x_7) + f(x_8) \right).$$

Substituting the evaluated $f(x_i)$ (calculator) gives $S_8 \approx 10.74159$.

Worked Example (Lecture Example 3.25): Error control for Simpson's Rule

Compute S_8 for $\int_1^3 \frac{1}{x} dx$ and bound the error; then find n for error $\leq 10^{-6}$.

Let $f(x) = \frac{1}{x}$, $a = 1$, $b = 3$, $n = 8$, so $\Delta x = 0.25$ and $x_i = 1 + 0.25i$.

Therefore,

$$S_8 = \frac{1}{3}(0.25) \left(\frac{1}{1} + \frac{4}{1.25} + \frac{2}{1.5} + \frac{4}{1.75} + \frac{2}{2} + \frac{4}{2.25} + \frac{2}{2.5} + \frac{4}{2.75} + \frac{1}{3} \right) \approx 1.09873.$$

Error bound.

For the error bound, compute $f^{(4)}(x)$: $f(x) = x^{-1}$, $f^{(4)}(x) = 24x^{-5}$.

On $[1, 3]$, $24x^{-5}$ decreases, so $|f^{(4)}(x)| \leq 24 = K$.

Thus $|E_S| \leq \frac{24 \cdot (3-1)^5}{180 \cdot n^4} = \frac{24 \cdot 32}{180 n^4} = \frac{64}{15 n^4}$.

For $n = 8$: $|E_S| \leq \frac{64}{15 \cdot 4096} \approx 0.001$.

Choosing n for $|E_S| \leq 10^{-6}$:

Solve $\frac{64}{15 n^4} \leq 10^{-6}$, i.e. $n^4 \geq \frac{64 \cdot 10^6}{15} \approx 4.267 \times 10^6$.

$n \geq (4.267 \times 10^6)^{1/4} \approx 45.45$. Since n must be **even**, take $n = 46$.

Worked Example (Tutorial 6, Q3): S_8 for $\int_1^5 \sqrt{x} dx$ and error control

Let $f(x) = \sqrt{x}$, $a = 1$, $b = 5$, $n = 8$, $\Delta x = 0.5$.

$S_8 = \frac{0.5}{3} (f(1) + 4f(1.5) + 2f(2) + 4f(2.5) + 2f(3) + 4f(3.5) + 2f(4) + 4f(4.5) + f(5)) \approx 4.046655$.

Error bound: $f(x) = x^{1/2}$, $f' = \frac{1}{2}x^{-1/2}$, $f'' = -\frac{1}{4}x^{-3/2}$, $f''' = \frac{3}{8}x^{-5/2}$, $f^{(4)} = -\frac{15}{16}x^{-7/2}$.

On $[1, 5]$: $|f^{(4)}(x)| = \frac{15}{16}x^{-7/2} \leq \frac{15}{16}$ at $x = 1$, so $K = \frac{15}{16}$.

$|E_S| \leq \frac{\frac{15}{16} \cdot 4^5}{180 \cdot 8^4} = \frac{\frac{15}{16} \cdot 1024}{180 \cdot 4096} \approx 0.00013$.

For $|E_S| \leq 10^{-6}$: $\frac{\frac{15}{16} \cdot 1024}{180 n^4} \leq 10^{-6} \Rightarrow n^4 \geq \frac{15 \cdot 1024 \cdot 10^6}{16 \cdot 180} \approx 5.33 \times 10^6$. So $n \geq (5.33 \times 10^6)^{1/4} \approx 48.1$. Take $n = 50$ (next even integer ≥ 49).

Exam Focus: Simpson's Rule computation checklist

- Verify n is **even** before starting.
- List all x_i values and compute $f(x_i)$ in a table for accuracy.
- Apply the coefficient pattern $1, 4, 2, 4, 2, \dots, 4, 1$.
- For the error bound: you need $f^{(4)}$, not f'' .
- When choosing n for a target error, remember to round up to the next **even** integer.

Extension (Beyond Syllabus: Simpson is exact for cubics)

Syllabus link: Chapter 3 — §3.6 Numerical Integration II: Simpson's Rule.

Proposition. If f is a polynomial of degree at most 3 on $[a, b]$, then Simpson's Rule is exact:

$$\int_a^b f(x) dx = S_n \quad \text{for any even } n.$$

Full working via the error-bound hypothesis.

For a cubic or lower-degree polynomial, the fourth derivative is identically zero: $f^{(4)}(x) \equiv 0$ for all x . Thus we may take $K = 0$ in the Simpson error theorem, obtaining

$$|E_S| \leq \frac{0 \cdot (b-a)^5}{180n^4} = 0.$$

Hence $E_S = 0$, i.e. Simpson's approximation equals the exact integral.

This explains why Simpson's Rule is significantly more accurate than trapezoidal/midpoint for smooth functions: its leading error term is controlled by $f^{(4)}$ rather than f'' .

Alternative proof (Tutorial 6, Q4): Direct verification.

By linearity, it suffices to check $S_2 = \int_a^b f(x) dx$ for $f(x) = 1$, $f(x) = x$, and $f(x) = x^2$ (the basis for polynomials up to degree 2; the cubic case then follows from the error-term argument). This is carried out explicitly in Tutorial 6, Question 4.

4 Sequences and Series

4.1 Limits of Sequences

Overview. This chapter shifts focus from continuous calculus (integrals, derivatives) to *discrete* mathematics: sequences and series. A sequence assigns a real number to each positive integer; a series asks whether the “infinite sum” of those numbers makes sense. Why care? Many quantities in science and engineering (power series, Fourier coefficients, numerical approximations) are defined as limits of sequences or sums of series. Before we can manipulate them, we need a rigorous definition of what “converges” means.

Definition 4.1: Sequence and notation

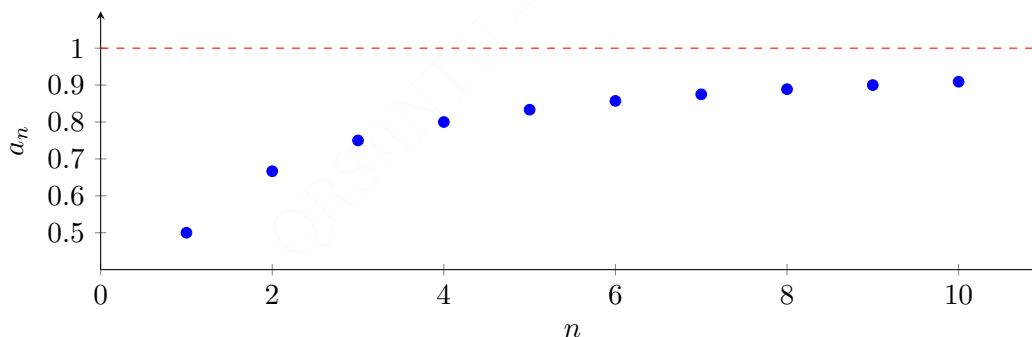
A (real) *sequence* is a function $a : \mathbb{N} \rightarrow \mathbb{R}$, written as $\{a_n\}_{n=1}^{\infty}$ (or simply $\{a_n\}$), where a_n is the n -th term.

The starting index does not have to be 1 (e.g. $\{a_n\}_{n=0}^{\infty}$ is also common).

Why “a function from $\mathbb{N}$ to $\mathbb{R}$ ”? At first glance, a sequence looks like a list: $a_1, a_2, a_3, \dots$. But mathematically, a list *is* a function: you input the position n and output the value a_n . Thinking of a sequence as a function is not pedantic; it lets us immediately apply function-theoretic tools (e.g. L’Hôpital’s Rule via Theorem ??) to compute sequence limits.

Geometric intuition. Think of a sequence as a list of points on the number line (or as plotted points (n, a_n)).

Saying “ $a_n \rightarrow L$ ” means that, as n grows, the points a_n are eventually trapped inside *any* narrow band around L .



Definition 4.2: Limit of a sequence (formal ε - N definition)

We say $\lim_{n \rightarrow \infty} a_n = L$ (or $a_n \rightarrow L$) if for every $\varepsilon > 0$ there exists an integer N such that

$$n > N \implies |a_n - L| < \varepsilon.$$

If such an L exists, $\{a_n\}$ is *convergent*; otherwise it is *divergent*.

Why this definition looks the way it does. The quantity $|a_n - L|$ is the distance from a_n to L . The phrase “for every $\varepsilon > 0$ ” means:

no matter how strict the tolerance is, the tail of the sequence eventually stays within that tolerance.

The index N may depend on ε (tighter tolerance $\implies$ larger N).

Unpacking the definition further (exam-ready). The definition has three quantifiers, and their *order* matters:

1. **“For every $\varepsilon > 0$ ”:** the challenger picks any positive tolerance (no matter how small).
2. **“There exists N ”:** you must respond with a cutoff index N (which may depend on ε).
3. **“For all $n > N$ ”:** every term past the cutoff must satisfy $|a_n - L| < \varepsilon$.

Think of it as a game: if you can always win (i.e. always find an N), the sequence converges to L .

Exam Focus: The ε - N proof template

For an ε - N proof, always follow this structure:

1. State “Let $\varepsilon > 0$.”
2. Compute $|a_n - L|$ and simplify it.
3. Find an upper bound for $|a_n - L|$ that is a simple decreasing function of n (e.g. $\frac{C}{n}$ or $\frac{C}{n^2}$).
4. Choose N so that the upper bound $< \varepsilon$ whenever $n \geq N$ (e.g. $N = \lceil C/\varepsilon \rceil$).
5. Conclude: “For all $n \geq N$, $|a_n - L| \leq \text{bound} \leq \varepsilon$.”

The “backward work” (finding the bound and choosing N) is *scratch work* that does not appear in the formal proof; the formal proof runs *forward* from the chosen N .

Theorem 4.1: Uniqueness and boundedness

- (i) A convergent sequence has a unique limit.
- (ii) Every convergent sequence is bounded.

Why uniqueness? If $a_n \rightarrow L$ and $a_n \rightarrow L'$, then by the triangle inequality $|L - L'| \leq |a_n - L| + |a_n - L'| < 2\varepsilon$ for n large enough. Since ε was arbitrary, $|L - L'| = 0$, so $L = L'$.

Why boundedness? If $a_n \rightarrow L$, take $\varepsilon = 1$: there exists N such that $|a_n - L| < 1$ for all $n > N$, i.e. $a_n \in (L - 1, L + 1)$. The finitely many terms $a_1, \dots, a_N$ are just finitely many real numbers, so they are bounded. Hence the *entire* sequence is bounded.

Common Mistake: Converse of boundedness

Theorem 4.1(ii) says convergent $\Rightarrow$ bounded, but the converse is **false**.

Example: $a_n = (-1)^n$ is bounded ($|a_n| \leq 1$) but divergent (oscillates between 1 and -1).

You need *monotonicity + boundedness* (Theorem ??) to guarantee convergence.

Worked Example (Tutorial 7, Question 1(a)): ε - N proof for $a_n = \frac{1}{3n}$

Show that $\lim_{n \rightarrow \infty} \frac{1}{3n} = 0$.

Solution.

Let $\varepsilon > 0$. We want $|a_n - 0| < \varepsilon$, i.e.

$$\left| \frac{1}{3n} \right| < \varepsilon \iff n > \frac{1}{3\varepsilon}.$$

Choose $N = \lceil \frac{1}{3\varepsilon} \rceil$. Then for all $n \geq N$,

$$|a_n - 0| = \frac{1}{3n} \leq \frac{1}{3N} < \varepsilon.$$

Hence $\lim_{n \rightarrow \infty} \frac{1}{3n} = 0$.

Worked Example (Tutorial 7, Question 1(c)): ε - N proof for a rational sequence

Show that $\lim_{n \rightarrow \infty} \frac{2n^2 - 3n}{3n^2 + 1} = \frac{2}{3}$.

Solution.

Let $\varepsilon > 0$. Compute the difference:

$$\left| \frac{2n^2 - 3n}{3n^2 + 1} - \frac{2}{3} \right| = \left| \frac{3(2n^2 - 3n) - 2(3n^2 + 1)}{3(3n^2 + 1)} \right| = \frac{9n + 2}{9n^2 + 3}.$$

Bounding step. For $n \geq 1$:

$$\frac{9n + 2}{9n^2 + 3} \leq \frac{9n + 2}{9n^2} = \frac{1}{n} + \frac{2}{9n^2} \leq \frac{1}{n} + \frac{2}{9n} = \frac{11}{9n}.$$

Choose $N = \lceil \frac{11}{9\varepsilon} \rceil$. Then for all $n \geq N$:

$$\left| \frac{2n^2 - 3n}{3n^2 + 1} - \frac{2}{3} \right| \leq \frac{11}{9n} \leq \frac{11}{9N} < \varepsilon.$$

Hence $\lim_{n \rightarrow \infty} \frac{2n^2 - 3n}{3n^2 + 1} = \frac{2}{3}$.

Exam note: The key step is replacing the exact expression $\frac{9n+2}{9n^2+3}$ with a simpler upper bound $\frac{11}{9n}$. Any valid bound works; you do *not* need to find the tightest one.

Definition 4.3: Subsequence

Given $\{a_n\}_{n=1}^{\infty}$, a *subsequence* is a sequence of the form

$$a_{n_1}, a_{n_2}, a_{n_3}, \dots \quad \text{where} \quad 1 \leq n_1 < n_2 < n_3 < \dots.$$

Why subsequences matter. The subsequence test (below) is the workhorse for *proving divergence*: if you can exhibit two subsequences with different limits, the original sequence cannot converge. This is typically much easier than a direct ε - N argument for divergence.

Theorem 4.2: Subsequence test

If $\{a_n\}$ converges to L , then *every* subsequence of $\{a_n\}$ also converges to L .

Why this is true (proof sketch). If $n > N$ implies $|a_n - L| < \varepsilon$, and $\{n_k\}$ is a strictly increasing sequence of indices, then $n_k \geq k$ for all k . So for $k > N$, we have $n_k \geq k > N$, giving $|a_{n_k} - L| < \varepsilon$.

Contrapositive (the useful direction for exams):

- If two subsequences converge to *different* limits $\Rightarrow \{a_n\}$ diverges.
- If any one subsequence diverges $\Rightarrow \{a_n\}$ diverges.

Worked Example: $\{(-1)^n\}$ diverges (Lecture Example 4.4)

The even subsequence $a_{2k} = (-1)^{2k} = 1 \rightarrow 1$.

The odd subsequence $a_{2k-1} = (-1)^{2k-1} = -1 \rightarrow -1$.

Since $1 \neq -1$, by the subsequence test, $\{(-1)^n\}$ diverges.

Worked Example (Tutorial 7, Question 2(vii)): subsequence test for $\frac{(-1)^n n^3}{n^3 + 2n^2 + 1}$

Write $a_n = (-1)^n \cdot \frac{1}{1 + \frac{2}{n} + \frac{1}{n^3}}$.

- Even subsequence: $a_{2k} = +\frac{1}{1 + \frac{2}{2k} + \frac{1}{(2k)^3}} \rightarrow 1$.
- Odd subsequence: $a_{2k-1} = -\frac{1}{1 + \frac{2}{2k-1} + \frac{1}{(2k-1)^3}} \rightarrow -1$.

Since $1 \neq -1$, the sequence diverges by the subsequence test.

Worked Example (Tutorial 7, Q3): If odd and even subsequences both converge to L , then $a_n \rightarrow L$

Claim. If $a_{2k-1} \rightarrow L$ and $a_{2k} \rightarrow L$, then $a_n \rightarrow L$.

Proof. Let $\varepsilon > 0$. Since $a_{2k} \rightarrow L$, there exists N_e such that $|a_{2k} - L| < \varepsilon$ for all $k \geq N_e$. Similarly, there exists N_o such that $|a_{2k-1} - L| < \varepsilon$ for all $k \geq N_o$.

Let $N = \max\{2N_e, 2N_o - 1\}$. For any $n \geq N$:

- If $n = 2k$ (even): $n \geq 2N_e$ implies $k \geq N_e$, so $|a_n - L| < \varepsilon$.
- If $n = 2k - 1$ (odd): $n \geq 2N_o - 1$ implies $k \geq N_o$, so $|a_n - L| < \varepsilon$.

Hence $a_n \rightarrow L$. $\square$

Exam use: This fact is useful for sequences defined piecewise (e.g. even and odd terms given by different formulas). If both halves converge to the same L , the full sequence converges.

Definition 4.4: Divergence to $\pm\infty$

We say $a_n \rightarrow \infty$ if for every $M > 0$ there exists N such that $n > N \Rightarrow a_n > M$.
Similarly define $a_n \rightarrow -\infty$.

Why a separate definition? The ε - N definition of $a_n \rightarrow L$ requires L to be a *real number*. Saying $a_n \rightarrow \infty$ is *not* the same as saying “ a_n converges to ∞ ”—it is a specific type of divergence where the terms grow without bound.

Quick implications for positive sequences

Assume $a_n > 0$ for all sufficiently large n .

- (i) If $a_n \rightarrow 0$, then $\frac{1}{a_n} \rightarrow \infty$.
- (ii) If $a_n \rightarrow \infty$, then $\frac{1}{a_n} \rightarrow 0$.
- (iii) If $b_n \geq a_n$ eventually and $a_n \rightarrow \infty$, then $b_n \rightarrow \infty$.

Why these hold (intuition). (i) If a_n is very small and positive, $1/a_n$ is very large. (ii) If a_n is very large, $1/a_n$ is very small. (iii) If a_n already exceeds every bound, and $b_n \geq a_n$, then b_n does too.

Common Mistake

Divergence does *not* behave well under algebra.
For instance, $a_n = (-1)^n$ is divergent, but $|a_n| = 1$ is convergent.
Always justify convergence before applying limit laws.

Exam Focus

To prove divergence quickly, the subsequence test is often the fastest:

- Try even/odd subsequences when $(-1)^n$, $\cos(n\pi)$, or oscillations appear.
- If the expression behaves differently along two simple subsequences (e.g. $n = 2k$ and $n = 2k + 1$), you can often conclude divergence immediately.

4.2 Finding Limits of Sequences

Motivation. The ε - N definition tells us what convergence *means*, but using it directly for every limit would be tedious. This section builds a toolkit of shortcuts: limit laws, the squeeze theorem, the function–sequence bridge (Theorem ??), and the continuity transfer (Theorem ??). Together, these let you compute most sequence limits algebraically.

Limit laws for sequences (require convergence)

Suppose $\{a_n\}$ and $\{b_n\}$ are convergent and $c \in \mathbb{R}$ is a constant. Then:

- (a) $\lim(a_n \pm b_n) = \lim a_n \pm \lim b_n$.
- (b) $\lim(ca_n) = c \lim a_n$.
- (c) $\lim(a_n b_n) = (\lim a_n)(\lim b_n)$.
- (d) $\lim \frac{a_n}{b_n} = \frac{\lim a_n}{\lim b_n}$, provided $\lim b_n \neq 0$.

Why the convergence hypothesis is non-negotiable. Without it, absurd conclusions follow. For example, $a_n = n$ and $b_n = -n$ are both divergent, but $a_n + b_n = 0 \rightarrow 0$. Writing “ $\lim(a_n + b_n) = \lim a_n + \lim b_n = \infty + (-\infty)$ ” is meaningless.

Method selection (high yield). When asked for $\lim_{n \rightarrow \infty} a_n$, try in this order:

1. **Algebraic simplification:** factor/divide by the dominant power of n ; rationalize radicals.
2. **Squeeze:** use boundedness ($|\sin|, |\cos| \leq 1$), or compare with simpler sequences.
3. **Recognise a function:** if $a_n = f(n)$ and $\lim_{x \rightarrow \infty} f(x)$ is known, transfer the result.
4. **Continuity:** if $a_n \rightarrow L$, then $f(a_n) \rightarrow f(L)$ for continuous f .

Exam Focus: When to use which method

- **Rational expressions in n :** Divide top and bottom by $n^{\text{highest power}}$. This always works and is the fastest approach.
- **Bounded oscillating factor $\times$ vanishing term:** Squeeze (e.g. $\frac{\sin n}{n}, \frac{\cos^2 n}{2n}$).
- **Expression involving $\ln$, e , $n!$:** Recognise as $f(n)$ and use L'Hôpital's Rule on the continuous version $f(x)$.
- **Composition with continuous function:** Use Theorem ?? (e.g. $\sin(\pi/n), e^{3n/(n+1)}$).

Theorem 4.3: Sequence defined by a function

Let $f : (A, \infty) \rightarrow \mathbb{R}$ be a function such that $\lim_{x \rightarrow \infty} f(x) = L$, and define $a_n = f(n)$ for integers n large enough.
Then $\lim_{n \rightarrow \infty} a_n = L$.

Why this works. The real limit $\lim_{x \rightarrow \infty} f(x) = L$ means: for every $\varepsilon > 0$, there exists M such that $x > M \Rightarrow |f(x) - L| < \varepsilon$. Integers are a subset of the reals, so choosing $N = \lceil M \rceil$ gives $|f(n) - L| < \varepsilon$ for all $n \geq N$.

Practical payoff: This lets you apply L'Hôpital's Rule (which requires differentiability, a function concept) to compute sequence limits.

Theorem 4.4: Squeeze theorem for sequences

If $a_n \leq b_n \leq c_n$ for all $n \geq N_0$ and $\lim a_n = \lim c_n = L$, then $\lim b_n = L$.

Why the squeeze works. Given $\varepsilon > 0$, for n large enough, $L - \varepsilon < a_n \leq b_n \leq c_n < L + \varepsilon$, so $|b_n - L| < \varepsilon$. The inequalities $a_n \leq b_n \leq c_n$ “squeeze” b_n into the ε -band around L .

Theorem 4.5: Continuity and limits of sequences

If $a_n \rightarrow L$ and f is continuous at L , then $f(a_n) \rightarrow f(L)$.

Why this is true. Continuity at L means: for every $\varepsilon > 0$, there exists $\delta > 0$ such that $|x - L| < \delta \Rightarrow |f(x) - f(L)| < \varepsilon$. Since $a_n \rightarrow L$, eventually $|a_n - L| < \delta$, so $|f(a_n) - f(L)| < \varepsilon$.

Worked Example: geometric sequence r^n

Determine $\lim_{n \rightarrow \infty} r^n$.

Solution.

- (i) If $|r| < 1$, then $|r|^n \rightarrow 0$, so $r^n \rightarrow 0$.
- (ii) If $r = 1$, then $r^n = 1$ for all n .
- (iii) If $r > 1$, then $r^n \rightarrow \infty$ (unbounded growth).
- (iv) If $r \leq -1$, the terms do not approach a single real number (they oscillate in sign and/or grow in magnitude), hence divergence.

Why the case $|r| < 1$ works. If $0 < r < 1$, write $r = \frac{1}{1+p}$ with $p > 0$. By Bernoulli's inequality, $(1+p)^n \geq 1+np$, so $0 < r^n = \frac{1}{(1+p)^n} \leq \frac{1}{1+np} \rightarrow 0$. For $-1 < r < 0$, we have $|r^n| = |r|^n \rightarrow 0$, and $|a_n| \rightarrow 0 \Rightarrow a_n \rightarrow 0$ by the squeeze theorem.

Worked Example: a bounded trigonometric factor (Squeeze)

Evaluate $\lim_{n \rightarrow \infty} \frac{\cos^2 n}{2n}$.

Solution.

Since $0 \leq \cos^2 n \leq 1$,

$$0 \leq \frac{\cos^2 n}{2n} \leq \frac{1}{2n}.$$

As $n \rightarrow \infty$, $\frac{1}{2n} \rightarrow 0$. Hence by Theorem ??,

$$\lim_{n \rightarrow \infty} \frac{\cos^2 n}{2n} = 0.$$

Worked Example: continuity trick

Evaluate $\lim_{n \rightarrow \infty} \sin\left(\frac{\pi}{n}\right)$.

Solution.

We know $\frac{\pi}{n} \rightarrow 0$. Since $\sin x$ is continuous at 0, Theorem ?? gives

$$\lim_{n \rightarrow \infty} \sin\left(\frac{\pi}{n}\right) = \sin 0 = 0.$$

Worked Example (Tutorial 7, Q2(viii)): continuity with tan

Evaluate $\lim_{n \rightarrow \infty} \tan\left(\frac{2n\pi}{1+8n}\right)$.

Solution. First find the “inner” limit:

$$\frac{2n\pi}{1+8n} = \frac{2\pi}{8 + \frac{1}{n}} \rightarrow \frac{2\pi}{8} = \frac{\pi}{4}.$$

Since $\tan$ is continuous at $\pi/4$ (which is not an odd multiple of $\pi/2$), Theorem ?? gives

$$\lim_{n \rightarrow \infty} \tan\left(\frac{2n\pi}{1+8n}\right) = \tan\left(\frac{\pi}{4}\right) = \boxed{1}.$$

Sanity check: Always verify the inner limit lands at a point where the outer function is continuous. If it landed at $\pi/2$, $\tan$ would be undefined and the argument would fail.

Worked Example: comparing growth rates

Show that $\lim_{n \rightarrow \infty} \frac{n!}{n^n} = 0$.

Solution.

$$\frac{n!}{n^n} = \frac{1 \cdot 2 \cdot 3 \cdots n}{n \cdot n \cdot n \cdots n} = \prod_{k=1}^n \frac{k}{n}.$$

For $2 \leq k \leq n$, $\frac{k}{n} \leq 1$, so

$$0 \leq \frac{n!}{n^n} = \frac{1}{n} \cdot \prod_{k=2}^n \frac{k}{n} \leq \frac{1}{n}.$$

Since $\frac{1}{n} \rightarrow 0$, the squeeze theorem yields $\lim_{n \rightarrow \infty} \frac{n!}{n^n} = 0$.

Worked Example: if $|a_n| \rightarrow 0$ then $a_n \rightarrow 0$

Assume $\lim_{n \rightarrow \infty} |a_n| = 0$. Show $\lim_{n \rightarrow \infty} a_n = 0$.

Solution.

For every n , $-|a_n| \leq a_n \leq |a_n|$.

Taking limits, the squeeze theorem gives $a_n \rightarrow 0$.

Worked Example (Lecture Example 4.14): L'Hôpital via the function bridge

Calculate $\lim_{n \rightarrow \infty} \frac{n + \ln n}{n^2}$.

Solution. Define $f(x) = \frac{x + \ln x}{x^2}$. This is an $\frac{\infty}{\infty}$ form as $x \rightarrow \infty$. By L'Hôpital's Rule:

$$\lim_{x \rightarrow \infty} \frac{x + \ln x}{x^2} = \lim_{x \rightarrow \infty} \frac{1 + \frac{1}{x}}{2x} = \lim_{x \rightarrow \infty} \frac{1}{2x} + \frac{1}{2x^2} = 0.$$

By Theorem ??, $\lim_{n \rightarrow \infty} \frac{n + \ln n}{n^2} = 0$.

Worked Example (Lecture Example 4.15): composing function bridge with continuity

Calculate $\lim_{n \rightarrow \infty} e^{3n/(n+1)}$.

Solution. Let $a_n = \frac{3n}{n+1} = \frac{3}{1+1/n} \rightarrow 3$.

Since $f(x) = e^x$ is continuous at 3, Theorem ?? gives

$$\lim_{n \rightarrow \infty} e^{3n/(n+1)} = e^{\lim a_n} = \boxed{e^3}.$$

Worked Example (Tutorial 7, Q5): $\sqrt[n]{a} \rightarrow 1$ for $a > 0$

Claim: $\lim_{n \rightarrow \infty} a^{1/n} = 1$ for all $a > 0$.

Proof (continuity method). Write $a^{1/n} = e^{\frac{\ln a}{n}}$. Since $\frac{\ln a}{n} \rightarrow 0$ and e^x is continuous at 0:

$$\lim_{n \rightarrow \infty} a^{1/n} = e^0 = 1.$$

Proof (squeeze for $a > 1$). Let $b = a - 1 > 0$. By Bernoulli's inequality, $(1 + b/n)^n \geq 1 + b = a$, so $a^{1/n} \leq 1 + b/n$. Also $a^{1/n} \geq 1$ since $a > 1$. Hence $1 \leq a^{1/n} \leq 1 + \frac{b}{n} \rightarrow 1$ by squeeze.

For $0 < a < 1$, apply the $a > 1$ case to $1/a$ and take reciprocals.

Extension (Beyond Syllabus: Stolz–Cesàro theorem for limits)

Syllabus link: Chapter 4 — §4.2 Finding Limits of Sequences.

Theorem (Stolz–Cesàro).

Let $\{b_n\}$ be a strictly increasing sequence with $b_n \rightarrow \infty$.

Let $\{a_n\}$ be any sequence.

If the limit

$$\lim_{n \rightarrow \infty} \frac{a_{n+1} - a_n}{b_{n+1} - b_n} = L$$

exists (finite), then $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = L$.

Application. Compute $\lim_{n \rightarrow \infty} \frac{\ln n}{n}$.

Take $a_n = \ln n$ and $b_n = n$. Then b_n is strictly increasing and $b_n \rightarrow \infty$.

Moreover,

$$\frac{a_{n+1} - a_n}{b_{n+1} - b_n} = \frac{\ln(n+1) - \ln n}{(n+1) - n} = \ln\left(1 + \frac{1}{n}\right).$$

Since $\ln(1+x) \rightarrow 0$ as $x \rightarrow 0$, we have $\ln(1 + \frac{1}{n}) \rightarrow 0$. By Stolz–Cesàro,

$$\lim_{n \rightarrow \infty} \frac{\ln n}{n} = 0.$$

Common Mistake

The limit laws require the individual limits to exist.

If $\{a_n\}$ or $\{b_n\}$ is divergent, statements like “ $\lim(a_n + b_n) = \lim a_n + \lim b_n$ ” are not valid.

Worked Example (Tutorial 7, Q4): True/false on divergent sequences

(a) “If $\{a_n\}$ and $\{b_n\}$ are divergent, then $\{a_n + b_n\}$ is divergent.” **FALSE.**

Counterexample: $a_n = (-1)^n$, $b_n = -(-1)^n$. Both diverge, but $a_n + b_n = 0 \rightarrow 0$.

(b) “If $\{a_n\}$ is divergent, then $\{|a_n|\}$ is divergent.” **FALSE.**

Counterexample: $a_n = (-1)^n$ diverges, but $|a_n| = 1 \rightarrow 1$.

(c) “If $a_n \rightarrow L$ and $b_n \rightarrow 0$, then $a_n b_n \rightarrow 0$.” **TRUE.**

Proof: Since $a_n \rightarrow L$, $\{a_n\}$ is bounded: $|a_n| \leq M$. Given $\varepsilon > 0$, choose N so that $|b_n| < \varepsilon/M$ for $n \geq N$. Then $|a_n b_n| \leq M|b_n| < \varepsilon$.

Exam Focus

For rational expressions in n :

- Divide numerator and denominator by the highest power of n present.
- For $\sqrt{\text{polynomial in } n}$, factor out the dominant power inside the root: $\sqrt{n^2(1 + \dots)} = n\sqrt{1 + \dots}$.
- If you rationalize, rationalize *once* and simplify before taking limits.

4.3 Monotonic Sequences

Motivation. Many important sequences (especially recursively defined ones) are hard to evaluate by direct algebraic manipulation. The Monotone Sequence Theorem gives an *existence* guarantee: if a sequence is monotonic and bounded, it converges—even if you cannot compute the limit directly. You then find the limit by solving the fixed-point equation $L = f(L)$. This three-step workflow (bound $\rightarrow$ monotone $\rightarrow$ fixed point) is one of the most frequently examined techniques in MH1101.

Definition 4.5: Monotonicity

A sequence $\{a_n\}$ is *increasing* if $a_n \leq a_{n+1}$ for all n (and *strictly increasing* if $a_n < a_{n+1}$).
 A sequence $\{a_n\}$ is *decreasing* if $a_n \geq a_{n+1}$ for all n (and *strictly decreasing* if $a_n > a_{n+1}$).
 A sequence is *monotonic* if it is increasing or decreasing.

Ways to check monotonicity.

- **Difference test:** Compute $a_{n+1} - a_n$ and determine its sign.
- **Ratio test (for positive sequences):** Compute a_{n+1}/a_n and check whether it is ≥ 1 or ≤ 1 .
- **Function derivative:** If $a_n = f(n)$, compute $f'(x)$ and check its sign.
- **For recursive sequences $a_{n+1} = f(a_n)$:** Show $f(x) \geq x$ (or $\leq x$) on the relevant interval.

Definition 4.6: Boundedness

A sequence $\{a_n\}$ is *bounded above* if there exists M such that $a_n \leq M$ for all n .
 It is *bounded below* if there exists m such that $m \leq a_n$ for all n .
 It is *bounded* if it is both bounded above and bounded below.

Theorem 4.6: Monotone sequence theorem

Every bounded, monotonic sequence is convergent.

Why this theorem is true (intuition). An increasing sequence that is bounded above has a least upper bound (supremum) L by the completeness of $\mathbb{R}$. Since L is the *least* upper bound, for any $\varepsilon > 0$, there must be some $a_N > L - \varepsilon$ (otherwise $L - \varepsilon$ would be a smaller upper bound). Since the sequence is increasing, all subsequent terms satisfy $a_n \geq a_N > L - \varepsilon$, and also $a_n \leq L < L + \varepsilon$. Hence $|a_n - L| < \varepsilon$ for all $n \geq N$.

Standard recursion workflow. For a recursively defined sequence $a_{n+1} = f(a_n)$:

1. **Invariant interval (boundedness):** find an interval $I = [m, M]$ such that $a_1 \in I$ and $f(I) \subseteq I$.
2. **Monotonicity:** show $a_{n+1} \geq a_n$ (or $\leq$) using algebra and the bounds from Step 1.
3. **Limit identification:** conclude convergence by Theorem ??; let $a_n \rightarrow L$ and solve $L = f(L)$.
4. **Reject extraneous roots:** keep only solutions consistent with the bounds (e.g. $L \in I$).

Exam Focus: Why you *must* prove convergence before solving $L = f(L)$

The equation $L = f(L)$ may have multiple solutions, and without knowing the sequence converges, you cannot even write $\lim a_{n+1} = f(\lim a_n)$. Example: define $a_1 = 0$, $a_{n+1} = a_n^2 + a_n$. The equation $L = L^2 + L$ gives $L = 0$, but the sequence is $0, 0, 0, \dots$ and $L = 0$ is the only fixed point—so it works. But if you had $a_1 = 2$, the sequence diverges to ∞ , and “solving $L = f(L)$ ” gives the wrong answer. Always establish convergence *first*, then solve the fixed-point equation.

Worked Example (Tutorial 8, Question 1): nested radicals

Find the limit of

$$\sqrt{2}, \sqrt{2\sqrt{2}}, \sqrt{2\sqrt{2\sqrt{2}}}, \dots$$

Define $a_1 = \sqrt{2}$ and $a_{n+1} = \sqrt{2a_n}$.

Bounded: show by induction that $0 < a_n < 2$ for all n .

Base: $a_1 = \sqrt{2} < 2$. ✓

Inductive step: if $a_n < 2$, then $a_{n+1} = \sqrt{2a_n} < \sqrt{2 \cdot 2} = 2$. ✓

Also $a_n > 0$ trivially (positive under square root).

Increasing: $a_{n+1} \geq a_n \iff \sqrt{2a_n} \geq a_n \iff 2a_n \geq a_n^2 \iff a_n(2 - a_n) \geq 0$.

Since $0 < a_n < 2$, both factors are positive. So increasing. ✓

Limit: by Theorem ??, $a_n \rightarrow L$ exists. Then $L = \sqrt{2L}$, so $L^2 = 2L$, giving $L(L - 2) = 0$.

Since $L > 0$, we conclude $\boxed{L = 2}$.

Worked Example (Tutorial 8, Question 2): $a_1 = 1$, $a_{n+1} = 3 - \frac{1}{a_n}$

Show the sequence is increasing and $0 < a_n < 3$, then find the limit.

Invariant interval: We show $a_n \in [1, 3)$ by induction.

Base: $a_1 = 1 \in [1, 3)$. ✓

Inductive step: if $1 \leq a_n < 3$, then $\frac{1}{a_n} \leq 1$, so $a_{n+1} = 3 - \frac{1}{a_n} \geq 2 \geq 1$.

Also $\frac{1}{a_n} > 0$, so $a_{n+1} < 3$. ✓

Increasing: $a_{n+1} - a_n = 3 - \frac{1}{a_n} - a_n = \frac{-a_n^2 + 3a_n - 1}{a_n}$.

The numerator $-a_n^2 + 3a_n - 1 \geq 0$ when $a_n \in [\alpha, \beta]$ where $\alpha = \frac{3-\sqrt{5}}{2} \approx 0.382$ and $\beta = \frac{3+\sqrt{5}}{2} \approx 2.618$.

Since $a_n \in [1, 3) \subset [\alpha, \beta]$ (check: $1 > \alpha$ and $3 > \beta$ but $a_n < 3$ and is increasing with $a_n \leq \beta$), the numerator is ≥ 0 and $a_n > 0$, so $a_{n+1} \geq a_n$. ✓

Limit: $L = 3 - \frac{1}{L} \Rightarrow L^2 - 3L + 1 = 0 \Rightarrow L = \frac{3 \pm \sqrt{5}}{2}$.

Since a_n is increasing from $a_1 = 1$ and bounded above by $\beta = \frac{3+\sqrt{5}}{2}$:

$$\boxed{L = \frac{3 + \sqrt{5}}{2}}.$$

Worked Example (Tutorial 8, Question 3): $a_1 = 2$, $a_{n+1} = \frac{1}{3-a_n}$

Invariant interval: Show $a_n \in (\alpha, \beta]$ where $\alpha = \frac{3-\sqrt{5}}{2}$ and $\beta = \frac{3+\sqrt{5}}{2}$. (Both are fixed points of $f(x) = \frac{1}{3-x}$.)

Since $a_1 = 2 \in (\alpha, \beta)$ and f is increasing on $(-\infty, 3)$ with $f(\alpha) = \alpha$, $f(\beta) = \beta$, the interval $[\alpha, \beta]$ is invariant.

Decreasing: $a_{n+1} \leq a_n \iff \frac{1}{3-a_n} \leq a_n \iff a_n^2 - 3a_n + 1 \leq 0$, which holds for $a_n \in [\alpha, \beta]$. ✓

Since $\{a_n\}$ is decreasing and bounded below, it converges. Solving $L = \frac{1}{3-L}$ gives $L \in \{\alpha, \beta\}$. Since the sequence decreases from $a_1 = 2$ and $a_2 = 1 < 2$, we have $L \leq 1 < \beta$, so:

$$L = \frac{3 - \sqrt{5}}{2}.$$

Common Mistake

For recursive sequences, solving $L = f(L)$ is *not* enough.

You must first justify that $\{a_n\}$ converges (typically by boundedness + monotonicity).

Exam Focus

When proving monotonicity, avoid expanding blindly.

A reliable pattern is:

$$a_{n+1} \geq a_n \iff f(a_n) \geq a_n,$$

then factor and use bounds on a_n to determine signs (e.g. $a_n(2 - a_n) \geq 0$).

- For the invariant-interval step, try $M =$ the larger fixed point of f (solve $L = f(L)$).
- For monotonicity, the sign of $f(x) - x$ on the invariant interval is the key.
- Common pitfall: assuming monotonicity from the first few terms without proof by induction.

Extension (Tutorial 8, Question 4: $(1 + \frac{1}{n})^n$ is increasing)

Syllabus link: Chapter 4 — §4.3 Monotonic Sequences.

Proposition. The sequence $e_n = (1 + \frac{1}{n})^n$ is increasing and bounded above (by 3), hence convergent. Its limit is e .

Proof of monotonicity (calculus approach). Define $\phi(x) = x \ln(1 + 1/x)$ for $x > 0$, so that $e_n = e^{\phi(n)}$. Differentiate:

$$\phi'(x) = \ln\left(1 + \frac{1}{x}\right) - \frac{1}{x+1}.$$

Using the inequality $\ln(1 + u) \geq \frac{u}{1+u}$ for $u > 0$ (with $u = 1/x$):

$$\ln\left(1 + \frac{1}{x}\right) \geq \frac{1/x}{1 + 1/x} = \frac{1}{x+1}.$$

Hence $\phi'(x) \geq 0$, so ϕ is increasing, and $e_n = e^{\phi(n)}$ is increasing.

Bounded: By the binomial theorem,

$$e_n = \sum_{k=0}^n \binom{n}{k} \frac{1}{n^k} \leq 1 + \sum_{k=1}^n \frac{1}{k!} \leq 1 + \sum_{k=1}^n \frac{1}{2^{k-1}} < 1 + 2 = 3.$$

By the Monotone Sequence Theorem, e_n converges. The limit is denoted $e \approx 2.71828$.

4.4 Series

From a sequence to an “infinite sum”. An infinite series is written $\sum_{n=1}^{\infty} a_n$.

Since we can only add finitely many numbers at a time, the rigorous meaning is via partial sums.

Why partial sums? We know how to add 2 numbers, 3 numbers, ..., k numbers. But “adding infinitely many numbers” is not defined by the arithmetic axioms. The only way to give $\sum_{n=1}^{\infty} a_n$ a meaning is to define it as a *limit*: form the finite sums $s_k = a_1 + \dots + a_k$ and ask whether $\{s_k\}$ converges. If it does, the limit *is* the “infinite sum.” If it doesn’t, the series has no sum.

This is analogous to how we defined the definite integral as a limit of Riemann sums: infinitely many infinitesimal areas added up via a limiting process.

Definition 4.7: Series and partial sums

Given a sequence $\{a_n\}_{n=1}^{\infty}$, define the k -th partial sum

$$s_k = \sum_{n=1}^k a_n.$$

- (i) If $\{s_k\}$ converges to a real number S , then the series $\sum_{n=1}^{\infty} a_n$ is *convergent* and we write

$$\sum_{n=1}^{\infty} a_n = S.$$

- (ii) If $\{s_k\}$ diverges, then the series is *divergent*.

Key conceptual point. The symbol $\sum_{n=1}^{\infty} a_n$ does **double duty**: it denotes both the formal expression (the “series”) and, when convergent, its numerical value $S = \lim_{k \rightarrow \infty} s_k$. Always be clear about which

meaning you intend.

Worked Example: a divergent series with oscillating partial sums

Consider $\sum_{n=1}^{\infty} (-1)^{n-1} = 1 - 1 + 1 - 1 + \dots$.

The partial sums are $s_1 = 1, s_2 = 0, s_3 = 1, s_4 = 0, \dots$, which do not converge. Hence the series diverges.

Why the “ $0 = 1$ ” paradox fails (Lecture Example 4.19). The “proof” that $0 = 1$ via $0 = (1 - 1) + (1 - 1) + \dots = 1 + (-1 + 1) + (-1 + 1) + \dots = 1$ fails because it *regroups* terms of a divergent series. Regrouping (associativity) is valid for finite sums but not for infinite series unless convergence is established first.

Common Mistake

Infinite series are *not* finite sums.

Operations such as regrouping terms, cancelling “ $\infty - \infty$ ”, or treating “ $\dots$ ” algebraically are not justified unless you can translate them into statements about the partial sums s_k .

Theorem 4.7: Geometric series

Let $a \neq 0$ and $r \in \mathbb{R}$. The geometric series

$$\sum_{n=1}^{\infty} ar^{n-1} = a + ar + ar^2 + \dots$$

converges iff $|r| < 1$, and in that case

$$\sum_{n=1}^{\infty} ar^{n-1} = \frac{a}{1-r}.$$

If $|r| \geq 1$, the series diverges.

Why the formula is $\frac{a}{1-r}$. The partial sum formula $s_k = \frac{a(1-r^k)}{1-r}$ is derived by the “multiply-and-subtract” trick (see Key Formula below). When $|r| < 1$, the term $r^k \rightarrow 0$ as $k \rightarrow \infty$, so $s_k \rightarrow \frac{a(1-0)}{1-r} = \frac{a}{1-r}$.

When $|r| \geq 1$, either r^k diverges ($|r| > 1$) or $s_k = ka \rightarrow \pm\infty$ ($r = 1$), or r^k oscillates without converging ($r = -1$).

Geometric partial sum (exam setup)

For $r \neq 1$, the k -th partial sum is

$$s_k = a + ar + \dots + ar^{k-1} = \frac{a(1-r^k)}{1-r}.$$

When $|r| < 1$, use $r^k \rightarrow 0$ to obtain the infinite sum $\frac{a}{1-r}$.

Derivation of the partial-sum formula. Write:

$$\begin{aligned} s_k &= a + ar + ar^2 + \dots + ar^{k-1}, \\ r \cdot s_k &= ar + ar^2 + ar^3 + \dots + ar^k. \end{aligned}$$

Subtract: $s_k - rs_k = a - ar^k$, so $s_k(1 - r) = a(1 - r^k)$, giving $s_k = \frac{a(1-r^k)}{1-r}$.

Exam Focus: Identifying geometric series

The first step is always to rewrite the general term in the form $a \cdot r^{n-1}$ (or equivalently $\text{const} \cdot r^n$).

- Identify a (first term) and r (common ratio).
- Check $|r| < 1$ for convergence.
- If the sum starts at $n = 0$ instead of $n = 1$: $\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$.
- If the sum starts at $n = k$ instead of $n = 1$: factor out r^{k-1} or compute as $S_{\text{total}} - s_{k-1}$.

Worked Example: sum of a geometric series

Evaluate $\sum_{n=1}^{\infty} \frac{(-3)^{n-1}}{4^n}$.

Solution.

Rewrite

$$\frac{(-3)^{n-1}}{4^n} = \frac{1}{4} \left(-\frac{3}{4} \right)^{n-1}.$$

This is geometric with $a = \frac{1}{4}$ and $r = -\frac{3}{4}$ (so $|r| = \frac{3}{4} < 1$).
Therefore,

$$\sum_{n=1}^{\infty} \frac{(-3)^{n-1}}{4^n} = \frac{\frac{1}{4}}{1 - (-\frac{3}{4})} = \frac{1/4}{7/4} = \frac{1}{7}.$$

Worked Example (Lecture Example 4.21): identifying divergent geometric

Is $\sum_{n=1}^{\infty} 2^{2n} 3^{1-n}$ convergent?

Solution. Rewrite: $2^{2n} 3^{1-n} = (2^2)^n \cdot 3 \cdot 3^{-n} = 3 \cdot \left(\frac{4}{3}\right)^n = 4 \cdot \left(\frac{4}{3}\right)^{n-1}$.

This is geometric with $r = \frac{4}{3} > 1$, so the series **diverges**.

Worked Example: telescoping via partial fractions

Evaluate $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$.

Solution.

$$\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}.$$

Then

$$s_k = \sum_{n=1}^k \left(\frac{1}{n} - \frac{1}{n+1} \right) = 1 - \frac{1}{k+1} \xrightarrow{k \rightarrow \infty} 1,$$

so $\boxed{\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = 1}.$

Why telescoping works. When partial fractions produce a difference $f(n) - f(n+1)$, consecutive terms in s_k cancel:

$$s_k = (f(1) - f(2)) + (f(2) - f(3)) + \cdots + (f(k) - f(k+1)) = f(1) - f(k+1).$$

Only the first and last terms survive—like a collapsing telescope.

Worked Example (Tutorial 8, Q5(iv)): telescoping plus geometric

Evaluate $\sum_{n=2}^{\infty} \left(\frac{1}{e^n} + \frac{2}{n^2 - 1} \right)$.

Solution. Split into two convergent series:

Geometric part: $\sum_{n=2}^{\infty} \frac{1}{e^n} = \frac{1/e^2}{1 - 1/e} = \frac{1}{e(e-1)}.$

Telescoping part: $\frac{2}{n^2-1} = \frac{2}{(n-1)(n+1)} = \frac{1}{n-1} - \frac{1}{n+1}.$

$$\sum_{n=2}^N \left(\frac{1}{n-1} - \frac{1}{n+1} \right) = \left(1 + \frac{1}{2} \right) - \left(\frac{1}{N} + \frac{1}{N+1} \right) \rightarrow \frac{3}{2}.$$

Total: $\boxed{\sum_{n=2}^{\infty} \left(\frac{1}{e^n} + \frac{2}{n^2 - 1} \right) = \frac{1}{e(e-1)} + \frac{3}{2}}.$

Worked Example: harmonic series diverges (grouping idea)

Show that $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges.

Solution.

Let $s_k = \sum_{n=1}^k \frac{1}{n}$. For $k = 2^m$,

$$s_{2^m} = 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \cdots + \frac{1}{8}\right) + \cdots + \left(\frac{1}{2^{m-1}+1} + \cdots + \frac{1}{2^m}\right).$$

Each block has 2^{j-1} terms, each at least $1/2^j$, so each block sum is at least $1/2$.

Thus $s_{2^m} \geq 1 + \frac{m}{2} \rightarrow \infty$, hence the harmonic series diverges.

Why the harmonic series is important. It is the canonical example showing that $a_n \rightarrow 0$ does *not* imply convergence: $\frac{1}{n} \rightarrow 0$, yet $\sum \frac{1}{n} = \infty$. This is why the n th-term test (below) can only prove *divergence*, never convergence.

Theorem 4.8: Necessary condition for series convergence

If $\sum_{n=1}^{\infty} a_n$ converges, then $\lim_{n \rightarrow \infty} a_n = 0$.

Why this is true. If $\sum a_n = S$, then $s_k \rightarrow S$ and $s_{k-1} \rightarrow S$. Since $a_k = s_k - s_{k-1}$, we get $a_k \rightarrow S - S = 0$.

Theorem 4.9: The n th-term test for divergence

If $\lim_{n \rightarrow \infty} a_n$ does not exist or $\lim_{n \rightarrow \infty} a_n \neq 0$, then $\sum_{n=1}^{\infty} a_n$ diverges.

Why this is just the contrapositive of Theorem ??. Theorem ?? says: convergent $\Rightarrow a_n \rightarrow 0$. Contrapositive: $a_n \not\rightarrow 0 \Rightarrow$ divergent. That's exactly the n th-term test.

Common Mistake: $a_n \rightarrow 0$ does NOT imply convergence

If $\lim a_n = 0$, the n th-term test is **inconclusive**—it tells you nothing.

The harmonic series ($a_n = 1/n \rightarrow 0$, but diverges) proves this.

You need a stronger test (comparison, ratio, root, integral, etc.) to establish convergence when $a_n \rightarrow 0$.

Exam Focus

A fast first check for any series $\sum a_n$:

- Compute $\lim_{n \rightarrow \infty} a_n$. If the limit is not 0 (or does not exist), the series *immediately* diverges by Theorem ??.
- If $\lim a_n = 0$, you cannot conclude convergence; you will need additional ideas (covered later in the course).

Worked Example (Tutorial 8, Q5 collection): quick divergence checks

$$(v) \sum_{n=1}^{\infty} \frac{n^3 + n^2}{n^3 - 2n + 5}.$$

$$a_n = \frac{n^3 + n^2}{n^3 - 2n + 5} \rightarrow \frac{1+0}{1-0+0} = 1 \neq 0. \text{ Diverges by the } n\text{th-term test.}$$

$$(iii) \sum_{n=1}^{\infty} \frac{e^n}{n^2}.$$

$$a_n = \frac{e^n}{n^2} \rightarrow \infty \text{ (exponential dominates polynomial). Diverges by the } n\text{th-term test.}$$

$$(vi) \sum_{n=1}^{\infty} \left(\frac{3}{5}n + \frac{2}{n} \right)^2.$$

$$a_n \sim \frac{9}{25}n^2 \rightarrow \infty. \text{ Diverges by the } n\text{th-term test.}$$

Theorem 4.10: Linearity of convergent series

If $\sum a_n$ and $\sum b_n$ are convergent and $c \in \mathbb{R}$, then:

- (i) $\sum(ca_n)$ converges and $\sum(ca_n) = c \sum a_n$.
- (ii) $\sum(a_n + b_n)$ converges and $\sum(a_n + b_n) = \sum a_n + \sum b_n$.
- (iii) $\sum(a_n - b_n)$ converges and $\sum(a_n - b_n) = \sum a_n - \sum b_n$.

Why linearity holds. These follow directly from the limit laws for sequences applied to partial sums: $\sum_{n=1}^k (a_n + b_n) = \sum_{n=1}^k a_n + \sum_{n=1}^k b_n$, and the limit of a sum is the sum of the limits (provided both converge).

Common Mistake

The linearity rules require that the involved series are already known to be convergent. It is *not* valid to use these identities to “prove” convergence without a separate convergence argument.

Worked Example (Lecture Example 4.26): combining series

Evaluate $\sum_{n=1}^{\infty} \left(\frac{3}{n(n+1)} + \frac{1}{2^n} \right).$

Solution.

Both component series converge:

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = 1 \text{ (telescoping) and } \sum_{n=1}^{\infty} \frac{1}{2^n} = \frac{1/2}{1-1/2} = 1 \text{ (geometric with } r = 1/2).$$

By linearity:

$$\sum_{n=1}^{\infty} \left(\frac{3}{n(n+1)} + \frac{1}{2^n} \right) = 3 \cdot 1 + 1 = \boxed{4}.$$

Exam Focus: Finite number of terms does not affect convergence

Adding or removing finitely many terms from a series does not change whether it converges or diverges (though it may change the sum).

For example, if $\sum_{n=4}^{\infty} \frac{n}{n^3+1}$ converges, then $\sum_{n=1}^{\infty} \frac{n}{n^3+1}$ also converges (it just adds three extra finite terms).

This is because the partial sums of the two series differ by a fixed constant (the sum of the finitely many added/removed terms), and shifting a sequence by a constant does not affect convergence.

Extension (Beyond Syllabus: p -series preview and the integral test connection)

Syllabus link: Chapter 4 — §4.4 Series.

The harmonic series $\sum \frac{1}{n}$ diverges, but what about $\sum \frac{1}{n^2}$ or $\sum \frac{1}{n^p}$ in general?

Theorem (p -series test). The series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges if and only if $p > 1$.

Intuition. For $p > 1$, the terms $1/n^p$ shrink fast enough that the partial sums stay bounded. For $p \leq 1$, the terms shrink too slowly (at most as fast as $1/n$), and the grouping argument from the harmonic series can be adapted.

This result is typically proved using the integral test (which may appear later in MH1101 or MH2100).

The key idea: $\sum \frac{1}{n^p}$ converges iff $\int_1^{\infty} \frac{1}{x^p} dx$ converges, and this integral equals $\frac{1}{p-1}$ when $p > 1$ and diverges when $p \leq 1$.

5 Convergence Tests

Overview and purpose. In Chapter 4 we defined what it means for a series $\sum a_n$ to converge: the partial sums $s_k = \sum_{n=1}^k a_n$ must tend to a finite limit. But computing partial sums explicitly is often impractical (e.g. try writing down s_k for $\sum 1/n^2$). Convergence tests bypass this by translating the question into a simpler criterion—typically a comparison, an integral, or a ratio—that yields a yes/no answer *without* finding the actual sum.

Why so many tests? No single test works for all series. Each test has its natural domain of applicability:

- **Integral Test:** series whose terms come from a “nice” (continuous, positive, decreasing) function $f(x)$.
- **Comparison / Limit Comparison:** series that are “close to” a known benchmark (p -series, geometric series).
- **Alternating Series Test:** series with terms that strictly alternate in sign.
- **Ratio Test:** series involving factorials, exponential terms, or products.
- **Root Test:** series where the n -th term is an n -th power.

Exam Focus: Which test to try first (decision flowchart)

Given a series $\sum a_n$:

1. **Step 0 (Term test):** Does $a_n \rightarrow 0$? If not, the series diverges immediately.
2. **Step 1 (Recognise known forms):** Is it geometric? Telescoping? A p -series? If so, use the known result directly.
3. **Step 2 (Factorials or n -th powers?):** If the term involves $n!$, $(n!)^2$, a^n , or $(\frac{\cdot}{\cdot})^n$, try the **Ratio** or **Root** Test.
4. **Step 3 (Rational/algebraic in n):** If the term is a ratio of polynomials or roots, try **Limit Comparison** with a p -series.
5. **Step 4 (Alternating sign?):** If $a_n = (-1)^n b_n$ with $b_n > 0$ decreasing to 0, use the **Alternating Series Test**.
6. **Step 5 (Integrable terms?):** If $a_n = f(n)$ for a continuous, positive, decreasing f , use the **Integral Test**.

5.1 The Integral Test

Why connect series to integrals? The terms $a_n = f(n)$ are the heights of rectangles of width 1. The area under $f(x)$ from 1 to ∞ and the total area of these rectangles are closely related: one bounds the other. If the integral converges, the rectangles’ total area is finite, so the series converges; if the integral diverges, so does the series. This is the geometric insight behind the Integral Test.

Theorem 5.1: The Integral Test

Suppose $f : [1, \infty) \rightarrow \mathbb{R}$ is continuous, positive, and decreasing, and let $a_n = f(n)$. Then:

- (i) If $\int_1^\infty f(x) dx$ converges, then $\sum_{n=1}^\infty a_n$ converges.
- (ii) If $\int_1^\infty f(x) dx$ diverges, then $\sum_{n=1}^\infty a_n$ diverges.

Why does this work? (Proof idea via bounding partial sums.) Since f is decreasing, for each integer $n \geq 1$:

$$f(n+1) \leq \int_n^{n+1} f(x) dx \leq f(n).$$

Summing from $n = 1$ to $k - 1$:

$$\sum_{n=2}^k f(n) \leq \int_1^k f(x) dx \leq \sum_{n=1}^{k-1} f(n).$$

- **(i) Integral converges $\Rightarrow$ series converges:** If $\int_1^\infty f(x) dx = I < \infty$, then $s_k = a_1 + \sum_{n=2}^k a_n \leq a_1 + I$, so $\{s_k\}$ is bounded. Since $a_n > 0$, $\{s_k\}$ is also increasing, so by the Monotone Sequence Theorem, $\{s_k\}$ converges.
- **(ii) Integral diverges $\Rightarrow$ series diverges:** If $\int_1^\infty f(x) dx = \infty$, then $s_{k-1} \geq \int_1^k f(x) dx \rightarrow \infty$.

Common Mistake: The Integral Test does NOT give the sum

The Integral Test tells you *whether* a series converges, not *what* it converges to. The value $\int_1^\infty f(x) dx$ is generally *not* equal to $\sum_{n=1}^\infty a_n$.

Exam Focus: Integral Test checklist

Before applying the Integral Test, you must verify **three** conditions on f :

1. **Continuous** on $[N, \infty)$ for some N .
2. **Positive:** $f(x) > 0$ on $[N, \infty)$.
3. **Decreasing:** $f'(x) < 0$ on $[N, \infty)$ (compute the derivative explicitly).

It is *not* necessary that f is decreasing from $x = 1$; it only needs to be *ultimately decreasing* (i.e. for all $x \geq N$ for some N). Finitely many initial terms do not affect convergence.

Worked Example (Lecture Example 5.1): $\sum \frac{1}{1+n^2}$ converges

Test $\sum_{n=1}^{\infty} \frac{1}{1+n^2}$ for convergence.

Solution. Let $f(x) = \frac{1}{1+x^2}$. This is continuous, positive, and decreasing on $[1, \infty)$ (since $f'(x) = \frac{-2x}{(1+x^2)^2} < 0$ for $x > 0$).

$$\int_1^{\infty} \frac{1}{1+x^2} dx = \lim_{t \rightarrow \infty} [\tan^{-1} x]_1^t = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4} < \infty.$$

By the Integral Test, the series $\sum_{n=1}^{\infty} \frac{1}{1+n^2}$ converges.

Sanity check: The integral evaluates to $\pi/4 \approx 0.785$, but the actual sum $\sum_{n=1}^{\infty} \frac{1}{1+n^2} \approx 1.077$. These are *not* equal—the integral is merely a tool for determining convergence.

Worked Example (Lecture Example 5.2): $\sum \frac{\ln n}{n}$ diverges

Determine whether $\sum_{n=1}^{\infty} \frac{\ln n}{n}$ converges or diverges.

Solution. Let $f(x) = \frac{\ln x}{x}$. This is continuous and positive for $x \geq 1$. Compute the derivative:

$$f'(x) = \frac{1 - \ln x}{x^2}.$$

So $f'(x) < 0$ when $\ln x > 1$, i.e. $x > e \approx 2.718$. Hence f is decreasing on $[3, \infty)$. Now evaluate the integral from 3 to ∞ :

$$\int_3^{\infty} \frac{\ln x}{x} dx = \lim_{t \rightarrow \infty} \left[\frac{(\ln x)^2}{2} \right]_3^t = \lim_{t \rightarrow \infty} \frac{(\ln t)^2}{2} - \frac{(\ln 3)^2}{2} = \infty.$$

By the Integral Test, $\sum_{n=3}^{\infty} \frac{\ln n}{n}$ diverges. Since adding finitely many terms does not affect convergence,

$\sum_{n=1}^{\infty} \frac{\ln n}{n}$ diverges.

Worked Example (Tutorial 9, Q1(c)): $\sum ne^{-n^2}$ converges

Determine whether $\sum_{n=1}^{\infty} ne^{-n^2}$ converges or diverges.

Solution. Let $f(x) = xe^{-x^2}$. This is continuous and positive for $x \geq 1$.

Monotonicity: $f'(x) = e^{-x^2} + x(-2x)e^{-x^2} = (1 - 2x^2)e^{-x^2} < 0$ for $x \geq 1$ (since $2x^2 \geq 2 > 1$).

Integral: Let $u = x^2$, $du = 2x dx$:

$$\int_1^{\infty} xe^{-x^2} dx = \frac{1}{2} \int_1^{\infty} e^{-u} du = \frac{1}{2} [-e^{-u}]_1^{\infty} = \frac{1}{2e} < \infty.$$

By the Integral Test, $\sum_{n=1}^{\infty} ne^{-n^2}$ converges.

The p -series: a fundamental benchmark. The most important application of the Integral Test is classifying p -series. These serve as the “benchmark” series against which almost all other series are compared.

Theorem 5.2: p -series convergence

The p -series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ is:

- **convergent** if $p > 1$,
- **divergent** if $p \leq 1$.

Why this threshold at $p = 1$? (Full proof.)

- **Case $p \leq 0$:** Then $1/n^p = n^{|p|} \geq 1$ for all $n \geq 1$, so $\lim_{n \rightarrow \infty} a_n \neq 0$. Diverges by the n th-term test.
- **Case $p = 1$:** This is the harmonic series $\sum 1/n$, which diverges (proved in Chapter 4).
- **Case $p > 0$, $p \neq 1$:** Let $f(x) = x^{-p}$. Then $f'(x) = -px^{-p-1} < 0$ for $x \geq 1$, so f is decreasing. Compute:

$$\int_1^{\infty} x^{-p} dx = \lim_{t \rightarrow \infty} \left. \frac{x^{-p+1}}{-p+1} \right|_1^t = \lim_{t \rightarrow \infty} \frac{t^{1-p} - 1}{1-p}.$$

If $p > 1$: the exponent $1 - p < 0$, so $t^{1-p} \rightarrow 0$. The integral equals $\frac{1}{p-1} < \infty$. Series converges.

If $0 < p < 1$: the exponent $1 - p > 0$, so $t^{1-p} \rightarrow \infty$. The integral diverges. Series diverges.

p -series reference table (memorise for exams)

Series	p -value	Verdict
$\sum 1/n^3$	$p = 3 > 1$	converges
$\sum 1/n^2$	$p = 2 > 1$	converges
$\sum 1/n^{3/2}$	$p = 3/2 > 1$	converges
$\sum 1/n$	$p = 1$	diverges
$\sum 1/\sqrt{n}$	$p = 1/2 < 1$	diverges
$\sum 1/n^{1/3}$	$p = 1/3 < 1$	diverges

Exam Focus

The p -series is your most important “test series” for the Comparison and Limit Comparison Tests. When you see a rational expression in n , identify its dominant behaviour and compare with the appropriate p -series.

For example, $\frac{n^2 + 3n}{n^5 + 7} \sim \frac{n^2}{n^5} = \frac{1}{n^3}$, so compare with $\sum 1/n^3$ (which converges).

5.2 The Comparison Tests

Why comparison? We often encounter series $\sum a_n$ that don't have a nice closed-form partial sum or a clean integral. The idea: if each term a_n is “smaller” than a term b_n of a known convergent series, then $\sum a_n$ converges too. Similarly, if a_n is “bigger” than terms of a known divergent series, then $\sum a_n$ diverges.

Theorem 5.3: The (Direct) Comparison Test

Suppose $\sum a_n$ and $\sum b_n$ are series with **positive** terms.

- (i) If $\sum b_n$ converges and $a_n \leq b_n$ for all n , then $\sum a_n$ converges.
- (ii) If $\sum b_n$ diverges (to ∞) and $a_n \geq b_n$ for all n , then $\sum a_n$ diverges (to ∞).

Why this works.

- (i) Both partial-sum sequences $s_k = \sum_{n=1}^k a_n$ and $t_k = \sum_{n=1}^k b_n$ are increasing (positive terms). Since $a_n \leq b_n$, we have $s_k \leq t_k$. Since $\sum b_n$ converges, $t_k \rightarrow t$ for some t , so $t_k \leq t$. Thus $s_k \leq t$ for all k , and $\{s_k\}$ is increasing and bounded $\Rightarrow$ convergent by the Monotone Sequence Theorem.
- (ii) Since $a_n \geq b_n$, $s_k \geq t_k$. Since $t_k \rightarrow \infty$, $s_k \rightarrow \infty$ as well.

Common Mistake: Wrong direction of comparison

To prove convergence, you need $a_n \leq b_n$ where $\sum b_n$ **converges** (bound above by something convergent). To prove divergence, you need $a_n \geq b_n$ where $\sum b_n$ **diverges** (bound below by something divergent). The reverse inequalities give no information:

- $a_n \geq b_n$ and $\sum b_n$ converges $\Rightarrow$ nothing (the larger series could converge or diverge).
- $a_n \leq b_n$ and $\sum b_n$ diverges $\Rightarrow$ nothing.

Worked Example (Lecture Example 5.4): direct comparison

Determine whether $\sum_{n=1}^{\infty} \frac{100}{2n^2 + 5n + 4}$ converges.

Solution. For $n \geq 1$: $2n^2 + 5n + 4 > 2n^2$, so

$$0 < \frac{100}{2n^2 + 5n + 4} < \frac{100}{2n^2} = \frac{50}{n^2}.$$

Since $\sum \frac{50}{n^2} = 50 \sum \frac{1}{n^2}$ converges (p -series with $p = 2 > 1$), the Comparison Test gives

$$\sum_{n=1}^{\infty} \frac{100}{2n^2 + 5n + 4} \text{ converges.}$$

Worked Example (Tutorial 9, Q2(e)): $\sum \sin(1/n)$ diverges

Show that $\sum_{n=1}^{\infty} \sin\left(\frac{1}{n}\right)$ diverges.

Solution. Use the standard limit $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$. Setting $x = 1/n$:

$$\lim_{n \rightarrow \infty} \frac{\sin(1/n)}{1/n} = 1 > 0.$$

Since $\sum 1/n$ diverges (harmonic series), the Limit Comparison Test implies $\sum \sin(1/n)$ diverges.

Alternative (direct comparison): For $x \in [0, \pi/2]$, $\sin x \geq \frac{2}{\pi}x$. Thus $\sin(1/n) \geq \frac{2}{\pi n}$, and divergence follows by comparison with $\sum \frac{1}{n}$.

The Limit Comparison Test: when direct comparison is fiddly. Direct comparison requires an *explicit* inequality $a_n \leq b_n$ (or $\geq$), which can be tedious to establish. The Limit Comparison Test replaces this with an *asymptotic* comparison: if a_n and b_n have the same “growth rate” (i.e. their ratio tends to a positive constant), they have the same convergence behaviour.

Theorem 5.4: The Limit Comparison Test (LCT)

Suppose $\sum a_n$ and $\sum b_n$ are series with **positive** terms.

- (i) If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = c$ with $0 < c < \infty$, then both series converge or both diverge.
- (ii) If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 0$ and $\sum b_n$ converges, then $\sum a_n$ converges.
- (iii) If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \infty$ and $\sum b_n$ diverges, then $\sum a_n$ diverges.

Why (i) works (proof sketch). If $a_n/b_n \rightarrow c > 0$, then for large n :

$$\frac{c}{2} b_n \leq a_n \leq \frac{3c}{2} b_n.$$

The right inequality gives: if $\sum b_n$ converges, then $\sum a_n$ converges (by comparison with $\frac{3c}{2} \sum b_n$). The left inequality gives: if $\sum b_n$ diverges, then $\sum a_n$ diverges (by comparison with $\frac{c}{2} \sum b_n$).

Exam Focus: Choosing the comparison series b_n for LCT

When a_n is a rational expression in n (possibly with roots), identify the **dominant terms** in the numerator and denominator:

$$a_n = \frac{\text{numerator}}{\text{denominator}} \sim \frac{\text{dominant numerator term}}{\text{dominant denominator term}} = b_n.$$

Then $\lim a_n/b_n = 1$ (or some nonzero constant), so $\sum a_n$ and $\sum b_n$ have the same behaviour.

Example: $a_n = \frac{2n^2+3n}{\sqrt{5+n^5}}$. Dominant: $\frac{2n^2}{n^{5/2}} = \frac{2}{n^{1/2}}$. Compare with $b_n = \frac{1}{\sqrt{n}}$ ($p = 1/2 < 1$, diverges).

Worked Example (Lecture Example 5.6): LCT with geometric comparison

Test $\sum_{n=1}^{\infty} \frac{1}{2^n - 1}$ for convergence.

Solution. Compare with $b_n = \frac{1}{2^n}$ (convergent geometric series). Compute:

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{2^n}{2^n - 1} = \lim_{n \rightarrow \infty} \frac{1}{1 - 1/2^n} = 1 > 0.$$

By the LCT, $\sum_{n=1}^{\infty} \frac{1}{2^n - 1}$ converges.

Worked Example (Lecture Example 5.7): LCT with p -series

Does $\sum_{n=1}^{\infty} \frac{2n^2 + 3n}{\sqrt{5 + n^5}}$ converge?

Solution. The dominant behaviour is $\frac{2n^2}{n^{5/2}} = \frac{2}{n^{1/2}}$. Set $b_n = \frac{2}{n^{1/2}}$.

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{2n^2 + 3n}{\sqrt{5 + n^5}} \cdot \frac{n^{1/2}}{2} = \lim_{n \rightarrow \infty} \frac{2n^{5/2} + 3n^{3/2}}{2\sqrt{5 + n^5}} = \lim_{n \rightarrow \infty} \frac{2 + 3/n}{2\sqrt{5/n^5 + 1}} = \frac{2}{2} = 1.$$

Since $\sum b_n = 2 \sum \frac{1}{n^{1/2}}$ diverges ($p = 1/2 < 1$), the LCT gives $\sum \frac{2n^2 + 3n}{\sqrt{5 + n^5}}$ diverges.

Worked Example (Tutorial 9, Q2(d)): $\sum \frac{n^2+n+1}{n^4+n^2}$ converges

Determine convergence of $\sum_{n=1}^{\infty} \frac{n^2+n+1}{n^4+n^2}$.

Solution. Dominant behaviour: $\frac{n^2}{n^4} = \frac{1}{n^2}$. Set $b_n = \frac{1}{n^2}$.

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{n^2(n^2+n+1)}{n^4+n^2} = \lim_{n \rightarrow \infty} \frac{n^4+n^3+n^2}{n^4+n^2} = 1 > 0.$$

Since $\sum 1/n^2$ converges ($p = 2 > 1$), the LCT gives $\sum \frac{n^2+n+1}{n^4+n^2}$ converges.

Worked Example (Tutorial 9, Q2(f)): $\sum \frac{1}{n \ln n}$ diverges

Determine convergence of $\sum_{n=2}^{\infty} \frac{1}{n \ln n}$.

Solution. Let $f(x) = \frac{1}{x \ln x}$ for $x \geq 2$. This is continuous and positive. Since $f'(x) = -\frac{1+\ln x}{(x \ln x)^2} < 0$ for $x \geq 2$, f is decreasing.

Integral Test: use substitution $u = \ln x$, $du = dx/x$:

$$\int_2^{\infty} \frac{1}{x \ln x} dx = \int_{\ln 2}^{\infty} \frac{1}{u} du = [\ln u]_{\ln 2}^{\infty} = \infty.$$

By the Integral Test, $\sum_{n=2}^{\infty} \frac{1}{n \ln n}$ diverges.

Exam note: This is a “borderline” series—it diverges, but very slowly (slower than $\sum 1/n$).

5.3 Absolute and Conditional Convergence

Why do we need a new concept? The Integral Test and Comparison Test only apply to series with **positive terms**. But many natural series have mixed signs, such as $\sum (-1)^n/n$. To handle these, we introduce *absolute convergence*: strip the signs (take $|a_n|$) and test the resulting positive series. If $\sum |a_n|$ converges, then $\sum a_n$ converges too.

Definition 5.1: Absolute convergence

A series $\sum a_n$ is called *absolutely convergent* if the series of absolute values $\sum |a_n|$ converges.

Why this definition is natural. Absolute convergence means the series converges “no matter what the signs are.” It is the strongest form of convergence. Absolutely convergent series can be rearranged in any order without changing the sum—conditionally convergent series cannot (this is the Riemann rearrangement theorem).

Theorem 5.5: Absolute convergence implies convergence

If $\sum a_n$ is absolutely convergent, then $\sum a_n$ converges.

Why this is true. For each n , $0 \leq a_n + |a_n| \leq 2|a_n|$. If $\sum |a_n|$ converges, then $\sum 2|a_n|$ converges, so by comparison $\sum (a_n + |a_n|)$ converges. But then

$$\sum a_n = \sum (a_n + |a_n|) - \sum |a_n|$$

is the difference of two convergent series, hence convergent.

Worked Example (Lecture Example 5.9): $\sum \frac{\cos n}{n^2}$ converges absolutely

Show that $\sum_{n=1}^{\infty} \frac{\cos n}{n^2}$ converges.

Solution. The signs change irregularly (not alternating), so we test absolute convergence.

$$\left| \frac{\cos n}{n^2} \right| = \frac{|\cos n|}{n^2} \leq \frac{1}{n^2}.$$

Since $\sum 1/n^2$ converges (p -series, $p = 2 > 1$), by comparison $\sum |\cos n|/n^2$ converges. Hence $\sum \cos n/n^2$ is absolutely convergent, and therefore convergent by Theorem ??.

Definition 5.2: Conditional convergence

A series $\sum a_n$ is called *conditionally convergent* if $\sum a_n$ converges but $\sum |a_n|$ diverges.

Why this distinction matters. Conditional convergence is “fragile”: the positive and negative terms each diverge to $+\infty$ and $-\infty$ respectively, but they nearly cancel to give a finite sum. Rearranging terms can change the sum (or make it diverge).

The Alternating Series Test. If a series is not absolutely convergent, how do we check whether it is conditionally convergent? For *alternating* series (signs strictly alternate: $+, -, +, -, \dots$), the following test applies.

Theorem 5.6: Alternating Series Test (AST)

Let $\{a_n\}$ be a decreasing positive sequence converging to 0:

$$a_1 \geq a_2 \geq a_3 \geq \dots \geq 0, \quad \text{and} \quad \lim_{n \rightarrow \infty} a_n = 0.$$

Then the alternating series $\sum_{n=1}^{\infty} (-1)^{n-1} a_n = a_1 - a_2 + a_3 - a_4 + \dots$ converges.

Why the AST works (geometric intuition). Consider the partial sums:

- $s_1 = a_1$ (overshoot),

- $s_2 = a_1 - a_2$ (undershoot below s_1 , but still positive since $a_1 \geq a_2$),
- $s_3 = s_2 + a_3$ (overshoot again, but less than s_1),
- $s_4 = s_3 - a_4$ (undershoot again, but more than s_2).

The even partial sums $s_2, s_4, s_6, \dots$ form an increasing sequence; the odd partial sums $s_1, s_3, s_5, \dots$ form a decreasing sequence. Both are bounded, so both converge. Since $s_{2k+1} - s_{2k} = a_{2k+1} \rightarrow 0$, they converge to the same limit.

Common Mistake: AST requires TWO conditions

Both conditions are needed:

1. $\{a_n\}$ is **decreasing** (you must show $a_{n+1} \leq a_n$ or show the corresponding function f has $f'(x) < 0$).
2. $a_n \rightarrow 0$.

If $a_n \rightarrow 0$ but $\{a_n\}$ is not eventually decreasing, the AST does not apply. (However, the converse rarely comes up in MH1101.)

Also, the AST says the series *converges*; it does *not* say whether the convergence is absolute or conditional. To classify fully, you must also test $\sum |a_n|$ separately.

Worked Example (Lecture Example 5.11): $\sum (-1)^{n-1}/\sqrt{n}$ is conditionally convergent

Show that $S = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{\sqrt{n}}$ is conditionally convergent.

Solution.

Step 1 (convergence): Let $a_n = 1/\sqrt{n}$. Then $\{a_n\}$ is decreasing and $a_n \rightarrow 0$. By the Alternating Series Test, S converges.

Step 2 (not absolute): $\sum |(-1)^{n-1}/\sqrt{n}| = \sum 1/\sqrt{n}$ diverges (p -series with $p = 1/2 < 1$).

Hence S is conditionally convergent.

Worked Example (Lecture Example 5.12): $\sum (-1)^{n+1} \frac{n^2}{n^3+1}$ converges by AST

Is $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{n^2}{n^3+1}$ convergent?

Solution. Let $a_n = \frac{n^2}{n^3+1}$.

Decreasing: Consider $f(x) = \frac{x^2}{x^3+1}$. Compute:

$$f'(x) = \frac{2x(x^3+1) - x^2(3x^2)}{(x^3+1)^2} = \frac{x(2-x^3)}{(x^3+1)^2}.$$

For $x > 2^{1/3} \approx 1.26$, we have $f'(x) < 0$, so f is decreasing on $(2^{1/3}, \infty)$. Thus $a_{n+1} < a_n$ for all $n \geq 2$. We also check $a_2 = 4/9 < 1/2 = a_1$ by direct computation, so $\{a_n\}$ is decreasing.

Limit: $\lim_{n \rightarrow \infty} \frac{n^2}{n^3+1} = \lim_{n \rightarrow \infty} \frac{1/n}{1+1/n^3} = 0$.

By the AST, the series converges.

Worked Example (Tutorial 10, Q1(a)): conditional convergence via AST + LCT

Classify $\sum_{n=0}^{\infty} \frac{(-1)^n}{5n+1}$ as absolutely convergent, conditionally convergent, or divergent.

Solution.

Test for absolute convergence: $\sum \left| \frac{(-1)^n}{5n+1} \right| = \sum \frac{1}{5n+1}$. By LCT with $b_n = 1/n$: $\lim \frac{1/(5n+1)}{1/n} = \lim \frac{n}{5n+1} = 1/5 > 0$. Since $\sum 1/n$ diverges, $\sum 1/(5n+1)$ diverges. So the series is *not* absolutely convergent.

Test for conditional convergence: Let $a_n = \frac{1}{5n+1}$. Then $a_{n+1} = \frac{1}{5n+6} < \frac{1}{5n+1} = a_n$ (since $5n+6 > 5n+1$), and $a_n \rightarrow 0$. By the AST, $\sum (-1)^n/(5n+1)$ converges.

Hence the series is conditionally convergent.

Exam Focus: Classification workflow (absolute / conditional / divergent)

Given a series $\sum a_n$ with mixed signs:

1. **First**, test $\sum |a_n|$ for convergence (using any test for positive series).
2. If $\sum |a_n|$ converges $\Rightarrow$ the series is **absolutely convergent** (done).
3. If $\sum |a_n|$ diverges $\Rightarrow$ test the original $\sum a_n$ for convergence (typically AST).
 - If $\sum a_n$ converges $\Rightarrow$ **conditionally convergent**.
 - If $\sum a_n$ diverges $\Rightarrow$ **divergent**.

5.4 The Ratio and Root Tests

Why these tests? The tests developed so far (Integral, Comparison) work well for algebraic and rational expressions in n , but struggle with:

- **Factorials:** $n!$, $(n!)^2$, $(2n)!$ — you cannot integrate these.
- **Exponential terms:** a^n — comparison with p -series is awkward.
- **n -th powers:** $\left(\frac{n}{2n+3}\right)^n$ — the exponent depends on n .

The Ratio Test and Root Test handle these naturally by comparing with a geometric series.

Theorem 5.7: The Ratio Test

Let $\{a_n\}$ be a sequence with $a_n \neq 0$ for all n , and assume

$$\rho = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$$

exists (or equals ∞).

- (i) If $\rho < 1$, then $\sum a_n$ converges absolutely.
- (ii) If $\rho > 1$ (or $\rho = \infty$), then $\sum a_n$ diverges.
- (iii) If $\rho = 1$, the Ratio Test is **inconclusive**.

Why the Ratio Test works (proof sketch for $\rho < 1$). Choose r with $\rho < r < 1$. Since $|a_{n+1}/a_n| \rightarrow \rho$, there exists N such that $|a_{n+1}/a_n| < r$ for all $n \geq N$. Then $|a_{N+k}| < r^k |a_N|$ for all $k \geq 0$. So:

$$\sum_{n=N}^{\infty} |a_n| = \sum_{k=0}^{\infty} |a_{N+k}| \leq |a_N| \sum_{k=0}^{\infty} r^k = \frac{|a_N|}{1-r} < \infty.$$

This is a comparison with a convergent geometric series.

For $\rho > 1$: By the same argument with reversed inequality, $|a_{N+k}| \geq r^k |a_N|$ with $r > 1$, so $|a_n| \rightarrow \infty$ and $a_n \not\rightarrow 0$. Divergence follows from the n th-term test.

Common Mistake: $\rho = 1$ does NOT mean divergence

When $\rho = 1$, both convergence and divergence are possible:

- $a_n = 1/n^2$: $\rho = \lim_{n \rightarrow \infty} \frac{n^2}{(n+1)^2} = 1$, but $\sum 1/n^2$ converges.
- $a_n = 1/n$: $\rho = \lim_{n \rightarrow \infty} \frac{n}{n+1} = 1$, but $\sum 1/n$ diverges.

When you get $\rho = 1$, you *must* use a different test (Comparison, Integral, etc.).

Exam Focus: When to use the Ratio Test

The Ratio Test is most effective when the general term involves:

- **Factorials:** $n!$, $(2n)!$ — the ratio a_{n+1}/a_n simplifies via $(n+1)!/n! = n+1$.
- **Exponential constants:** c^n — the ratio gives $c^{n+1}/c^n = c$.
- **Mixed factorial–exponential:** $\frac{n!}{2^n}$, $\frac{3^n}{n!}$ — these are the Ratio Test’s “sweet spot.”

The Ratio Test is **not** useful for rational expressions in n (you always get $\rho = 1$).

Worked Example (Lecture Example 5.15): $\sum 1/n!$ converges

Show that $\sum_{n=1}^{\infty} \frac{1}{n!}$ converges.

Solution. Let $a_n = \frac{1}{n!}$. Then:

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{1/(n+1)!}{1/n!} = \frac{n!}{(n+1)!} = \frac{1}{n+1}.$$

So $\rho = \lim_{n \rightarrow \infty} \frac{1}{n+1} = 0 < 1$.

By the Ratio Test, $\sum_{n=1}^{\infty} \frac{1}{n!}$ converges.

Worked Example (Lecture Example 5.16): $\sum n^2/2^n$ converges

Does $\sum_{n=1}^{\infty} \frac{n^2}{2^n}$ converge?

Solution. Let $a_n = \frac{n^2}{2^n}$. Then:

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{(n+1)^2}{2^{n+1}} \cdot \frac{2^n}{n^2} = \frac{1}{2} \cdot \frac{(n+1)^2}{n^2} = \frac{1}{2} \left(1 + \frac{1}{n} \right)^2.$$

So $\rho = \frac{1}{2} \cdot 1 = \frac{1}{2} < 1$.

By the Ratio Test, $\sum \frac{n^2}{2^n}$ converges.

Worked Example (Lecture Example 5.17): $\sum (-1)^n n!/100^n$ diverges

Determine whether $\sum_{n=0}^{\infty} \frac{(-1)^n n!}{100^n}$ converges.

Solution. Let $a_n = \frac{(-1)^n n!}{100^n}$. Then:

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{(n+1)!}{100^{n+1}} \cdot \frac{100^n}{n!} = \frac{n+1}{100}.$$

So $\rho = \lim_{n \rightarrow \infty} \frac{n+1}{100} = \infty > 1$.

By the Ratio Test, the series diverges.

Intuition: Factorials grow faster than any exponential. Hence $n!/100^n \rightarrow \infty$, and the terms do not approach 0.

Worked Example (Tutorial 10, Q2): $\sum (n!)^2 / (kn)!$ and parameter k

For which positive integers k does $\sum_{n=1}^{\infty} \frac{(n!)^2}{(kn)!}$ converge?

Solution. Let $a_n = \frac{(n!)^2}{(kn)!}$. Compute:

$$\frac{a_{n+1}}{a_n} = \frac{((n+1)!)^2}{(k(n+1))!} \cdot \frac{(kn)!}{(n!)^2} = (n+1)^2 \cdot \frac{(kn)!}{(kn+k)!} = \frac{(n+1)^2}{\prod_{j=1}^k (kn+j)}.$$

$k = 1$: $\frac{a_{n+1}}{a_n} = \frac{(n+1)^2}{n+1} = n+1 \rightarrow \infty$. Diverges.

$k = 2$: $\frac{a_{n+1}}{a_n} = \frac{(n+1)^2}{(2n+1)(2n+2)} = \frac{n+1}{2(2n+1)} \rightarrow \frac{1}{4} < 1$. Converges.

$k \geq 3$: The denominator $\prod_{j=1}^k (kn+j) \sim (kn)^k = k^k n^k$, so $\frac{a_{n+1}}{a_n} \sim \frac{n^2}{k^k n^k} = \frac{1}{k^k} n^{2-k} \rightarrow 0 < 1$. Converges.

The series converges for all $k \geq 2$, and diverges for $k = 1$.

Theorem 5.8: The Root Test

Let $\{a_n\}$ be a sequence and assume

$$L = \lim_{n \rightarrow \infty} \sqrt[n]{|a_n|}$$

exists (or equals ∞).

- (i) If $L < 1$, then $\sum a_n$ converges absolutely.
- (ii) If $L > 1$ (or $L = \infty$), then $\sum a_n$ diverges.
- (iii) If $L = 1$, the Root Test is **inconclusive**.

The Root Test.

Why the Root Test works (same idea as Ratio). If $L < 1$, pick r with $L < r < 1$. For large n , $\sqrt[n]{|a_n|} < r$, i.e. $|a_n| < r^n$. Comparing with the convergent geometric series $\sum r^n$ gives convergence.

If $L > 1$, then for large n , $|a_n| > r^n$ with $r > 1$, so $|a_n| \not\rightarrow 0$.

Exam Focus: When to use the Root Test

Use the Root Test when the general term is an n -th power:

$$a_n = (\text{expression in } n)^n.$$

Then $\sqrt[n]{|a_n|} = |\text{expression in } n|$, and you just compute its limit.

The Root Test is also useful for piecewise-defined terms (e.g. odd/even) where the Ratio Test may not have a limit.

Worked Example (Lecture Example 5.18): Root Test on n -th power

Does $\sum_{n=1}^{\infty} \left(\frac{n}{2n+3} \right)^n$ converge?

Solution. $\sqrt[n]{|a_n|} = \frac{n}{2n+3} \rightarrow \frac{1}{2} < 1$.

By the Root Test, the series converges.

Worked Example (Tutorial 10, Q1(c)): Root Test for $\left(\frac{1-n}{2+3n} \right)^n$

Classify $\sum_{n=1}^{\infty} \left(\frac{1-n}{2+3n} \right)^n$.

Solution.

$$\sqrt[n]{|a_n|} = \left| \frac{1-n}{2+3n} \right| = \frac{n-1}{3n+2} \rightarrow \frac{1}{3} < 1.$$

By the Root Test, the series converges absolutely.

Absolutely convergent.

Worked Example (Tutorial 10, Q3): $\lim_{n \rightarrow \infty} x^n/n! = 0$ via Ratio Test

Claim: $\lim_{n \rightarrow \infty} \frac{x^n}{n!} = 0$ for all $x \in \mathbb{R}$.

Proof. Let $u_n = |x|^n/n!$. Then:

$$\frac{u_{n+1}}{u_n} = \frac{|x|}{n+1} \rightarrow 0 < 1.$$

By the Ratio Test (applied to the sequence: if ratios tend to $L < 1$, terms tend to 0), $u_n \rightarrow 0$.

Hence $|x^n/n!| = u_n \rightarrow 0$, so $x^n/n! \rightarrow 0$ for all $x \in \mathbb{R}$.

Interpretation: Factorials grow faster than *any* exponential. This is why the Taylor series $e^x = \sum x^n/n!$ converges for all x .

Extension (Tutorial 10, Q5): Bounding the remainder of a convergent series

Syllabus link: Chapter 5 — §5.4 Ratio Test.

Theorem (Tail bound under ratio assumptions). Let $\sum a_n$ be a series with positive terms, with ratio $r_n = a_{n+1}/a_n$. Suppose $\lim r_n = L < 1$, $\{r_n\}$ is decreasing, and $r_{n+1} < 1$. Then the tail

$$R_n = a_{n+1} + a_{n+2} + \cdots$$

satisfies

$$R_n \leq \frac{a_{n+1}}{1 - r_{n+1}}.$$

Proof. Since $\{r_m\}$ is decreasing, for all $m \geq n+1$: $r_m \leq r_{n+1} < 1$. Then for $k \geq 1$:

$$a_{n+1+k} = a_{n+1} \prod_{j=n+1}^{n+k} r_j \leq a_{n+1} \cdot r_{n+1}^k.$$

Summing: $R_n = a_{n+1} + \sum_{k=1}^{\infty} a_{n+1+k} \leq a_{n+1} \sum_{k=0}^{\infty} r_{n+1}^k = \frac{a_{n+1}}{1 - r_{n+1}}.$

Application: This is useful for estimating how many terms are needed to approximate $\sum a_n$ to a given accuracy.

Master convergence test summary (exam reference)

Test	Best for	Pitfall
n th-term test	Quick divergence check	Cannot prove convergence
Integral Test	$f(n)$ with nice integral	Must verify f is decreasing
Comparison Test	Explicit inequality	Wrong direction gives nothing
Limit Comparison	Asymptotic match	Both series must have positive terms
Alternating Series	$(-1)^n b_n$ form	Must verify b_n decreasing and $\rightarrow 0$
Ratio Test	Factorials, c^n	Inconclusive when $\rho = 1$
Root Test	n -th powers	Inconclusive when $L = 1$

6 Power Series

6.1 Power series and radius of convergence (§6.1)

Definition 6.1: Power series and partial sums

Fix a centre $x_0 \in \mathbb{R}$ and coefficients $(c_n)_{n \geq 0}$. A *power series* (about x_0) is

$$\sum_{n=0}^{\infty} c_n (x - x_0)^n,$$

with *partial sums* $S_N(x) = \sum_{n=0}^N c_n (x - x_0)^n$.

Why “infinite polynomials”? A polynomial $p(x) = a_0 + a_1x + \cdots + a_Nx^N$ is the simplest class of function to evaluate, differentiate, and integrate. A power series generalises this by allowing *infinitely many* terms, giving us the expressive power to represent transcendental functions (e^x , $\sin x$, $\ln(1+x)$, ...) while retaining the easy term-by-term algebra of polynomials. The price we pay is that we must track *where* the infinite sum converges.

A useful mental model. Think of $S_N(x)$ as the “ N th polynomial snapshot” of the full series. As $N \rightarrow \infty$, more and more terms are included; the radius of convergence R tells you the region where these snapshots settle down to a well-defined function. Outside $|x - x_0| > R$ the snapshots blow up.

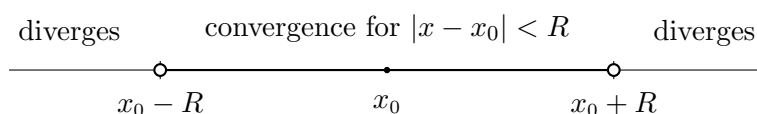
A power series behaves like a polynomial inside its convergence region, but the region itself must be determined. The key object is the *radius of convergence* $R \in [0, \infty]$.

Theorem 6.1: Radius of convergence (interval form)

For the power series $\sum_{n=0}^{\infty} c_n (x - x_0)^n$, there exists $R \in [0, \infty]$ such that:

- it converges absolutely for $|x - x_0| < R$;
- it diverges for $|x - x_0| > R$;
- when $0 < R < \infty$, convergence at the endpoints $x_0 \pm R$ must be checked separately.

Why exactly three cases? At the centre x_0 the series trivially converges (every term with $n \geq 1$ vanishes). Moving away from x_0 , the terms $|c_n||x - x_0|^n$ grow or shrink depending on the interplay between $|c_n|$ and $|x - x_0|^n$. If $|c_n|$ decays fast enough (e.g. $c_n = 1/n!$), the series converges for all x ($R = \infty$). If $|c_n|$ grows too fast (e.g. $c_n = n!$), it diverges for every $x \neq x_0$ ($R = 0$). In between, there is a critical distance R where the behaviour switches from convergence to divergence.



Computing R in practice. Apply the ratio or root test to $\sum |c_n||x - x_0|^n$ and solve for $|x - x_0|$.

- **Ratio test route:** Compute $L = \lim_{n \rightarrow \infty} \left| \frac{c_{n+1}}{c_n} \right|$. Then $R = 1/L$ (with conventions $1/0 = \infty$, $1/\infty = 0$).
- **Root test route:** Compute $A = \lim_{n \rightarrow \infty} \sqrt[n]{|c_n|}$. Then $R = 1/A$.

Both methods give the same R whenever both limits exist.

Worked Example: finding R and the interval (Lecture Ex. 6.3)

Find the radius and interval of convergence of $\sum_{n=1}^{\infty} \frac{(-1)^n}{(2n-1)2^n} (x-1)^n$.

Step 1 (Ratio test). Let $a_n = \frac{(-1)^n}{(2n-1)2^n} (x-1)^n$. Then

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{2n-1}{2n+1} \cdot \frac{|x-1|}{2} \xrightarrow{n \rightarrow \infty} \frac{|x-1|}{2}.$$

Convergence when $\frac{|x-1|}{2} < 1$, i.e. $|x-1| < 2$, so $R = 2$.
Endpoints:

- At $x = 3$: $(x-1)^n = 2^n$, so the series becomes $\sum \frac{(-1)^n}{2n-1}$, which converges by the alternating series test.
- At $x = -1$: $(x-1)^n = (-2)^n$, so the series becomes $\sum \frac{1}{2n-1}$, which diverges.

Therefore the interval of convergence is $(-1, 3]$.

Worked Example: endpoint behaviour (Lecture Ex. 6.4)

Find R and the interval of convergence for $\sum_{n=0}^{\infty} \frac{(-3)^n x^n}{\sqrt{n+1}}$.

Step 1 (Ratio test).

$$\left| \frac{a_{n+1}}{a_n} \right| = 3|x| \sqrt{\frac{n+1}{n+2}} \xrightarrow{n \rightarrow \infty} 3|x|.$$

Converges when $3|x| < 1$, i.e. $|x| < \frac{1}{3}$. So $R = \frac{1}{3}$.

Step 2 (Endpoints).

- $x = -\frac{1}{3}$: series becomes $\sum \frac{1}{\sqrt{n+1}}$, which diverges (p -series, $p = \frac{1}{2} < 1$).
- $x = \frac{1}{3}$: series becomes $\sum \frac{(-1)^n}{\sqrt{n+1}}$, which converges (AST: $\frac{1}{\sqrt{n+1}} \downarrow 0$).

Interval of convergence: $(-\frac{1}{3}, \frac{1}{3}]$.

Common Mistake: forgetting that endpoints can go either way

The ratio/root test is *inconclusive* at $|x-x_0| = R$. You must substitute each endpoint individually and apply a separate convergence test (AST, p -series, comparison, n th-term test, ...). The four possible shapes $()$, $[)$, $(]$, $[]$ are all achievable.

Exam Focus: a fast workflow for R

1. Identify the centre x_0 (from $(x - x_0)^n$).
2. Compute R via ratio/root test *ignoring* endpoints.
3. Test endpoints using quick divergence/alternating/comparison logic.
4. State both **radius** and **interval** clearly.

Exam Shortcut: radius from nearest singularity (Tutorial 12, Q1 Method 2)

For a Taylor series of an analytic function f centred at a , the radius of convergence equals the distance from a to the *nearest singularity* (real or complex) of f .

- $\ln x$ at $a = 2$: singularity at $x = 0$, so $R = |2 - 0| = 2$.
- e^{2x} at $a = 3$: entire function, so $R = \infty$.
- $\sqrt{x}$ at $a = 16$: branch point at $x = 0$, so $R = |16 - 0| = 16$.
- $\frac{1}{7x-13}$ at $a = 2$: pole at $x = \frac{13}{7}$, so $R = |2 - \frac{13}{7}| = \frac{1}{7}$.

This is not a formal exam method (it relies on complex analysis) but is a powerful quick check.

6.2 Representing functions by power series (§6.2)

A major use-case is to rewrite a function into a form where a known power series applies, then transfer the convergence condition.

The big idea. We already know one convergent power series by heart — the geometric series. The strategy of this entire section is: *start from the geometric series, then adapt it using algebra (substitution, multiplication) and calculus (term-by-term differentiation / integration) to represent new functions.* Every example below follows this pattern.

Theorem 6.2: Geometric series

For $|x| < 1$,

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n.$$

More generally, for constants $A \neq 0$ and B ,

$$\frac{1}{A - B(x - x_0)} = \frac{1}{A} \cdot \frac{1}{1 - \frac{B}{A}(x - x_0)} = \frac{1}{A} \sum_{n=0}^{\infty} \left(\frac{B}{A}\right)^n (x - x_0)^n,$$

valid when $|\frac{B}{A}(x - x_0)| < 1$.

Rewriting recipe (step by step). Given a rational function $\frac{C}{\alpha + \beta x}$:

1. Factor out the constant from the denominator so you have $\frac{C}{\alpha} \cdot \frac{1}{1 + \frac{\beta}{\alpha}x}$.
2. Recognise $\frac{1}{1 + \frac{\beta}{\alpha}x} = \frac{1}{1 - (-\frac{\beta}{\alpha}x)}$.

3. Apply the geometric series with $u = -\frac{\beta}{\alpha}x$.
4. The convergence condition is $|u| < 1$, i.e. $|x| < \frac{|\alpha|}{|\beta|}$.

Key Formula: “rewrite to geometric form” pattern

Aim to express $f(x)$ as

$$f(x) = \frac{C}{1 - u(x)}, \quad \text{so that} \quad f(x) = C \sum_{n=0}^{\infty} u(x)^n \quad \text{when } |u(x)| < 1.$$

Then **the inequality** $|u(x)| < 1$ **becomes your interval condition.**

Worked Example: rational function to power series

Find a power series for $f(x) = \frac{3}{2+x}$ about $x_0 = 0$ and its interval of convergence.

Rewrite:

$$\frac{3}{2+x} = \frac{3}{2} \cdot \frac{1}{1 + \frac{x}{2}} = \frac{3}{2} \cdot \frac{1}{1 - (-\frac{x}{2})}.$$

Using Theorem ??,

$$\frac{3}{2+x} = \frac{3}{2} \sum_{n=0}^{\infty} \left(-\frac{x}{2}\right)^n = \sum_{n=0}^{\infty} \frac{3(-1)^n}{2^{n+1}} x^n,$$

valid when $|\frac{x}{2}| < 1$, i.e. $|x| < 2$. Thus the radius is $R = 2$.

Worked Example: expansion about a non-zero centre (Lecture Ex. 6.7)

Find a power series representation of $\frac{1}{3x+1}$ at $x = 1$.

Rewrite:

$$\frac{1}{3x+1} = \frac{1}{3(x-1)+4} = \frac{1}{4} \cdot \frac{1}{1 - (-\frac{3}{4}(x-1))}.$$

For $|\frac{3}{4}(x-1)| < 1$, i.e. $|x-1| < \frac{4}{3}$,

$$\frac{1}{3x+1} = \frac{1}{4} \sum_{n=0}^{\infty} \left(-\frac{3(x-1)}{4}\right)^n = \sum_{n=0}^{\infty} \frac{(-1)^n 3^n}{4^{n+1}} (x-1)^n.$$

The interval of convergence is $(-\frac{1}{3}, \frac{7}{3})$ and $R = \frac{4}{3}$.

Worked Example: substituting $u = -x^2$ (Lecture Ex. 6.6)

Express $f(x) = \frac{1}{1+x^2}$ as a power series and find its interval of convergence.

Replace x by $-x^2$ in the geometric series:

$$\frac{1}{1+x^2} = \frac{1}{1-(-x^2)} = \sum_{n=0}^{\infty} (-x^2)^n = \sum_{n=0}^{\infty} (-1)^n x^{2n}.$$

Convergence when $|-x^2| < 1$, i.e. $|x| < 1$. So $R = 1$ and the interval is $(-1, 1)$.

Sanity check. At $n = 0, 1, 2$: $1 - x^2 + x^4 - \dots$. Substituting $x = 0$ gives $1 = \frac{1}{1+0}$. ✓

Common Mistake: losing the convergence condition

It is not enough to write down the series; you must also state where it is valid. Always carry the condition $|u(x)| < 1$ through your substitutions.

Building new series via calculus. The term-by-term theorem (Theorem ?? in §6.3 below) lets us differentiate and integrate a known series to get a *new* series with the *same* radius R . Two standard patterns:

- **Differentiation:** $\frac{d}{dx} \frac{1}{1-x} = \frac{1}{(1-x)^2} = \sum_{n=1}^{\infty} nx^{n-1}$ for $|x| < 1$.
Differentiating again: $\frac{2}{(1-x)^3} = \sum_{n=2}^{\infty} n(n-1)x^{n-2}$ for $|x| < 1$.
- **Integration:** $\int \frac{1}{1-x} dx = -\ln(1-x) = \sum_{n=1}^{\infty} \frac{x^n}{n} + C$ for $|x| < 1$.
Setting $C = 0$ (since $\ln 1 = 0$ at $x = 0$) gives $\ln(1-x) = -\sum_{n=1}^{\infty} \frac{x^n}{n}$.

Worked Example: $\tan^{-1} x$ via integration (Lecture Ex. 6.10)

Prove that $\tan^{-1} x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1}$ for $|x| < 1$.

Start from $\frac{1}{1+x^2} = \sum_{n=0}^{\infty} (-1)^n x^{2n}$ (valid for $|x| < 1$). Integrate both sides from 0 to x :

$$\tan^{-1} x = \int_0^x \frac{1}{1+t^2} dt = \sum_{n=0}^{\infty} (-1)^n \int_0^x t^{2n} dt = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1}.$$

The constant of integration is zero because $\tan^{-1} 0 = 0$. The radius remains $R = 1$ by Theorem ??.

Worked Example: differentiating twice (Tutorial 11, Q5(b))

Find a power series for $\left(\frac{x}{2-x}\right)^3$ and its radius of convergence.

Step 1. Rewrite: $\left(\frac{x}{2-x}\right)^3 = \frac{x^3}{8} \cdot \frac{1}{\left(1-\frac{x}{2}\right)^3}$.

Step 2. From $\frac{1}{1-u} = \sum u^n$, differentiate twice:

$$\frac{2}{(1-u)^3} = \sum_{n=2}^{\infty} n(n-1)u^{n-2} \implies \frac{1}{(1-u)^3} = \sum_{n=0}^{\infty} \binom{n+2}{2} u^n.$$

Step 3. Substitute $u = x/2$ and multiply by $x^3/8$:

$$\left(\frac{x}{2-x}\right)^3 = \sum_{n=0}^{\infty} \binom{n+2}{2} \frac{x^{n+3}}{2^{n+3}}, \quad |x| < 2.$$

So $R = 2$.

Exam Focus: choosing the right manipulation

- If f has $\frac{1}{\text{linear in } x}$: use geometric series directly.
- If f has $\frac{1}{(\text{linear})^k}$: differentiate the geometric series $k-1$ times.
- If f involves $\ln$, $\tan^{-1}$: integrate the geometric series (after a substitution).
- If f has an extra x^m factor: find the series for the rest, then multiply by x^m .

Always state the convergence condition after each manipulation.

6.3 Taylor and Maclaurin series (§6.3)

Power series become especially structured when the coefficients come from derivatives of a function.

Definition 6.2: Taylor polynomial and Taylor series

Let f be $(n+1)$ times differentiable near a . The n th Taylor polynomial of f at a is

$$P_n(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k.$$

If the infinite series converges, the Taylor series of f at a is

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!} (x-a)^k.$$

When $a = 0$ this is called the Maclaurin series.

Why these coefficients? A polynomial P_n can be made to *match f and its derivatives at a* by enforcing

$$P_n^{(k)}(a) = f^{(k)}(a) \quad (k = 0, 1, \dots, n).$$

Solving these $n+1$ matching conditions yields exactly the coefficients $\frac{f^{(k)}(a)}{k!}$. In other words, P_n is the unique degree- $\leq n$ polynomial that shares the same “local data” (value, slope, curvature, ...) as f at a .

Deriving the formula step by step (from the lecture notes). Suppose $f(x) = c_0 + c_1(x-a) + c_2(x-a)^2 + c_3(x-a)^3 + \dots$ for $|x-a| < R$.

- Setting $x = a$: $f(a) = c_0$.
- Differentiating once and setting $x = a$: $f'(a) = c_1$.
- Differentiating twice and setting $x = a$: $f''(a) = 2c_2$, so $c_2 = \frac{f''(a)}{2!}$.
- Differentiating three times and setting $x = a$: $f'''(a) = 3!c_3$, so $c_3 = \frac{f'''(a)}{3!}$.
- In general: $f^{(n)}(a) = n!c_n$, giving $c_n = \frac{f^{(n)}(a)}{n!}$.

The factor $k!$ in the denominator arises because differentiating $(x-a)^k$ exactly k times produces $k!$ (by the power rule applied repeatedly). This is why the Taylor coefficient formula has $k!$ in the denominator — it “undoes” the factorial that differentiation creates.

Theorem 6.3: Taylor’s theorem and remainder (Lagrange form)

Assume f has continuous derivatives up to order $n+1$ on an interval containing a and x . Then

$$f(x) = P_n(x) + R_n(x),$$

where P_n is the n th Taylor polynomial at a , and there exists c between a and x such that

$$R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}.$$

Why does the remainder look like that? Taylor’s theorem is fundamentally an iterated application of the **Fundamental Theorem of Calculus** (FTC). Here is the key idea:

Base case ($n = 0$): The FTC says

$$f(x) - f(a) = \int_a^x f'(u) du.$$

This is the remainder $R_0(x) = \int_a^x f'(u) du$ — the difference between $f(x)$ and the 0th Taylor polynomial $P_0(x) = f(a)$.

General case: Repeating integration by parts gives the *integral form* of the remainder:

$$R_n(x) = \frac{1}{n!} \int_a^x (x-u)^n f^{(n+1)}(u) du.$$

The Lagrange form then follows by applying the Mean Value Theorem for integrals: since $(x-u)^n$ does not change sign on $[a, x]$, there exists some c between a and x such that

$$\frac{1}{n!} \int_a^x (x-u)^n f^{(n+1)}(u) du = \frac{f^{(n+1)}(c)}{n!} \int_a^x (x-u)^n du = \frac{f^{(n+1)}(c)}{(n+1)!} (x-a)^{n+1}.$$

Proof of the integral form via IBP (from lecture notes, Theorem 5). Define $I_n = \frac{1}{n!} \int_a^x (x-u)^n f^{(n+1)}(u) du$. Use integration by parts with $U = \frac{(x-u)^n}{n!}$ and $V' = f^{(n+1)}(u)$:

$$I_n = \left[\frac{(x-u)^n}{n!} f^{(n)}(u) \right]_a^x + \frac{1}{(n-1)!} \int_a^x (x-u)^{n-1} f^{(n)}(u) du = -\frac{f^{(n)}(a)}{n!} (x-a)^n + I_{n-1}.$$

Rearranging: $I_{n-1} = \frac{f^{(n)}(a)}{n!} (x-a)^n + I_n$. Applying this recurrence n times starting from $f(x) = f(a) + I_0$ yields

$$f(x) = f(a) + \underbrace{f'(a)(x-a) + \cdots + \frac{f^{(n)}(a)}{n!} (x-a)^n}_{P_n(x)} + I_n,$$

proving $R_n(x) = I_n$.

Key Formula: Error bound for Taylor approximation

If $|f^{(n+1)}(t)| \leq M$ for all t between a and x , then

$$|R_n(x)| \leq \frac{M}{(n+1)!} |x-a|^{n+1}.$$

This is the standard tool to prove $R_n(x) \rightarrow 0$ as $n \rightarrow \infty$ for a fixed x .

Why $\frac{|x-a|^{n+1}}{(n+1)!} \rightarrow 0$ always. This fact, used in almost every “prove Taylor series equals f ” problem, deserves a short justification. Fix any real number $r = |x-a|$. For $n \geq 2r$, each new factor satisfies $\frac{r}{n+1} \leq \frac{r}{2r+1} < \frac{1}{2}$. So beyond that point, multiplying by $\frac{r}{n+1}$ halves the term each time, driving it to 0. More precisely, $\frac{r^{n+1}}{(n+1)!} \leq \frac{r^{\lceil 2r \rceil}}{\lceil 2r \rceil!} \cdot \left(\frac{1}{2}\right)^{n+1-\lceil 2r \rceil} \rightarrow 0$.

Theorem 6.4: When a Taylor series equals the function

If $\lim_{n \rightarrow \infty} R_n(x) = 0$ for a given x , then

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!} (x-a)^k$$

at that x . If this holds for all x in an interval, then the Taylor series represents f on that interval.

Worked Example: proving e^x equals its Maclaurin series (Lecture Ex. 6.12)

Step 1 (Series). Since $f^{(n)}(x) = e^x$ for all n , we have $f^{(n)}(0) = 1$, giving the Maclaurin series $\sum_{n=0}^{\infty} \frac{x^n}{n!}$ with $R = \infty$ (by ratio test).

Step 2 (Remainder $\rightarrow 0$). Pick any $d > 0$. For $|x| \leq d$, we have $|f^{(n+1)}(t)| = e^t \leq e^d$. So

$$|R_n(x)| \leq \frac{e^d}{(n+1)!} |x|^{n+1} \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Since d is arbitrary, $R_n(x) \rightarrow 0$ for *all* $x \in \mathbb{R}$.

Conclusion. By Theorem ??, $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$ for all x .

Worked Example: Maclaurin series for $\sin x$ and $\cos x$

Using derivatives of $\sin x$:

$$\sin x, \quad \cos x, \quad -\sin x, \quad -\cos x, \quad \sin x, \quad \dots$$

At $x = 0$:

$$\sin 0 = 0, \quad \cos 0 = 1, \quad -\sin 0 = 0, \quad -\cos 0 = -1, \quad \dots$$

Hence the Maclaurin series is

$$\sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots \quad (R = \infty).$$

Term-by-term differentiation (justified by Theorem ?? below) yields

$$\cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots \quad (R = \infty).$$

Proving $\sin x$ equals its Maclaurin series (Lecture Ex. 6.13). Since $f^{(n+1)}(x) = \pm \sin x$ or $\pm \cos x$, we have $|f^{(n+1)}(x)| \leq 1$ for all x . The error bound gives $|R_n(x)| \leq \frac{|x|^{n+1}}{(n+1)!} \rightarrow 0$. By the Squeeze Theorem, $R_n(x) \rightarrow 0$ for all x , so the Maclaurin series represents $\sin x$ on all of $\mathbb{R}$. The same argument applies to $\cos x$ (or use term-by-term differentiation of the $\sin x$ series).

Common Mistake: “Taylor series exists” vs “Taylor series equals f ”

Writing down $\sum \frac{f^{(k)}(a)}{k!} (x-a)^k$ does *not* automatically mean it equals $f(x)$. You must either:

- use Theorem ?? by showing $R_n(x) \rightarrow 0$ (often via the error bound), or
- identify it with a known series representation (e.g. reduce to $\sin$, $\cos$, e^x , geometric).

Exam Focus: proving $R_n(x) \rightarrow 0$ quickly

A standard pattern is:

- bound $|f^{(n+1)}(t)| \leq M$ by a constant (often 1, $e^{|t|}$, or a simple maximum);
- apply $|R_n(x)| \leq \frac{M}{(n+1)!} |x - a|^{n+1}$;
- conclude $\frac{|x-a|^{n+1}}{(n+1)!} \rightarrow 0$ as $n \rightarrow \infty$ (factorial dominates powers).

Worked Example: Taylor series at $a \neq 0$ via substitution (Tutorial 12, Q1(a))

Find the Taylor series for $\ln x$ centred at $a = 2$.

Key idea: reduce to a known Maclaurin series by substitution. Write $x = 2 + (x - 2)$, so

$$\ln x = \ln(2 + (x - 2)) = \ln 2 + \ln\left(1 + \frac{x - 2}{2}\right).$$

Let $u = \frac{x-2}{2}$. Using $\ln(1 + u) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{u^n}{n}$ for $|u| < 1$:

$$\ln x = \ln 2 + \sum_{n=1}^{\infty} (-1)^{n+1} \frac{(x - 2)^n}{n \cdot 2^n}, \quad |x - 2| < 2.$$

So $R = 2$. (Alternatively: nearest singularity of $\ln x$ is $x = 0$, distance $|2 - 0| = 2$.)

Exam tip. The substitution method (reduce to a known Maclaurin series) is almost always faster than computing $f^{(n)}(a)/n!$ by hand, especially for $\ln$, e^x , and trig functions.

Worked Example: first three nonzero Maclaurin terms (Tutorial 12, Q3)

Find the first three nonzero terms of the Maclaurin series for:

(a) $\sec x$. Since $\sec x$ is even, write $\sec x = 1 + Ax^2 + Bx^4 + O(x^6)$. Use $(\sec x)(\cos x) = 1$ and $\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{24} + O(x^6)$:

$$(1 + Ax^2 + Bx^4)\left(1 - \frac{x^2}{2} + \frac{x^4}{24}\right) = 1.$$

Matching x^2 : $A - \frac{1}{2} = 0 \Rightarrow A = \frac{1}{2}$. Matching x^4 : $B - \frac{A}{2} + \frac{1}{24} = 0 \Rightarrow B = \frac{5}{24}$. So $\sec x = 1 + \frac{x^2}{2} + \frac{5x^4}{24} + O(x^6)$.

(b) $e^x \ln(1+x)$. Multiply truncated series: $e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + O(x^4)$ and $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} + O(x^4)$. Collecting: $e^x \ln(1+x) = x + \frac{x^2}{2} + \frac{x^3}{3} + O(x^4)$.

Theorem 6.5: Term-by-term differentiation/integration of power series

Let $\sum_{n=0}^{\infty} c_n(x - x_0)^n$ have radius of convergence $R > 0$. For $|x - x_0| < R$:

1. (Differentiation) $\frac{d}{dx} \sum_{n=0}^{\infty} c_n(x - x_0)^n = \sum_{n=1}^{\infty} n c_n(x - x_0)^{n-1}$, with the same radius R .
2. (Integration) $\int \sum_{n=0}^{\infty} c_n(x - x_0)^n dx = C + \sum_{n=0}^{\infty} \frac{c_n}{n+1} (x - x_0)^{n+1}$, with the same radius R .

Why does the radius stay the same? Differentiation multiplies the n th coefficient by n , and integration divides it by $n+1$. Neither operation changes the growth rate of $|c_n|$ fast enough to alter $R = 1/\limsup \sqrt[n]{|c_n|}$. (Formally: $\sqrt[n]{n} \rightarrow 1$, so $\limsup \sqrt[n]{n|c_n|} = \limsup \sqrt[n]{|c_n|}$.)

Important caveat. The radius R is preserved, but endpoint convergence may change. For example, $\sum \frac{x^n}{n}$ converges at $x = -1$, but its derivative $\sum x^{n-1}$ (the geometric series) diverges at $x = -1$. Always re-check endpoints after differentiation or integration.

Key Formula: Important Maclaurin series (memorise for exams)

Function	Maclaurin series	Radius
$\frac{1}{1-x}$	$\sum_{n=0}^{\infty} x^n$	$R = 1$
e^x	$\sum_{n=0}^{\infty} \frac{x^n}{n!}$	$R = \infty$
$\sin x$	$\sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}$	$R = \infty$
$\cos x$	$\sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}$	$R = \infty$
$\tan^{-1} x$	$\sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1}$	$R = 1$
$\ln(1+x)$	$\sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n}$	$R = 1$
$(1+x)^k$	$\sum_{n=0}^{\infty} \binom{k}{n} x^n$	$R = 1$

6.4 The binomial series (§6.4)

Motivation. The classical Binomial Theorem $(a+b)^n = \sum_{i=0}^n \binom{n}{i} a^{n-i} b^i$ works only when n is a *non-negative integer* (the sum is finite). What if n is a fraction like $\frac{1}{2}$ (for square roots) or a negative integer like -1 (for reciprocals)? The *binomial series* answers this: it extends the expansion to *any* real exponent k , at the price of becoming an infinite series that converges only for $|x| < 1$.

Definition 6.3: Generalised binomial coefficient

For any real k and non-negative integer $n \geq 1$,

$$\binom{k}{n} = \frac{k(k-1)(k-2)\cdots(k-n+1)}{n!}.$$

By convention $\binom{k}{0} = 1$. When k is a positive integer, this reduces to the usual $\frac{k!}{n!(k-n)!}$.

Why this definition? The standard binomial coefficient $\binom{n}{i} = \frac{n!}{i!(n-i)!}$ only makes sense when n is a non-negative integer (because of the factorials). The “falling factorial” form $\frac{k(k-1)\cdots(k-n+1)}{n!}$ makes sense for *any* real k , because it only involves n factors in the numerator. When k is a positive integer and $n > k$, one of the factors in the numerator is zero, so $\binom{k}{n} = 0$ and the series terminates (recovering the classical theorem). For non-integer k , none of these factors is ever zero, so the series is genuinely infinite.

Theorem 6.6: The binomial series

If k is any real number and $|x| < 1$, then

$$(1+x)^k = \sum_{n=0}^{\infty} \binom{k}{n} x^n.$$

The radius of convergence is $R = 1$ (when k is not a non-negative integer).

Why $R = 1$? Let $a_n = \binom{k}{n} x^n$. The ratio test gives

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{|k-n|}{n+1} |x| = \frac{|1-k/n|}{1+1/n} |x| \xrightarrow{n \rightarrow \infty} |x|.$$

So the series converges when $|x| < 1$ and diverges when $|x| > 1$, giving $R = 1$. Convergence at $x = \pm 1$ depends on the specific value of k .

Worked Example: binomial series for $1/\sqrt{4-x}$ (Lecture Ex. 6.18)

Rewrite:

$$\frac{1}{\sqrt{4-x}} = \frac{1}{2} \left(1 - \frac{x}{4}\right)^{-1/2}.$$

Apply the binomial series with $k = -\frac{1}{2}$ and substitute $x \mapsto -\frac{x}{4}$:

$$\frac{1}{\sqrt{4-x}} = \frac{1}{2} \sum_{n=0}^{\infty} \binom{-1/2}{n} \left(-\frac{x}{4}\right)^n, \quad |x| < 4.$$

Using $\binom{-1/2}{n} = \frac{(-1)^n \cdot 1 \cdot 3 \cdot 5 \cdots (2n-1)}{2^n n!}$, the series simplifies to

$$\frac{1}{\sqrt{4-x}} = \frac{1}{2} + \frac{1}{2} \sum_{n=1}^{\infty} \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{n! 8^n} x^n, \quad R = 4.$$

Worked Example: cube root (Tutorial 12, Q5)

Find the Maclaurin series for $f(x) = \sqrt[3]{8+x}$ and its radius of convergence.

Rewrite: $(8+x)^{1/3} = 2\left(1 + \frac{x}{8}\right)^{1/3}$. Apply binomial series with $k = \frac{1}{3}$ and $u = \frac{x}{8}$:

$$\sqrt[3]{8+x} = 2 \sum_{n=0}^{\infty} \binom{1/3}{n} \left(\frac{x}{8}\right)^n, \quad |x| < 8.$$

Explicitly:

$$\sqrt[3]{8+x} = 2 + \frac{x}{12} - \frac{x^2}{288} + \frac{5x^3}{20736} + \dots$$

The radius is $R = 8$ (the substitution $u = x/8$ requires $|u| < 1$, i.e. $|x| < 8$).

Common Mistake: wrong form for binomial series

The binomial series applies to $(1+x)^k$, **not** $(a+x)^k$ directly. You must first factor out the constant:

$$(a+x)^k = a^k \left(1 + \frac{x}{a}\right)^k,$$

then apply the series with $u = x/a$. The convergence condition becomes $|x/a| < 1$, i.e. $|x| < |a|$.

Exam Focus: when to use the binomial series

Use the binomial series when you see:

- Square roots: $\sqrt{1+x} = (1+x)^{1/2}$, $\frac{1}{\sqrt{1+x}} = (1+x)^{-1/2}$.
- Cube roots: $\sqrt[3]{1+x} = (1+x)^{1/3}$.
- Negative integer powers: $\frac{1}{(1+x)^2} = (1+x)^{-2}$.

Always rewrite into the form $(1+u)^k$ first, then state $|u| < 1$.

6.5 Finding limits using power series (§6.5)

Why use power series for limits? Some limits of the form $\lim_{x \rightarrow 0} \frac{f(x)}{g(x)}$ are $0/0$ indeterminate forms that would require multiple rounds of L'Hôpital's rule. Power series provide a faster alternative: expand numerator and denominator, cancel the leading power of x , and read off the limit directly. This is especially efficient when the Maclaurin series are already known.

Key Formula: series limit technique

To evaluate $\lim_{x \rightarrow 0} \frac{f(x)}{g(x)}$ (a $0/0$ form):

1. Replace $f(x)$ and $g(x)$ by their Maclaurin series (keep enough terms).
2. Cancel the lowest common power of x from numerator and denominator.
3. Set $x = 0$ in what remains.

The number of terms you need equals the number of cancellations plus one.

Worked Example: series limit (Lecture Ex. 6.20)

Evaluate $\lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2}$.

Expand: $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$. Then

$$\frac{e^x - 1 - x}{x^2} = \frac{\frac{x^2}{2!} + \frac{x^3}{3!} + \dots}{x^2} = \frac{1}{2} + \frac{x}{6} + \dots \xrightarrow{x \rightarrow 0} \frac{1}{2}.$$

Worked Example: series limit with $\ln$ (Tutorial 12, Q4(a))

Evaluate $\lim_{x \rightarrow 0} \frac{2x - \ln(1 + 2x)}{x^2}$.

Use $\ln(1 + u) = u - \frac{u^2}{2} + \frac{u^3}{3} + O(u^4)$ with $u = 2x$:

$$\ln(1 + 2x) = 2x - 2x^2 + \frac{8}{3}x^3 + O(x^4).$$

Then

$$2x - \ln(1 + 2x) = 2x^2 - \frac{8}{3}x^3 + O(x^4),$$

so

$$\frac{2x - \ln(1 + 2x)}{x^2} = 2 - \frac{8}{3}x + O(x^2) \xrightarrow{x \rightarrow 0} 2.$$

Worked Example: binomial series limit (Tutorial 12, Q4(b))

Evaluate $\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1 - \frac{1}{2}x}{x^2}$.

Use $(1+x)^{1/2} = 1 + \frac{1}{2}x - \frac{1}{8}x^2 + O(x^3)$. Then

$$\sqrt{1+x} - 1 - \frac{1}{2}x = -\frac{1}{8}x^2 + O(x^3),$$

so the limit is $-\frac{1}{8}$.

Worked Example: $\lim_{x \rightarrow 0} \frac{1 - \cos x}{1 + x - e^x}$ (Lecture Ex. 6.21)

Expand:

$$1 - \cos x = \frac{x^2}{2} - \frac{x^4}{24} + \cdots,$$

$$1 + x - e^x = 1 + x - \left(1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \cdots\right) = -\frac{x^2}{2} - \frac{x^3}{6} - \cdots.$$

Divide:

$$\frac{1 - \cos x}{1 + x - e^x} = \frac{\frac{x^2}{2} + O(x^4)}{-\frac{x^2}{2} + O(x^3)} = \frac{\frac{1}{2} + O(x^2)}{-\frac{1}{2} + O(x)} \xrightarrow{x \rightarrow 0} \frac{1/2}{-1/2} = -1.$$

Summing numerical series by recognising a power series. A common exam question gives you a numerical series and asks you to find its sum. The technique: identify the series as a known power series evaluated at a specific x .

Worked Example: summing via standard series (Tutorial 12, Q6)

(a) Find $\sum_{n=0}^{\infty} (-1)^n \frac{\pi^{2n+1}}{4^{2n+1} (2n+1)!}$.

This is $\sum_{n=0}^{\infty} (-1)^n \frac{z^{2n+1}}{(2n+1)!}$ with $z = \pi/4$, which equals $\sin(\pi/4) = \frac{1}{\sqrt{2}}$.

(b) Find $\frac{1}{1 \cdot 2} - \frac{1}{3 \cdot 2^3} + \frac{1}{5 \cdot 2^5} - \cdots$.

Write it as $\sum_{n=0}^{\infty} (-1)^n \frac{(1/2)^{2n+1}}{2n+1} = \tan^{-1}(1/2)$.

Worked Example: $\sum n(n-1)/2^n$ via generating function (Tutorial 11, Q6(b))

Find $\sum_{n=2}^{\infty} \frac{n(n-1)}{2^n}$.

Step 1. From $\frac{1}{1-x} = \sum x^n$, differentiate twice: $\frac{2}{(1-x)^3} = \sum_{n=2}^{\infty} n(n-1)x^{n-2}$.

Step 2. Multiply by x^2 : $\frac{2x^2}{(1-x)^3} = \sum_{n=2}^{\infty} n(n-1)x^n$.

Step 3. Substitute $x = \frac{1}{2}$: $\frac{2 \cdot (1/4)}{(1/2)^3} = \frac{1/2}{1/8} = 4$.

Exam Focus: series limits vs L'Hôpital

- Use series when you know the Maclaurin expansions and the limit involves $x \rightarrow 0$. For $x \rightarrow a \neq 0$, substitute $h = x - a$ and expand in h .
- Use L'Hôpital when the function does not have a convenient series expansion.
- Both methods can verify each other; in exams, series is usually faster.

Extension (Beyond Syllabus: Fibonacci generating function — Tutorial 12, Q8)

Syllabus link: Chapter 6 — §6.2 Representing Functions by Power Series.

The function $f(x) = \frac{x}{1-x-x^2}$ has Maclaurin series $\sum_{n=1}^{\infty} f_n x^n$, where f_n is the n th Fibonacci number ($f_1 = f_2 = 1$, $f_n = f_{n-1} + f_{n-2}$).

Proof. Assume $f(x) = \sum_{n=1}^{\infty} a_n x^n$. From $(1-x-x^2)f(x) = x$, equate coefficients: $a_1 = 1$, $a_2 - a_1 = 0$ (so $a_2 = 1$), and for $n \geq 3$: $a_n - a_{n-1} - a_{n-2} = 0$. This is exactly the Fibonacci recurrence.

Binet formula. Factor $1-x-x^2 = -(x-\alpha)(x-\beta)$ where $\alpha = \frac{-1+\sqrt{5}}{2}$, $\beta = \frac{-1-\sqrt{5}}{2}$. Partial fractions give $f(x) = \frac{1}{\sqrt{5}} \left(\frac{1}{1-x/\alpha^{-1}} - \frac{1}{1-x/\beta^{-1}} \right)$. Expanding as geometric series and comparing yields the Binet formula:

$$f_n = \frac{1}{\sqrt{5}} (\varphi^n - \psi^n), \quad \varphi = \frac{1+\sqrt{5}}{2}, \quad \psi = \frac{1-\sqrt{5}}{2}.$$

Extension (Beyond Syllabus: coefficient-ratio formula for R — Tutorial 11, Q2)

Syllabus link: Chapter 6 — §6.1 Power Series (Radius of Convergence).

Theorem. Suppose $\sum_{n=0}^{\infty} c_n(x-a)^n$ satisfies $c_n \neq 0$ for all n , and $L = \lim_{n \rightarrow \infty} \left| \frac{c_n}{c_{n+1}} \right|$ exists. Then $R = L$.

Proof sketch. Fix $x \neq a$ and set $u_n = |c_n| |x-a|^n > 0$. Then

$$\frac{u_{n+1}}{u_n} = \frac{|c_{n+1}|}{|c_n|} |x-a| \rightarrow \frac{|x-a|}{L}.$$

By the ratio test:

- if $|x-a| < L$, then $\frac{|x-a|}{L} < 1$, so $\sum u_n$ converges and thus $\sum c_n(x-a)^n$ converges absolutely;
- if $|x-a| > L$, then $\frac{|x-a|}{L} > 1$, so $\sum u_n$ diverges and thus $\sum c_n(x-a)^n$ diverges.

Therefore the radius of convergence is $R = L$.

Why this matters (exam-safe). This gives a one-line radius computation when $\left| \frac{c_n}{c_{n+1}} \right|$ has a clean limit, avoiding repeated ratio/root test algebra.

Extension (Beyond Syllabus: Cauchy–Hadamard radius formula)**Syllabus link:** Chapter 6 — §6.1 Power Series (Radius of Convergence).**Theorem (Cauchy–Hadamard).** For $\sum_{n=0}^{\infty} c_n(x - x_0)^n$, define

$$A := \limsup_{n \rightarrow \infty} \sqrt[n]{|c_n|} \in [0, \infty].$$

Then the radius of convergence is

$$R = \frac{1}{A} \quad (\text{with } 1/0 = \infty, \ 1/\infty = 0).$$

Proof (full working). Fix x and set $r = |x - x_0|$. Consider $a_n = |c_n|r^n \geq 0$. Then

$$\limsup_{n \rightarrow \infty} \sqrt[n]{a_n} = \limsup_{n \rightarrow \infty} (\sqrt[n]{|c_n|} r) = r \cdot \limsup_{n \rightarrow \infty} \sqrt[n]{|c_n|} = rA.$$

By the root test:

- if $rA < 1$, i.e. $r < \frac{1}{A}$, then $\sum a_n$ converges, hence the power series converges absolutely;
- if $rA > 1$, i.e. $r > \frac{1}{A}$, then $\sum a_n$ diverges, hence the power series diverges.

Thus the radius is $R = 1/A$.**Why this matters (exam-safe).** It packages the radius computation into a single lim sup, and it explains why ratio/root tests lead to the same R .**Extension (Beyond Syllabus: uniqueness of power series coefficients)****Syllabus link:** Chapter 6 — §6.3 Taylor & Maclaurin Series (Power series representation).**Proposition.** Suppose $\sum_{n=0}^{\infty} c_n(x - a)^n$ converges for $|x - a| < R$ with $R > 0$, and

$$\sum_{n=0}^{\infty} c_n(x - a)^n = 0 \quad \text{for all } |x - a| < R.$$

Then $c_n = 0$ for all $n \geq 0$.**Proof (inductive).** Setting $x = a$ gives $c_0 = 0$. Subtracting c_0 and dividing by $(x - a)$: $\sum_{n=1}^{\infty} c_n(x - a)^{n-1} = 0$ for $0 < |x - a| < R$. By continuity of power series the sum is also 0 at $x = a$, giving $c_1 = 0$. Repeating yields $c_n = 0$ for all n .**Why this matters.** It guarantees that the Taylor series of f at a is the *only* power series representation of f near a . If you find *any* power series equal to f , its coefficients must be $\frac{f^{(n)}(a)}{n!}$. This is why the substitution method (e.g. reducing $\ln x$ to $\ln(1 + u)$) and the direct derivative method must give the same answer.

Extension (Beyond Syllabus: radius under $x \mapsto x^2$ — Tutorial 11, Q3)

Syllabus link: Chapter 6 — §6.1 Power Series (Radius of Convergence).

Proposition. If $\sum c_n x^n$ has radius R , then $\sum c_n x^{2n}$ has radius $\sqrt{R}$.

Proof. Let $y = x^2$. Then $\sum c_n x^{2n} = \sum c_n y^n$, which converges iff $|y| < R$, i.e. $|x^2| < R$, i.e. $|x| < \sqrt{R}$.

A Formula & Key Results Reference

Ch1: Definite integrals, FTC, substitution, improper integrals

FTC / Leibniz / MVT (use: evaluate or differentiate integrals)

FTC I (use: differentiate $\int_a^x$): $f \in C[a, b]$, $F(x) := \int_a^x f(t) dt \Rightarrow F'(x) = f(x)$.

FTC II (use: evaluate definite integrals): $f \in C[a, b]$, $\exists F$ s.t. $F' = f \Rightarrow \int_a^b f(x) dx = F(b) - F(a)$.

Leibniz, variable limits (use: $\frac{d}{dx} \int_{g(x)}^{h(x)}$): $f \in C$, $g, h \in C^1$, $\frac{d}{dx} \int_{g(x)}^{h(x)} f(t) dt = f(h(x))h'(x) - f(g(x))g'(x)$.

Leibniz, integrand depends on x (use: parameter inside integrand): $F, \partial_x F$ cont, $g, h \in C^1$,

$$\frac{d}{dx} \int_{g(x)}^{h(x)} F(x, t) dt = F(x, h(x))h'(x) - F(x, g(x))g'(x) + \int_{g(x)}^{h(x)} \partial_x F(x, t) dt.$$

MVT for integrals (use: average value / existence): $f \in C[a, b] \Rightarrow \exists c \in (a, b) : \int_a^b f(x) dx = f(c)(b - a)$,

$$f_{\text{avg}} = \frac{1}{b - a} \int_a^b f(x) dx, \quad \exists c : f(c) = f_{\text{avg}}.$$

u -substitution (use: “inner” + missing derivative)

$$u = g(x) (C^1), \quad du = g'(x) dx : \quad \int f(g(x))g'(x) dx = \int f(u) du.$$

Definite form (use: change bounds): $g \in C^1[a, b]$, f cont on $\text{range}(g)$, $\int_a^b f(g(x))g'(x) dx = \int_{g(a)}^{g(b)} f(u) du$.

Improper integrals (use: ∞ bounds or vertical asymptote)

Type I (infinite interval): $\int_a^\infty f := \lim_{t \rightarrow \infty} \int_a^t f$, $\int_{-\infty}^b f := \lim_{t \rightarrow -\infty} \int_t^b f$.

Type II (blow-up at endpoint): $\int_a^b f := \lim_{t \rightarrow a^+} \int_t^b f$ ($f \rightarrow \infty$ at a^+), similarly at b^- ;

if singular at $c \in (a, b)$: $\int_a^b f = \int_a^c f + \int_c^b f$ (both limits).

p -tests (use: compare to powers): $\int_1^\infty x^{-p} dx$ conv $\Leftrightarrow p > 1$; $\int_0^1 x^{-p} dx$ conv $\Leftrightarrow p < 1$.

Comparison (use: positive integrands): $0 \leq f \leq g$, $\int g$ conv $\Rightarrow \int f$ conv; $0 \leq g \leq f$, $\int g$ div $\Rightarrow \int f$ div.

Limit comparison (use: asymptotic match): $f, g > 0$, $\lim_{x \rightarrow \alpha} \frac{f}{g} = c \in (0, \infty) \Rightarrow \int f, \int g$ same conv/div ($\alpha = \infty$),

Ch2: Applications of integration (setup templates)

Area (use: region in xy -plane)

$$\text{w.r.t. } x : A = \int_{x=a}^b (y_{\text{top}}(x) - y_{\text{bot}}(x)) dx \quad (\text{split at intersections}).$$

$$\text{w.r.t. } y : A = \int_{y=c}^d (x_{\text{right}}(y) - x_{\text{left}}(y)) dy.$$

Volumes (use: solids; choose washers vs shells)

$$\text{Slicing: } V = \int A(x) dx \quad \text{or} \quad V = \int A(y) dy.$$

$$\text{Washers/disks about } y = k \text{ } (\perp \text{ axis}) : V = \pi \int_a^b (R_{\text{out}}(x)^2 - R_{\text{in}}(x)^2) dx,$$

$$R_{\text{out}} = |y_{\text{top}} - k|, \quad R_{\text{in}} = |y_{\text{bot}} - k|.$$

$$\text{Washers/disks about } x = k \text{ } (\perp \text{ axis}) : V = \pi \int_c^d (R_{\text{out}}(y)^2 - R_{\text{in}}(y)^2) dy,$$

$$R_{\text{out}} = |x_{\text{right}} - k|, \quad R_{\text{in}} = |x_{\text{left}} - k|.$$

$$\text{Shells about } x = k \text{ } (\parallel \text{ axis}) : V = 2\pi \int_a^b r(x)h(x) dx, \quad r = |x - k|, \quad h = y_{\text{top}} - y_{\text{bot}}.$$

$$\text{Shells about } y = k \text{ } (\parallel \text{ axis}) : V = 2\pi \int_c^d r(y)h(y) dy, \quad r = |y - k|, \quad h = x_{\text{right}} - x_{\text{left}}.$$

Arc length / surface area (use: curve length / area of revolution)

$$\text{Arc length } y = f(x) : f' \text{ cont on } [a, b], \quad L = \int_a^b \sqrt{1 + (f'(x))^2} dx.$$

$$\text{Arc length } x = g(y) : g' \text{ cont on } [c, d], \quad L = \int_c^d \sqrt{1 + (g'(y))^2} dy.$$

$$\text{Surface area about } y = k, \quad y = f(x) : S = 2\pi \int_a^b |f(x) - k| \sqrt{1 + (f'(x))^2} dx.$$

$$\text{Surface area about } x = k, \quad x = g(y) : S = 2\pi \int_c^d |g(y) - k| \sqrt{1 + (g'(y))^2} dy.$$

Work / fluids (use: slice $\Rightarrow$ density $\times$ area $\times$ distance)

$$\text{Work (1D force): } W = \int_a^b F(x) dx.$$

$$\text{Spring (Hooke): } F = kx \Rightarrow W = \int_{x=a}^b kx dx.$$

$$\text{Pumping (vertical slices at height } y) : dW = \rho g A(y) D(y) dy, \quad W = \int dW.$$

$$\text{Hydrostatic force (vertical plate): } p(y) = \rho g h(y), \quad dF = p(y) w(y) dy, \quad F = \int dF.$$

Ch3: Techniques & numerical integration

Integration by parts (use: product; reduce algebraic degree)

$$\text{Indef: } \int u \, dv = uv - \int v \, du.$$

$$\text{Def: } \int_a^b u \, dv = [uv]_a^b - \int_a^b v \, du.$$

Trig integrals / trig substitution (decision rules)

$$\int \sin^m x \cos^n x \, dx : \quad n \text{ odd} \Rightarrow u = \sin x; \quad m \text{ odd} \Rightarrow u = \cos x; \quad \text{both even} \Rightarrow (\text{see formula sheet}).$$

$$\int \tan^m x \sec^n x \, dx : \quad n \text{ even} \Rightarrow u = \tan x; \quad m \text{ odd} \Rightarrow u = \sec x.$$

$$\text{Trig-sub mapping (use: radicals): } \sqrt{a^2 - x^2} \Rightarrow x = a \sin \theta; \quad \sqrt{a^2 + x^2} \Rightarrow x = a \tan \theta; \quad \sqrt{x^2 - a^2} \Rightarrow x = a \sec \theta.$$

Partial fractions (use: rational P/Q ; do long division if $\deg P \geq \deg Q$)

$$\text{Distinct linear: } \frac{P(x)}{\prod_i (a_i x + b_i)} = \sum_i \frac{A_i}{a_i x + b_i}.$$

$$\text{Repeated linear } (ax + b)^r : \quad \sum_{k=1}^r \frac{A_k}{(ax + b)^k}.$$

$$\text{Irreducible quadratic } (ax^2 + bx + c)^r : \quad \sum_{k=1}^r \frac{A_k x + B_k}{(ax^2 + bx + c)^k}.$$

$$\text{Fast split for quadratic: } Ax + B = \lambda(2ax + b) + \mu \Rightarrow \lambda \ln(ax^2 + bx + c) + (\arctan \text{ term; see formula sheet}).$$

Numerical rules + error (use: approximate $\int_a^b f$)

$$\text{Let } \Delta x = \frac{b-a}{n}, \quad x_i = a + i\Delta x.$$

$$L_n = \sum_{i=1}^n f(x_{i-1})\Delta x, \quad R_n = \sum_{i=1}^n f(x_i)\Delta x, \quad M_n = \sum_{i=1}^n f\left(\frac{x_{i-1} + x_i}{2}\right)\Delta x,$$

$$T_n = \frac{\Delta x}{2} \left(f(x_0) + 2 \sum_{i=1}^{n-1} f(x_i) + f(x_n) \right) = \frac{L_n + R_n}{2},$$

$$S_n = \frac{\Delta x}{3} \left(f(x_0) + 4 \sum_{\substack{1 \leq i \leq n-1 \\ i \text{ odd}}} f(x_i) + 2 \sum_{\substack{2 \leq i \leq n-2 \\ i \text{ even}}} f(x_i) + f(x_n) \right) \quad (n \text{ even}).$$

Error bounds (need smoothness):

$$f'' \text{ cont, } |f''| \leq K : \quad |E_T| \leq \frac{K(b-a)^3}{12n^2}, \quad |E_M| \leq \frac{K(b-a)^3}{24n^2}.$$

$$f^{(4)} \text{ cont, } |f^{(4)}| \leq K \text{ (} n \text{ even)} : \quad |E_S| \leq \frac{K(b-a)^5}{180n^4}.$$

Over/under-estimate cues:

$$f \uparrow : L_n \leq \int_a^b f \leq R_n \quad (\text{reverse if } f \downarrow), \quad f'' > 0 : T_n \geq \int, \quad M_n \leq \int \quad (\text{reverse if } f'' < 0).$$

Ch4: Sequences & series (definitions + core tests)

Sequences (use: limit / monotone bounded)

$$a_n \rightarrow L \iff \forall \varepsilon > 0 \exists N : n \geq N \Rightarrow |a_n - L| < \varepsilon, \quad a_n \text{ monotone \& bounded} \Rightarrow a_n \text{ converges.}$$

Series basics (use: partial sums + remainder)

$$\sum_{n=1}^{\infty} a_n \text{ converges} \iff S_n := \sum_{k=1}^n a_k \rightarrow S \in \mathbb{R}, \quad R_n := S - S_n = \sum_{k=n+1}^{\infty} a_k.$$

Necessary condition: $\sum a_n$ convergent $\Rightarrow a_n \rightarrow 0$ (not sufficient).

Geometric / p -series (high frequency)

$$\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r} \quad (|r| < 1) \text{ else diverges,} \quad \sum_{n=1}^{\infty} \frac{1}{n^p} \text{ conv} \Leftrightarrow p > 1.$$

Convergence tests (applicability cues)

Assume $a_n \geq 0$ when required.

Comparison (use: $0 \leq a_n \leq b_n$): $\sum b_n \text{ conv} \Rightarrow \sum a_n \text{ conv}$; $\sum a_n \text{ div} \Rightarrow \sum b_n \text{ div}$.

Limit comparison (use: asymptotic match): $a_n, b_n > 0$, $\lim \frac{a_n}{b_n} = c \in (0, \infty) \Rightarrow \sum a_n, \sum b_n \text{ same}$.

Integral test (use: $a_n = f(n)$, $f \downarrow$, $f > 0$): $\sum_{n=1}^{\infty} a_n \text{ conv} \Leftrightarrow \int_1^{\infty} f(x) dx \text{ conv}$,

$$\int_{n+1}^{\infty} f \leq R_n \leq \int_n^{\infty} f \quad (\text{remainder bound}).$$

Alternating (Leibniz; use: $(-1)^n b_n$): $b_n \downarrow$, $b_n \rightarrow 0 \Rightarrow \sum (-1)^n b_n \text{ conv}$, $|R_n| \leq b_{n+1}$.

Absolute vs conditional (use: take abs): $\sum |a_n| \text{ conv} \Rightarrow \sum a_n \text{ conv}$ (absolute); else may be conditional.

Ratio (use: factorial/exponential): $L = \limsup \left| \frac{a_{n+1}}{a_n} \right|$; $L < 1 \Rightarrow \sum a_n \text{ abs conv}$, $L > 1 \Rightarrow \text{div}$.

Root (use: n -th powers): $L = \limsup \sqrt[n]{|a_n|}$; $L < 1 \Rightarrow \text{abs conv}$, $L > 1 \Rightarrow \text{div}$.

Ch5: Taylor / Maclaurin polynomials (approximation)

Definitions (use: local approximation near

a)

$$T_n(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k, \quad \text{Maclaurin: } a = 0.$$

Taylor's theorem (Lagrange remainder; need $f^{(n+1)}$ cont near a):

$$f(x) = T_n(x) + R_n(x), \quad R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} (x-a)^{n+1} \text{ for some } \xi \text{ between } x, a.$$

Error bound (use: $|f^{(n+1)}| \leq M$ on interval between a and x):

$$|R_n(x)| \leq \frac{M}{(n+1)!} |x-a|^{n+1}.$$

Common Maclaurin expansions: (see formula sheet).

Ch6: Power series (radius/interval + termwise ops)

Definitions & radius/interval workflow (use: ratio test)

$$\sum_{n=0}^{\infty} c_n (x-a)^n \quad (\text{power series about } a), \quad \text{converges for } |x-a| < R, \text{ diverges for } |x-a| > R.$$

Compute R via ratio test (typical):

$$L(x) = \limsup_{n \rightarrow \infty} \left| \frac{c_{n+1}(x-a)^{n+1}}{c_n(x-a)^n} \right| = \limsup_{n \rightarrow \infty} \left| \frac{c_{n+1}}{c_n} \right| \cdot |x-a|, \quad L(x) < 1 \Rightarrow \text{conv.}$$

Endpoints $x = a \pm R$: test separately (often comparison / alternating / integral test).

Term-by-term differentiation/integration (use: inside radius)

For $|x-a| < R$,

$$\frac{d}{dx} \sum_{n=0}^{\infty} c_n (x-a)^n = \sum_{n=1}^{\infty} n c_n (x-a)^{n-1}, \quad \int \sum_{n=0}^{\infty} c_n (x-a)^n dx = C + \sum_{n=0}^{\infty} \frac{c_n}{n+1} (x-a)^{n+1},$$

with the *same* radius of convergence R (check endpoints separately after integrating/differentiating).

B Notation

Symbol	Meaning	Where / typical use
$\int f(x) dx$	Indefinite integral / antiderivative $F'(x) = f(x)$	Find families of primitives; IBP, u -sub, PF
$\int_a^b f(x) dx$	Definite integral (net area)	FTC II evaluation; area/volume/work templates
t (dummy var.) $u = g(x)$	Placeholder variable in integrals/series Substitution variable	FTC I: $F(x) = \int_a^x f(t) dt$ u -sub: match $du = g'(x) dx$; change bounds in definite form
$\int_a^\infty f, \int_{-\infty}^b f$	Improper (Type I) as limits of proper integrals	Convergence via p -test, comparison, limit comparison
$\int_a^b f, f \rightarrow \infty$ at end-point	Improper (Type II) limit at singularity	Split at interior singularities; evaluate by limits
f_{avg}	Average value $\frac{1}{b-a} \int_a^b f$	MVT for integrals; average-value problems
A	Area	Area between curves: $A = \int (\text{top} - \text{bot}) dx$ or $\int (\text{right} - \text{left}) dy$
V	Volume	Washers/shells/slicing formulas
L	Arc length	$L = \int \sqrt{1 + (f')^2} dx$ or $\int \sqrt{1 + (g')^2} dy$
S	Surface area of revolution	$S = \int 2\pi(\text{radius}) ds$
W	Work	$W = \int F(x) dx$; fluids: $dW = \rho g A(y) D(y) dy$
ρ, g	Density / gravitational accel.	Hydrostatic force, pumping work
$\sum_{n=1}^\infty a_n$	Infinite series	Convergence tests; compare to geometric / p -series
$S_n = \sum_{k=1}^n a_k$	Partial sums	$\sum a_n$ converges iff $S_n \rightarrow S$
S	Series sum (if convergent)	$S = \lim_{n \rightarrow \infty} S_n$
$R_n = S - S_n$	Remainder (tail)	Error estimate; alternating: $ R_n \leq b_{n+1}$
$\rightarrow, \lim$	Convergence / limit	$a_n \rightarrow L, S_n \rightarrow S$
$\sum a_n $	Absolute series	Absolute vs conditional convergence
$\sim$	Asymptotic equivalence: $f \sim g \iff f/g \rightarrow 1$	Limit comparison for integrals/series; simplify large- n behavior
$O(g(x))$	Big-O: $ f \leq C g $ near point / ∞	Error bounds, asymptotic estimates, test setup
$o(g(x))$	Little-o: $f/g \rightarrow 0$	Sharper asymptotics; remainder comparisons
$\sum_{n=0}^\infty c_n(x-a)^n$	Power series about a	Radius/interval of convergence; termwise ops
R	Radius of convergence	Usually from ratio/root test
IOC	Interval of convergence	$(a-R, a+R)$ plus endpoint checks
$T_n(x)$	Taylor polynomial degree n about a	Approximation; series generation
$R_n(x)$	Taylor remainder	Lagrange form; error bounds
$n!$	Factorial ($n \cdot (n-1) \cdots 1$; $0! = 1$)	Taylor coefficients; ratio test with factorial terms
$\binom{n}{k}$	Binomial coefficient $\frac{n!}{k!(n-k)!}$	Binomial expansions; series manipulations
$\approx$	Numerical approximation	Taylor / numerical integration approximations

C Exam & Assessment Information

Assessment Breakdown (Course-Specific Summary)

- Assignment 1: 5% (released Week 4; due late Week 7).
- Wooclap class participation: 5% (full marks typically requires meeting a completion threshold across in-class activities).
- Midterm test: 20% (Week 8).
- Assignment 2: 10% (released Week 9; due mid-semester).
- Final exam: 60%.

Midterm / Final Notes (Placeholders for Personalisation)

- Bring writing instruments, student card, and any permitted materials.
- Timing strategy: allocate time per question; reserve a buffer for checks.
- Show working clearly; make substitutions and limit steps explicit.

Allowed Reference Sheet Reminder

- You are allowed **one double-sided A4 reference sheet** with handwritten or typed text.
- Do **not** attach sticky notes/post-its to the sheet.

Revision Checklist

- I can set up a definite integral correctly (bounds, variable, integrand).
- I can choose between substitution vs. integration by parts vs. partial fractions.
- I can classify an integral as improper and apply the correct limit definition.
- I can distinguish sequence vs. series and use an appropriate convergence test.
- I can compute an interval of convergence for a power series and test endpoints.